Derivatives – Study Notes

Conceptual edition from the 2026 course material

12 October 2026

This concept-organized study edition complements the source-faithful `Derivatives_Notes`. The theory chapters retain the supplied technical content, technical figures, tables and examples alongside explanations. The full question sheet is interleaved with clearly labelled generated answers. Source-page references and the coverage manifest record the reorganisation and explicit administrative and contextual visual exclusions. OCR screening and coverage checks do not certify every original formula. No recordings were supplied.

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Part I
Theory

1 Market mechanics and compounding

1.1 What a derivative is—and what it is for

A *derivative* is a contract between two parties whose value depends on something else, called the *underlying*. The underlying may be a stock, bond, exchange rate, index, metal, or commodity. An option gives a conditional right, a forward or future fixes a later exchange, and a swap exchanges specified cash-flow streams. The contract itself is not the underlying: it specifies who receives what, and when, as the underlying changes.

The same contract can serve different aims. A hedger offsets an existing exposure; a speculator chooses an exposure; and a trader may use a derivative to change exposure with fewer transactions in the underlying. The course contrasts exchange-traded and over-the-counter markets with historical charts of contract counts and notional amounts. These measures describe different things: the count is the number of open contracts, whereas notional is the reference quantity used to calculate payments. Neither is, by itself, the amount at risk or the current market value of those contracts.

What is a Derivative?

Definition

An agreement between two parties which has a value determined by the price of something else:

A stock like Apple

- A bond such as a T-Bond
- A currency such as the EUR/CHF rate
- An index such as the S&P500
- A metal like Gold
- A commodity like Soy beans

Types

- Options, futures, and swaps.

Uses

Risk management

- Speculation
- Reduce transaction costs

Exchange Traded Derivatives: Contracts Outstanding

source: Bank for International Settlements, Quarterly Review, June 2013.

Exchange Traded Derivatives: Notional Amount

source: Bank for International Settlements, Quarterly Review, June 2012.

Over-the-Counter (OTC) Derivatives

source: Bank for International Settlements, Quarterly Review, June 2012.

1.2 The price you see is not always the price you trade

The *ask* (offer) is the market-maker's selling price, so a buyer pays it. The *bid* is the market-maker's buying price, so a seller receives it. Normally bid $\leq$ ask; their difference is the *spread*. Brokerage commissions add to a purchase and reduce sale proceeds. For N shares, proportional commission rate c, bid B, and ask A,

$$\begin{aligned} \text{purchase cash paid} &= NA(1+c), \\ \text{sale cash received} &= NB(1-c), \\ \text{immediate round-trip cost} &= N\big[A(1+c) - B(1-c)\big]. \end{aligned}$$

Thus an unchanged quoted market can still produce a loss on an immediate buy–sell round trip. The spread and both commissions matter; applying the bid to a purchase reverses the market-maker's perspective.

Worked example

The lecture's ABC quote is $B = \$40.95$, $A = \$41.05$, $N = 100$, and $c = 0.003$. Buying costs $100(41.05)(1.003) = \$4{,}117.315$, displayed to cents as \$4,117.32. Selling returns $100(40.95)(0.997) = \$4{,}082.715$, displayed as \$4,082.72. The round-trip cost is \$34.60, calculated from the unrounded values. Of this, \$10 is the quoted spread on 100 shares and \$24.60 is commission.

Buying and Selling a Financial Asset

Brokers: commissions

Market-makers: bid-ask spread (reflects the perspective of the market-maker)

Investor action	Quotation	Market-maker perspective
The price at which you can buy	ask (offer)	What the market-maker will sell for
The price at which you can sell	bid	What the market-maker pays

Example: Buy and sell 100 shares of XYZ

- XYZ: bid=\$49.75, ask=\$50, commission=\$15
  Buy: $(100 \times 50) + 15 = 5{,}015$
- Sell: $(100 \times 49.75) - 15 = 4{,}960$
- Transaction cost: \$5,015 – \$4960 = \$55

Problem 1

Problem 1: ABC stock has a bid price of \$40.95 and an ask price of \$41.05. Assume that the brokerage fee is quoted as 0.3% of the bid or ask price.

a. What amount will you pay to buy 100 shares?
b. What amount will you receive for selling 100 shares?
c. Suppose you buy 100 shares, then immediately sell 100 shares. What is your round-trip transaction cost?

Problem 1

Problem 1: ABC stock has a bid price of \$40.95 and an ask price of \$41.05. Assume that the brokerage fee is quoted as 0.3% of the bid or ask price.

a. What amount will you pay to buy 100 shares?

$$(\$41.05 \times 100) + (\$41.05 \times 100) \times 0.003 = \$4,117.32$$

b. What amount will you receive for selling 100 shares?

$$(\$40.95 \times 100) - (\$40.95 \times 100) \times 0.003 = \$4,082.72$$

c. Suppose you buy 100 shares, then immediately sell 100 shares. What is your round-trip transaction cost?

$$\$4,117.32 - \$4,082.72 = \$34.6$$

1.3 Short selling: a sale followed by a purchase

To take a short position in a share, a trader borrows it and sells it now, then later buys a share to return to the lender. Ignoring collateral interest, borrow fees, commissions and dividends, the cash profit per share is $S_0 - S_T$. A fall in price helps the short seller; a rise hurts. Unlike the maximum gain on a stock bought outright, the loss from a rising stock price is not capped by the initial price.

If a dividend D is paid while the share is on loan, the short seller must compensate the lender. A simple cash-flow ledger for one share is:

	Opening sale	Dividend date	Close and return
Share movement	Borrow, then sell	—	Buy, then return
Cash flow	$+S_0$	$-D$	$-S_T$

The broker locates the borrowed shares, may obtain them from its inventory or another account, and manages the lending and return. This operational step matters: the ability to short an asset is conditional on locating a borrow. If many short sellers need to buy scarce shares to close at once, their covering demand can drive a *short squeeze*. The lecture uses Volkswagen's 2008 price surge as an illustration, not as a general pricing rule.

Worked example

In the course example, short 300 XYZ shares at \$30.19, cover at \$29.87, and pay a 0.5% commission on both trades. Initial cash received is $300(30.19)(0.995) = \$9{,}011.715$; the purchase to cover costs $300(29.87)(1.005) = \$9{,}005.805$. The profit before any borrow fee or dividend is \$5.91. The raw price move would have yielded $300(0.32) = \$96$; commissions consume \$90.09 of that amount.

Short-Selling

When price of an asset is expected to fall

- First: borrow and sell the asset (get \$\$)
  Then: buy back and return the asset (pay \$)
  If price fell in the mean time: Profit \$ = \$\$ - \$
  The lender must be compensated for dividends received

Example: Cash flows associated with short-selling a share of IBM for 90 days. Note that the short-seller must pay the dividend, D, to the share-lender.

Source table, row by row:

- Day 0; Dividend Ex-Day; Day 90
- Action; Borrow shares; —; Return shares
- Action; Sell shares; —; Purchase shares
- Cash; +S0; -D; -S90

Short selling form the perspective of a broker

- A trader places a short sale order

The broker searches its own inventory, another trader's margin account, or even another brokerage firm's inventory to locate the shares that the client wants to borrow

If the stock is located, the short sale order is filled and the trader sells the shares in the market

Once the transaction is placed, the broker does the lending $\Rightarrow$ any benefit (interest for lending out the shares) belongs to the broker

The broker is responsible for returning the shares (not a big risk due to margin requirements)

VW's 348% Two-Day Gain Is Pain for Hedge Funds

From the Wall Street Journal, 2008:

In short squeezes, investors who borrowed and sold stock expecting its value to fall exit from the trades by buying those shares, or covering their positions. That can send a stock upward if shares are hard to come by. When shares are scarce, that can push a company-s market capitalization well beyond a reasonable valuation. [...] Indeed, the recent stock gains left Volkswagen's market value at about $346 billion, just below that of the world's largest publicly traded corporation, Exxon Mobil Corp.

SHARE PRICE IN GERMANY
Volkswagen
Tuesday: EUR945 ($1,200), up 82%
October change: up 244%
Source: Thomson Reuters

Problem 2

Problem 2: Suppose you short-sell 300 shares of XYZ stock at $30.19 with a commission charge of 0.5%.

Supposing you pay commission charges for purchasing the security to cover the short-sale, how much profit have you made if you close the short-sale at a price of $29.87?

Problem 2

Problem 2: Suppose you short-sell 300 shares of XYZ stock at $30.19 with a commission charge of 0.5%.

Supposing you pay commission charges for purchasing the security to cover the short-sale, how much profit have you made if you close the short-sale at a price of $29.87?

Initially, we will receive the proceeds form the sale of the asset, less the proportional commission charge:

$$300 \times (30.19) - 300 \times (30.19) \times 0.005 = 9,011.72$$

When we close out the position, we will again incur the commission charge, which is added to the purchasing cost:

$$300 \times (29.87) + 300 \times (29.87) \times 0.005 = 9,005.81$$

Finally, we receive total profits of: $9,011.72 - $9,005.81 = $5.91.

1.4 One cash flow, several rate conventions

An interest rate is incomplete without its compounding convention. If \$1 grows for T years, an effective annual rate r_{eff}, a nominal annual rate r_n compounded n times a year, and a continuously compounded annual rate r_c produce, respectively,

$$(1 + r_{\text{eff}})^T, \qquad \left(1 + \frac{r_n}{n}\right)^{nT}, \qquad e^{r_c T}.$$

These are alternative descriptions of the same accumulation only when their numerical rates are converted consistently. Equating one-year growth gives

$$r_c = \ln(1 + r_{\text{eff}}), \qquad r_{\text{eff}} = e^{r_c} - 1, \qquad r_c = n \ln\left(1 + \frac{r_n}{n}\right).$$

For a zero-coupon bond priced P today and paying face value F at time T, discounting and its inverse are

$$P = Fe^{-r_c T} = \frac{F}{(1 + r_{\text{eff}})^T}, \qquad r_c = \frac{1}{T} \ln\left(\frac{F}{P}\right).$$

Worked example

The lecture's five-year zero-coupon bond costs \$62.092 and pays \$100. Its effective annual return is $(100/62.092)^{1/5} - 1 \approx 10\%$. The continuously compounded quote is $\ln(100/62.092)/5 \approx 9.531\%$. The two figures do *not* disagree: $e^{0.09531} \approx 1.10$.

Continuous Compounding

- Terms often used to refer to interest rates:
- Effective annual rate r: if you invest \$1 today, T years later you will have

$$(1 + r)^T$$

- Annual rate r, compounded n times per year: if you invest \$1 today, T years later you will have

$$\left(1 + \frac{r}{n}\right)^{nT}$$

- Annualized continuously compounded rate r: if you invest \$1 today, T years later you will have

$$e^{rT} \equiv \lim_{n \to \infty} \left(1 + \frac{r}{n}\right)^{nT}$$

Continuous Compounding: Example

- Suppose you have a zero-coupon bond that matures in 5 years. The price today is \$62.092 for a bond that pays \$100.
  The effective annual rate of return is

$$\left(\frac{100}{62.092}\right)^{1/5} - 1 = 0.10$$

  The continuously compounded rate of return is

$$\frac{\ln(\$100/\$62.092)}{5} = \frac{0.47655}{5} = 0.09531$$

The continuously compounded rate of return of 9.53% corresponds to the effective annual rate of return of 10%. To verify this, observe that

$$e^{0.09531} = 1.10$$

or

$$\ln(1.10) = \ln\left(e^{0.09531}\right) = 0.09531$$

1.5 Why log returns add across time

The one-period simple return from S_0 to S_1 is $R = S_1/S_0 - 1$. The corresponding log return is $\ell = \ln(S_1/S_0)$. Over successive periods,

$$\ell_{0,2} = \ln\frac{S_2}{S_0} = \ln\frac{S_1}{S_0} + \ln\frac{S_2}{S_1} = \ell_{0,1} + \ell_{1,2}.$$

Simple percentage returns do not add this way: gross returns $1 + R$ multiply. This is the practical advantage of logs when aggregating across time; it does not mean log returns and simple returns are numerically interchangeable.

Worked example

A share goes from \$100 to \$200, then back to \$100. Simple returns are $+100\%$ and -50%, not opposites, although the final price equals the initial price. Log returns are $\ln 2 \approx +69.31\%$ and $\ln(1/2) \approx -69.31\%$; they sum to zero. A log return may be below -100% because it is the logarithm of a positive price ratio, whereas a simple return on a limited-liability asset cannot fall below -100%.

Why it matters

Keep the cash-flow viewpoint throughout this course. First identify *what* is exchanged and *when*; then choose the relevant transaction price, fees and dividend obligations; finally use a consistent interest-rate convention to compare cash flows at different dates.

Continuous Compounding

When we multiply exponentials, exponents add. So we have

$$e^x e^y = e^{x+y}$$

This makes calculations of average rate of return easier.
When using continuous compounding, increases and decreases are symmetric.
Moreover, continuously compounded returns can be less than -100%

Problem 3

Problem 3: Suppose that over 1 year a stock price increases from \$100 to \$200. Over the subsequent year it falls back to \$100.

What is the effective return over the first year? What is the continuously compounded return?
What is the effective return over the second year? The continuously compounded return?
What do you notice when you compare the first- and second-year returns computed arithmetically and continuously?

Problem 3

Problem 3: Suppose that over 1 year a stock price increases from \$100 to \$200. Over the subsequent year it falls back to \$100.

What is the effective return over the first year? What is the continuously compounded return?

$$\text{effectivereturn} = \frac{\$200 - \$100}{\$100} = 100\%$$

$$\text{continuouslycompoundedreturn} = \ln\left(\frac{\$200}{\$100}\right) = 69.31\%$$

What is the effective return over the second year? The continuously compounded return?

$$\text{effectivereturn} = \frac{\$100 - \$200}{\$200} = -50\%$$

$$\text{continuouslycompoundedreturn} = \ln\left(\frac{\$100}{\$200}\right) = -69.31\%$$

What do you notice when you compare the first- and second-year returns computed arithmetically and continuously?

2 Forwards, futures and the cost of carrying an asset

2.1 Start with the contractual cash flows

A *forward* is a binding agreement made today to exchange an underlying asset at a specified future date T for a delivery price K. It fixes the asset and quantity, delivery date and place, and payment. A long forward is obliged to buy; a short forward is obliged to sell. An ordinary newly agreed forward typically has zero value at inception, but it has a payoff at delivery:

$$\Pi_T^{\text{long}} = S_T - K, \qquad \Pi_T^{\text{short}} = K - S_T.$$

Here S_T is the spot price at delivery. The two payoffs sum to zero before counterparty, funding and other costs. *Payoff* at expiration is not the same as today's contract value; nor does it include the initial cash flows of a separate stock position.

Worked example

If the agreed six-month delivery price is $K = \$50$, the long's payoff is:

S_T	\$40	\$45	\$50	\$55	\$60
$S_T - K$	−\$10	−\$5	\$0	\$5	\$10

Its payoff is a straight line crossing zero at $S_T = K$; the short's line is its mirror image. In the lecture's separate index example, $K = \$1{,}020$ and $S_T = \$1{,}050$ give +\$30 to the long and −\$30 to the short per index unit.

Forward Contracts

Definition: a binding agreement (obligation) to buy/sell an underlying asset in the future, at a price set today.

- A forward contract specifies:

1. The features and quantity of the asset to be delivered
2. The delivery logistics, such as time, date, and place
3. The price the buyer will pay at the time of delivery

Payoff (Value at Expiration) of a Forward Contract

- Every forward contract has both a buyer and a seller.
  The term long is used to describe the buyer and short is used to describe the seller.
  Payoff for
- Long forward = Spot price at expiration – Forward price
- Short forward = Forward price – Spot price at expiration
  Example: S&P index:
  Today: Spot price = \$1,000. 6-month forward price = \$1,020
  In 6 months at contract expiration: Spot price = \$1,050
- Long position payoff = \$1,050 - \$1,020 = \$30
  Short position payoff = \$1,020 - \$1,050 = -\$30

Payoff Diagram for a Forward

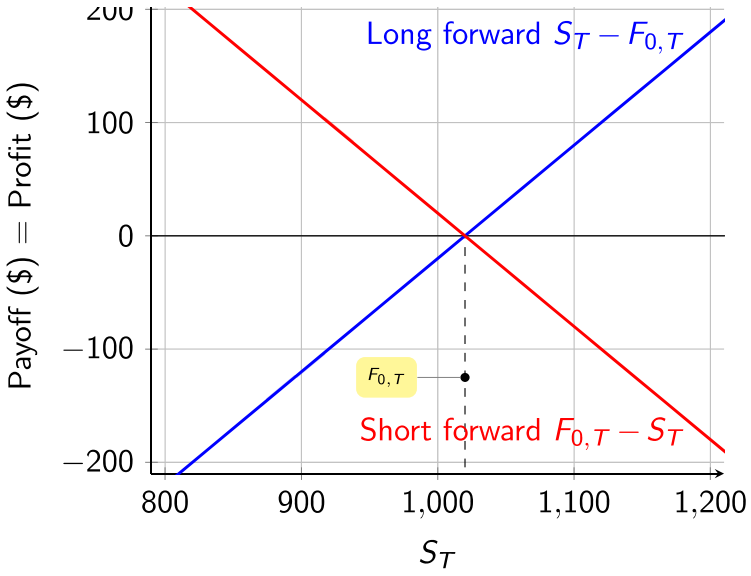


Figure 1: Forward payoffs: long $S_T - F_{0,T}$ and short $F_{0,T} - S_T$; both cross zero at the delivery price.

Problem 4

Problem 4: Suppose you enter in a long 6-month forward position at a forward price of \$50. What is the payoff in 6 months for prices of \$40, \$45, \$50, \$55, and \$60?

Problem 4

Problem 4: Suppose you enter in a long 6-month forward position at a forward price of \$50. What is the payoff in 6 months for prices of \$40, \$45, \$50, \$55, and \$60?
The payoff to a long forward at expiration is equal to:
Payoff to long forward = Spot price at expiration – Forward price
Therefore, we can construct the following table:

Price of asset in 6 months	Payoff of the long forward
40	-10
45	-5
50	0
55	5
60	10

2.2 Derive the fair delivery price by replication

Consider a stock index with spot level S_0, continuously compounded funding rate r, and continuous dividend yield δ. Assume, for this classroom derivation, that borrowing and lending use the same rate, dividends are reinvestable at the stated yield, and trading, taxes, margin and credit frictions can be ignored. To deliver one unit at T, buy $e^{-\delta T}$ units today and reinvest dividends: the holding grows to one unit. Its present cost is $S_0 e^{-\delta T}$. Financing that cost to T requires $S_0 e^{(r-\delta)T}$. Hence the no-arbitrage forward delivery price is

$$F_{0,T} = S_0 e^{(r-\delta)T}.$$

The exponent is the *net carry*: financing raises the forward price and dividends lower it. In the special case of no income, $\delta = 0$ and $F_{0,T} = S_0 e^{rT}$.

The following ledger makes the arbitrage argument concrete. If the quoted $F_{0,T}$ is *above* fair value, buy the dividend-reinvested asset position, finance it, and short the forward:

Position	At 0	At T
Buy $e^{-\delta T}$ units; reinvest dividends	$-S_0 e^{-\delta T}$	$+S_T$
Borrow the purchase cost	$+S_0 e^{-\delta T}$	$-S_0 e^{(r-\delta)T}$
Short forward	0	$F_{0,T} - S_T$
Net	0	$F_{0,T} - S_0 e^{(r-\delta)T} > 0$

If the quoted forward is *below* fair value, reverse every trade. The terminal net becomes $S_0 e^{(r-\delta)T} - F_{0,T} > 0$. The forward-pricing formula is therefore a statement about equivalent cash flows under those assumptions, not a prediction of S_T.

Alternative ways to buy a stock

1. Outright Purchase:
   Pay S_0
- Receive security
   Time 0
2. Forward:
   Pay $F_{0,T}$ =?
- Receive security
   Time T
   A forward contract is an arrangement in which you both pay for the stock and receive it at time T, with the time T price specified at time 0.
   What should you pay for the stock in this case?
- Arbitrage ensures that there is a very close relationship between prices and forward prices

Pricing a Forward Contract

Let S_0 be the spot price of an asset at time 0, and r the continuously compounded interest rate. Assume that dividends are continuous and paid at a rate δ.

Then the forward price at a future time T must satisfy

$$F_{0,T} = S_0 e^{(r-\delta)T} \tag{1}$$

- Suppose that $F_{0,T} > S_0 e^{(r-\delta)T}$. Then an investor can execute the following trades at time 0 (buy low and sell high) and obtain an arbitrage profit:

Transaction	Time 0	Time T (expiration)
Buy tailed position in stock ($e^{-\delta T}$ units)	$-S_0 e^{-\delta T}$	S_T
Borrow $S_0 e^{-\delta T}$	$+S_0 e^{-\delta T}$	$-S_0 e^{(r-\delta)T}$
Short forward	0	$F_{0,T} - S_T$
Total	0	$F_{0,T} - S_0 e^{(r-\delta)T} > 0$

Pricing a Forward Contract (cont'd)

- Suppose that $F_{0,T} < S_0 e^{(r-\delta)T}$. Then an investor can execute the following trades at time 0 (buy low and sell high) and obtain once again an arbitrage profit:

Transaction	Time 0	Time T (expiration)
Short tailed stock position ($e^{-\delta T}$ units)	$S_0 e^{-\delta T}$	$-S_T$
Lend $S_0 e^{-\delta T}$	$-S_0 e^{-\delta T}$	$S_0 e^{(r-\delta)T}$
Long forward	0	$S_T - F_{0,T}$
Total	0	$S_0 e^{(r-\delta)T} - F_{0,T} > 0$

Consequently, and assuming that the non-arbitrage condition holds, we have

$$F_{0,T} = S_0 e^{(r-\delta)T}$$

2.3 What changes when a forward becomes a future?

Futures are standardized, exchange-traded contracts with specified settlement procedures. Their distinguishing cash-flow feature here is *marking to market*: gains and losses are settled periodically, normally daily, rather than being left entirely until maturity. The clearing system requires margin as collateral. The *notional value* of a position is the current futures price times its contract multiplier and number of contracts; it is not the cash paid to purchase the underlying.

The lecture uses a historical S&P 500 futures contract with multiplier \$250 per index point, quarterly contract months, and cash settlement. The corresponding Bloomberg screenshots also show a natural-gas contract with a different unit, tick and physical-delivery schedule. Those dated specifications illustrate why one must read the actual contract terms; they are not current trading specifications. In the historical index example, eight contracts at 1,100 points represent $8(250)(1100) = \$2.2$ million of notional exposure, even though a 10% illustrative initial margin is only \$220,000.

Bloomberg: CTM

The historical contract monitor lists futures categories. HELP gives explanations and MENU provides related functions.

Bloomberg: SPX CT

The historical index-contract table illustrates that each expiry has its own price, bid, ask, volume and open interest.

Bloomberg: SPZ3 DES

The historical S&P contract-description screen provides the contract multiplier, tick, expiry, settlement and margin specifications.

Bloomberg: NGX3 DES

The historical natural-gas screen describes a physically delivered contract of 10,000 MMBtu. The quoted tick is \$0.001 per MMBtu, or \$10 per contract. Its displayed margin requirements are:

Margin requirement	Speculator	Hedger
Initial	2,310	2,100
Secondary	2,100	2,100

Forward Contracts vs Futures Contracts

Forward and futures contracts are essentially the same except for the daily resettlement feature of futures contracts, called marking-to-market.

Because futures are exchange-traded, they are standardized and have specified delivery dates, locations, and procedures.

- Plenty of information is available from: www.cmegroup.com

The S&P 500 Futures Contract

Specifications for the S&P 500 index futures contract

- Underlying: S&P 500 index
- Where traded: Chicago Mercantile Exchange
- Size: \$250 $\times$ S&P 500 index
- Months: Mar, Jun, Sep, Dec
- Trading ends: Business day prior to determination of settlement price
- Settlement: Cash-settled, based upon opening price of S&P 500 on third Friday of expiration month
- Suppose the futures price is 1100 and you wish to enter into 8 long futures contracts.
  The notional value of 8 contracts is

$$8 \times 250 \times 1100 = 2,000 \times 1100 = 2.2 \text{ million}$$

2.4 Margin magnifies cash-flow pressure, not the underlying payoff

For a long futures position with N contracts and multiplier m, a settlement-price change $F_t - F_{t-1}$ adds $Nm(F_t - F_{t-1})$ to the margin account. In the course's simplified weekly example, the balance also earns interest at a continuously compounded annual rate of 6%:

$$M_t = M_{t-1}e^{0.06/52} + Nm(F_t - F_{t-1}), \qquad Nm = 8(250) = 2{,}000.$$

The lecture uses weekly settlement for exposition; real exchange settlement is daily. Starting from $M_0 = \$220{,}000$, a fall from 1,100 to 1,027.99 produces $2{,}000(-72.01) = -\$144{,}020$ of variation margin. The week-one balance is

$$\$220{,}000e^{0.06/52} - \$144{,}020 = \$76{,}233.99.$$

The futures price fell about 6.55%, but the margin balance fell about 65% because margin was one tenth of notional. The account is collateral, not a price paid for the contract; this is why percentage changes in the account should not be confused with the return on the underlying.

A broker can demand a top-up when the balance falls below *maintenance margin*. The displayed lecture path continues without modelling the cash contribution following such a call, so it is best read as an illustrative no-top-up balance path, not as a fully executable account history. At week ten, its balance is \$44,990.57; subtracting the future value of the initial margin gives an illustrative financing-adjusted profit of

$$\$44{,}990.57 - \$220{,}000e^{0.06(10/52)} = -\$177{,}562.60.$$

For a single forward with the same entry and terminal prices, but no intervening settlement, the terminal payoff is $2{,}000(1{,}011.65 - 1{,}100) = -\$176{,}700$. The difference here comes from the timing and financing of interim futures cash flows. The exact relationship between a futures price and a forward price depends on rate uncertainty and settlement assumptions; periodic settlement alone does not justify declaring them universally unequal.

The S&P 500 Futures Contract (cont'd)

- Suppose that there is 10% margin and weekly settlement (in practice settlement is daily). The margin on futures contracts with a notional value of \$2.2 million is \$220,000.

  The margin balance today from long position in 8 S&P 500 futures contracts is

Week	Multiplier (\$)	Futures price	Price change	Margin balance (\$)
0	2000.00	1100.00	—	220,000.00

Over the first week, the futures price drops 72.01 points to 1027.99. On a mark-to-market basis, we have lost

$$\$2,000 \times (-72.01) = -\$144,020$$

Thus, if the continuously compounded interest rate is 6%, our margin balance after one week is

$$\$220,000 \times e^{0.06\times 1/52} - \$144,020 = \$76,233.99$$

The S&P 500 Futures Contract (cont'd)

Because we have a 10% margin, a 6.5% decline in the futures price results in a 65% decline in margin. The margin balance after the first week is

Week	Multiplier (\$)	Futures price	Price change	Margin balance (\$)
0	2000.00	1100.00	—	220,000.00
1	2000.00	1027.99	-72.01	76,233.99

The decline in margin balance means the broker has significantly less protection should we default. For this reason, participants are required to maintain the margin at a minimum level, called the maintenance margin. This is often set at 70% to 80% of the initial margin level.

In this example, the broker would make a margin call, requesting additional margin.

We can go on for a period of 10 weeks, assuming weekly marking-to-market and a continuously compounded risk-free rate of 6%.

The S&P 500 Futures Contract (cont'd)

The margin balance after a period of 10 weeks is

Week	Multiplier ($)	Futures price	Price change	Margin balance ($)
0	2000.00	1100.00	—	220,000.00
1	2000.00	1027.99	-72.01	76,233.99
2	2000.00	1037.88	9.89	96,102.01
3	2000.00	1073.23	35.35	166,912.96
4	2000.00	1048.78	-24.45	118,205.66
5	2000.00	1090.32	41.54	201,422.13
6	2000.00	1106.94	16.62	234,894.67
7	2000.00	1110.98	4.04	243,245.86
8	2000.00	1024.74	-86.24	71,046.69
9	2000.00	1007.30	-17.44	36,248.72
10	2000.00	1011.65	4.35	44,990.57

The 10-week profit on the position is obtained by subtracting from the final margin balance the future value of the original margin investment:

$$\$44,990.57 - \$220,000 \times e^{0.06\times 10/52} = -\$177,562.60$$

The S&P 500 Futures Contract (cont'd)

What if the position had been forwarded rather than a futures position, but with prices the same? In that case, after 10 weeks our profit would have been

$$(1011.65 - 1100) \times 2,000 = -176,700$$

The futures and forward profits differ because of the interest earned on the mark-to-market proceeds (in the present cases, we have founded losses as they occurred and not at expiration, which explains the loss).

2.5 Why use index futures?

An index future can change broad market exposure without trading every constituent stock. In the lecture's temporary shift from equities to bonds, shorting an index future can offset some equity exposure while the stock portfolio remains owned. That may reduce transaction costs and simplify a temporary allocation decision. The hedge is not automatically perfect: a particular portfolio may behave differently from the index, which creates *basis risk* and motivates the cross-hedging chapter. Choosing the number and direction of contracts requires the portfolio's actual sensitivity, the contract multiplier and the target exposure.

Why it matters

Forwards and futures share the basic linear exposure $S_T - K$, but their cash-flow timing differs. Pricing begins with a replicating stock-and-financing strategy; risk management begins with the actual contract multiplier, settlement schedule and margin requirements.

Uses Of Index Futures

Why buy an index futures contract instead of synthesizing it using the stocks in the index? — Lower transaction costs

Asset allocation: switching investments among asset classes. Example: invested in the S&P 500 index and wish to temporarily invest in bonds instead of the index. What to do?

- Alternative #1: sell all 500 stocks and invest in bonds
- Alternative #2: take a short forward position in S&P 500 index
  Cross-hedging: hedge portfolios that are not exactly the index
  Risk management for stock-pickers
- More in Chapter 5, Section 5.5 of McDonald (2009)

3 Options: asymmetric payoffs, parity and insurance

3.1 A right is different from an obligation

A call gives its buyer the right, but not the obligation, to buy the underlying at strike K; a put gives the right to sell it. The option seller must perform if the buyer validly exercises. European exercise occurs only at expiration, American exercise can occur up to expiration, and Bermudan exercise is permitted on specified dates. For a European option at expiration,

$$C_T = (S_T - K)^+ = \max(S_T - K, 0), \qquad P_T = (K - S_T)^+ = \max(K - S_T, 0).$$

The writer's payoff is the negative of the buyer's payoff. A call is *in the money* when $S > K$, a put when $S < K$; at the money means $S = K$. Out-of-the-money options have *zero intrinsic value*, not a negative exercised payoff: their unprotected exercise expression would be negative, so the holder simply does not exercise. These definitions concern exercise value, not whether a trade is profitable after the premium.

Call Options

A non-binding agreement (right but not an obligation) to buy an asset into the future, at a price set today

Preserves the upside potential, while at the same time eliminating the downside

The seller of a call option is obligated to deliver if asked

Definition and terminology

A call option gives the owner the right but not the obligation to buy the underlying asset at a predetermined price during a predetermined time period

Strike (or exercise) price: the amount paid by the option buyer for the asset if he/she decides to exercise

Exercise: the act of paying the strike price to buy the asset

Expiration: the date by which the option must be exercised or becomes worthless

Exercise style: specifies when the option can be exercised

European-style: can be exercised only at expiration date

American-style: can be exercised at any time before expiration

Bermudan-style: can be exercised during specified periods

Moneyness

In-the-money option: positive payoff if exercised immediately

At-the-money option: zero payoff if exercised immediately

Source wording: "Out-of-the-money option: negative payoff if exercised immediately". This refers to the negative raw exercise expression. Because the holder can decline to exercise, the option has zero intrinsic value, not a negative option payoff.

Bloomberg: WFC US OMON

The dated option-chain screen records strikes, expiries and market quotes; it is a historical example, not current market data.

Put Options

A put option gives the owner the right but not the obligation to sell the underlying asset at a predetermined price during a predetermined time period.

The payoff of the put option is

$$P_T = \max[K - S_T, 0] \tag{4}$$

The option profit is computed as

$$\text{Put profit} = \max[K - S_T, 0] - \text{future value of premium} \tag{5}$$

3.2 Payoff versus profit

An option buyer pays a premium at inception. To compare all cash flows at expiration, carry that premium forward at the relevant funding rate. Thus a purchased call's terminal *profit* is $(S_T - K)^+ - FV(c_0)$, and a purchased put's is $(K - S_T)^+ - FV(p_0)$. A written option receives the premium first and has the opposite terminal option payoff; its terminal profit includes the future value of that received premium. The buyer's maximum loss on a purchased option is its funded premium, while a written call can have an unbounded loss as S_T rises.

Worked example

For the lecture's six-month call with $K = \$1{,}000$ and future premium value \$95.68, profit is zero when $(S_T - 1{,}000)^+ = 95.68$, namely at $S_T = \$1{,}095.68$. At $S_T = \$900$, the call expires unused: payoff is zero but profit is $-\$95.68$.

For its separate one-year, 40-strike put, an initial \$3.26 premium at an 8% effective annual rate becomes \$3.5208 at maturity. The long put's payoff is $(40 - S_T)^+$; its profit is $(40 - S_T)^+ - 3.5208$, with break-even $S_T = 36.4792$.

Call Option Example

Consider a call option on the S&R index with 6 months to expiration and strike price of \$1,000.

In six months at contract expiration: if spot price is

\$1,100 $\Rightarrow$ call buyer's payoff = \$1,100 - \$1,000 = \$100, call seller's payoff = -\$100

\$900 $\Rightarrow$ call buyer's payoff = \$0, call seller's payoff = \$0

The payoff of a call option is then

$$C_T = \max[S_T - K, 0] \tag{2}$$

where K is the strike price, and S_T is the spot price at expiration.

The option profit is computed as

$$\text{Call profit } = \max[S_T - K, 0] - \text{ future value of premium} \tag{3}$$

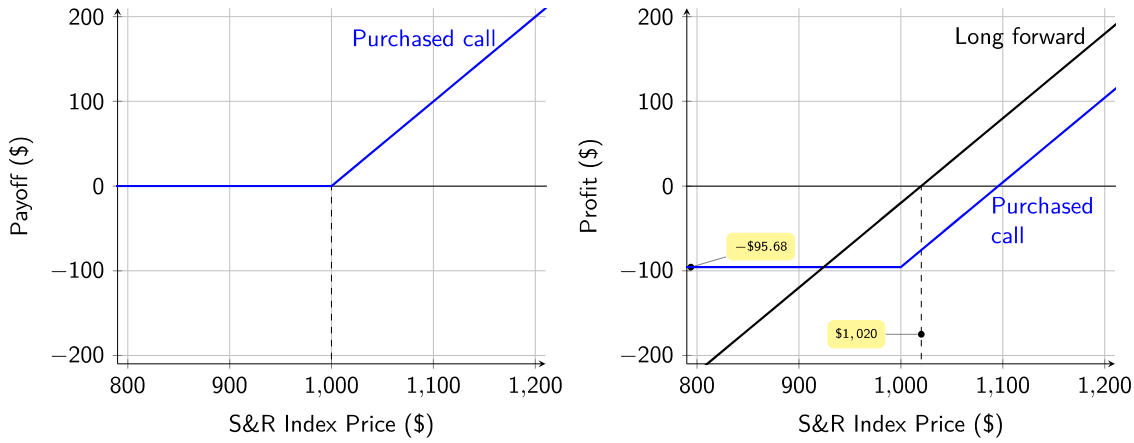


Figure 2: Long call: the left curve is $(S_T - K)^+$; the profit curve on the right subtracts the financed premium. Exercise begins at K; break-even is above K.

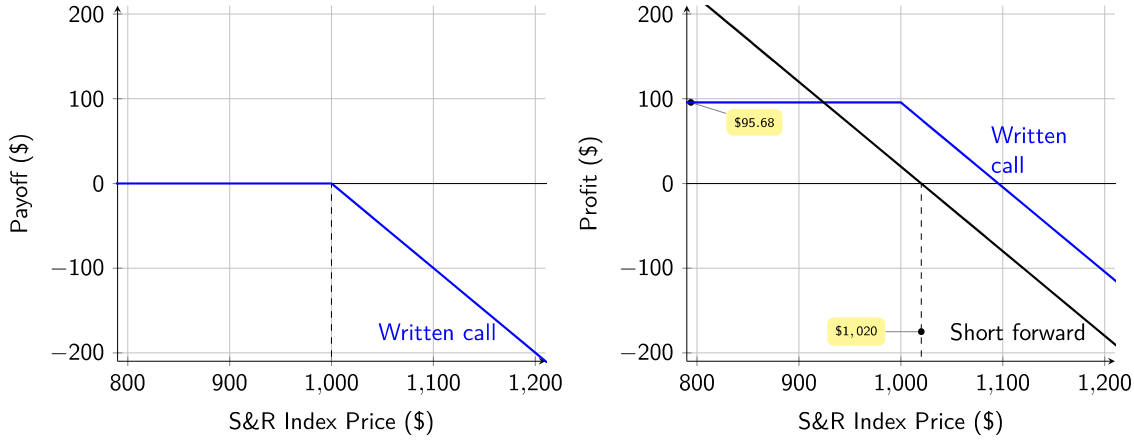


Figure 3: Short call: the premium creates a positive profit plateau below K, but losses are unbounded as S_T rises.

Diagrams for Purchased Call

Diagrams for Written Call

Problem 5

Problem 5: Consider a call option on the S&R index with 6 months to expiration and strike price of $1,000. The future value of the option premium is $95.68. For the figure below, which plots the profit on a purchased call, find the S&R index price at which the call option diagram intersects the x-axis.

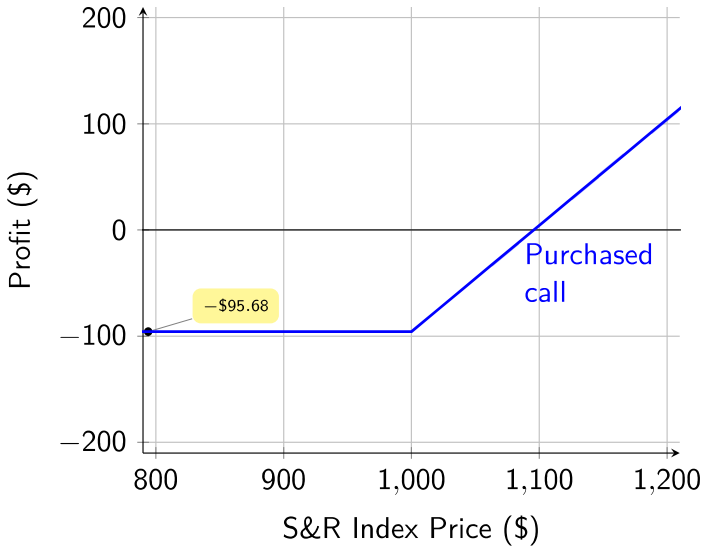


Figure 4: Call-profit exercise: the premium future value is 95.68, so the zero-profit point is $1000 + 95.68 = 1095.68$.

Problem 5

Problem 5: Consider a call option on the S&R index with 6 months to expiration and strike price of $1,000. The future value of the option premium is $95.68. For the figure below, which plots the profit on a purchased call, find the S&R index price at which the call option diagram intersects the x-axis.

The profit of the long call option is:

$$\max[0, S_T - 1,000] - 95.68$$

To find the S&R index price at which the call option diagram intersects the x-axis, we have to set the above equation equal to zero. We get

$$S_T = 1,095.68$$

Bloomberg: WFC 10/19/13 C41 OV

The dated valuation screen illustrates the contract inputs, model valuation and sensitivities for the specified call.

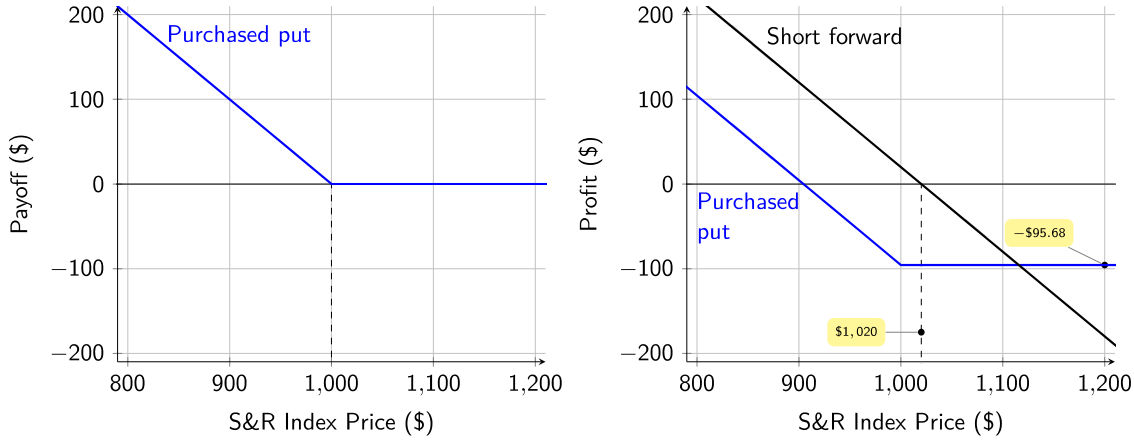


Figure 5: Long put: payoff decreases until $S_T = K$ and is then zero; financed premium shifts profit downward.

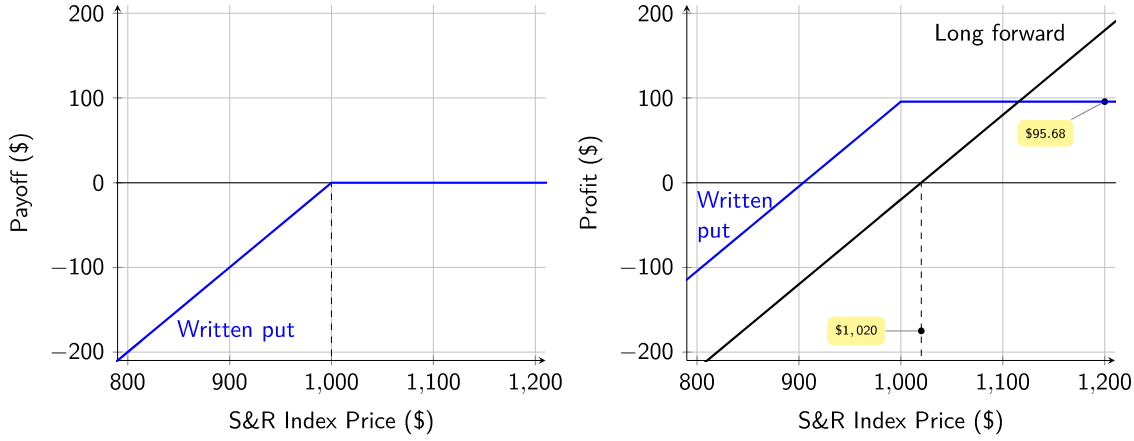


Figure 6: Short put: profit is capped at the financed premium; the downward-price risk lies below the strike.

Diagrams for Purchased Put

Diagrams for Written Put

Problem 6

Problem 6: Suppose the stock price is \$40 and the effective annual interest rate is 8%. Draw payoff and profit diagrams for a 40-strike put with a premium of \$3.26 and maturity of 1 year.

Problem 6

Problem 6: Suppose the stock price is \$40 and the effective annual interest rate is 8%. Draw payoff and profit diagrams for a 40-strike put with a premium of \$3.26 and maturity of 1 year.

In order to be able to draw the profit diagram, we need to find the future value of the put premium:

$$\begin{aligned} FV(\text{ premium }) &= 3.26 \times (1 + 0.08) \\ &= 3.5208 \end{aligned}$$

We get the following payoff and profit diagram:

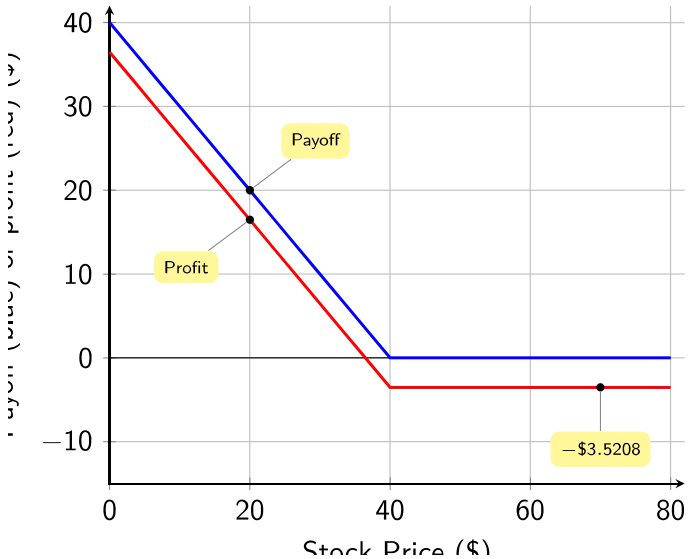


Figure 7: Put exercise: profit is payoff minus $3.26(1.08) = 3.5208$. The separation between the two curves is constant.

3.3 Put–call parity comes from equal terminal payoffs

Buy a European call and write a European put with the same stock, strike and maturity. Whatever S_T, the combined terminal payoff is

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K.$$

The same payoff arises from owning one share and borrowing the present value of K: the share is worth S_T and the loan repayment is K. If the stock pays no dividends before maturity and the frictionless assumptions apply, equal payoffs must have equal current prices:

$$c_0 - p_0 = S_0 - PV(K), \qquad PV(K) = Ke^{-rT}$$

for a continuously compounded rate r. Equivalently, $c_0 = p_0 + S_0 - PV(K)$ and $p_0 = c_0 - S_0 + PV(K)$. This is a replication identity, not a statement that calls and puts should have equal

premiums. For an at-the-money, non-dividend-paying stock, $S_0 = K$, so $c_0 - p_0 = K - PV(K) > 0$ when interest is positive.

Worked example

In the lecture's six-month index exercise, $S_0 = \$1{,}000$, $c_0 = \$109.20$, $p_0 = \$60.18$, and the *effective six-month* interest rate is 2%. Hence $109.20 - 60.18 = 1000 - K/1.02$, so $K = \$970.00$. The 2% is already a six-month rate; it must not be compounded for another half-year.

When the stock pays known dividends before expiration, the share in the replication delivers those dividends while the options do not. Subtract their present value: $c_0 - p_0 = S_0 - PV(\text{Div}) - PV(K)$. These exact equalities are for European options under their stated assumptions; early exercise changes the American-option relationships.

Put-Call Parity

- Suppose you are buying a call option and selling a put option on a non-dividend paying stock. Both options have maturity T and strike price K:

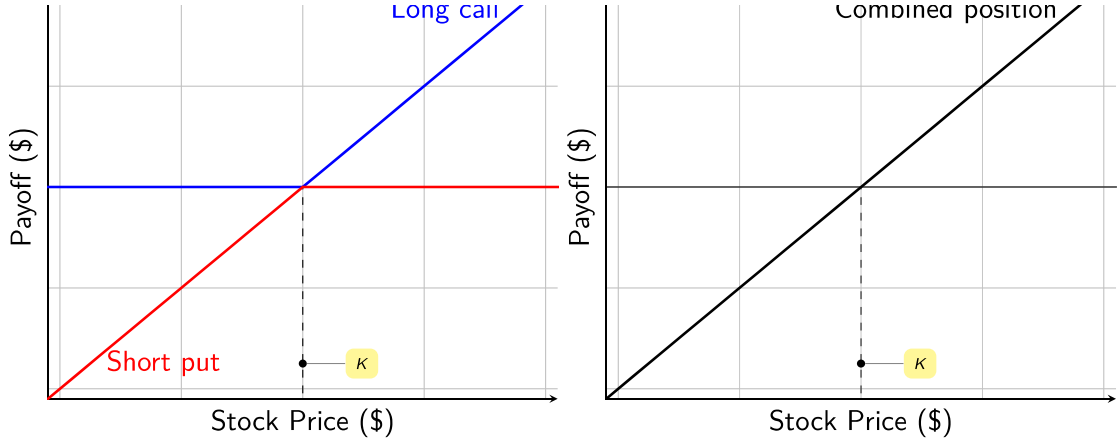


Figure 8: Long call plus short put: the two signed legs add to the linear payoff $S_T - K$, which explains put–call parity.

Put-Call Parity (cont'd)

- Your payoff at maturity is

$$\begin{aligned} C_T - P_T &= \max[S_T - K, 0] - \max[K - S_T, 0] \\ &= \max[S_T - K, 0] + \min[S_T - K, 0] \\ &= S_T - K \end{aligned}$$

  We have two strategies with the same payoff at maturity:
  Buy a call and sell a put, thus paying a premium of $C_t - P_t$ today
  Buy a share of the stock and borrow $PV(K)$, thus paying a premium of $S_t - PV(K)$ today
- Positions that have the same payoff should have the same cost (Law of one price):

$$C_t - P_t = S_t - PV(K) \tag{6}$$

  Equation (6) is known as put-call parity, and one of the most important relations in options.

Put-Call Parity (cont'd)

Parity provides a cookbook for the synthetic creation of options. It tells us that

$$C_t = P_t + S_t - PV(K) \tag{7}$$

and that

$$P_t = C_t - S_t + PV(K) \tag{8}$$

The first relation says that a call is equivalent to a leveraged position on the underlying asset, which is insured by the purchase of a put. The second relation says that a put is equivalent to a short position on the stock, insured by the purchase of a call

Parity generally fails for American-style options, which may be exercised prior to maturity.

Why Does the Price of an At-the-Money call Exceed the Price of an At-the-Money put?

Parity shows that the reason for the call being more expensive is the time value of money:

$$C_t - P_t = K - PV(K) > 0 \tag{9}$$

Problem 7

Problem 7: The S&R index price is \$1,000 and the effective 6-month interest rate is 2%. Suppose the premium on a 6-month S&R call is \$109.20 and the premium on a 6-month put with the same strike price is \$60.18. What is the strike price?

Problem 7

Problem 7: The S&R index price is \$1,000 and the effective 6-month interest rate is 2%. Suppose the premium on a 6-month S&R call is \$109.20 and the premium on a 6-month put with the same strike price is \$60.18. What is the strike price?

This question is a direct application of the Put-Call Parity:

$$C_t - P_t = S_t - PV(K)$$

$$\$109.20 - \$60.18 = \$1,000 - \frac{K}{1.02}$$

$$K = 970.00$$

Put-Call Parity for Dividend Paying Stocks

If the stock is paying dividends over the lifetime of the option, the put-call parity becomes

$$C_t - P_t = [S_t - PV(\text{Div})] - PV(K) \tag{10}$$

where $PV(\text{Div})$ is the present value of the stream of dividends paid on the stock until maturity.

- Hence, we can write

$$C_t = P_t + [S_t - PV(\text{ Div })] - PV(K) \tag{11}$$

$$P_t = C_t - [S_t - PV(\text{ Div })] + PV(K) \tag{12}$$

Equations (11)-(12) help us to find maximum and minimum option prices.

3.4 Bounds are quick arbitrage checks

A non-dividend-paying European call cannot cost more than its stock, and a European put cannot cost more than the discounted strike: $0 \le c_0 \le S_0$, $0 \le p_0 \le PV(K)$. Nonnegative option prices plus parity also imply

$$c_0 \ge \max\{0, S_0 - PV(K)\}, \qquad p_0 \ge \max\{0, PV(K) - S_0\}.$$

These are bounds, not valuation formulas. They can expose impossible quotes even before choosing a stochastic pricing model.

Worked example

The lecture asks whether a one-month European put with $K = \$50$, $S_0 = \$46$, premium \$2.50, and 10% interest is too cheap. Interpreting the unspecified 10% as an annual continuously compounded rate, $PV(K) = 50e^{-0.10/12} \approx \$49.585$, so the lower bound is approximately \$3.585. The put is underpriced by about \$1.085. Buy the put and share, and borrow $PV(K)$: the initial net cash is positive, while the terminal package pays $\max(S_T, 50) - 50 \ge 0$. The conclusion is unchanged under an effective-annual 10% convention, but the exact arbitrage amount changes slightly; the source does not specify the convention.

Maximum and Minimum Option Prices: Call Price

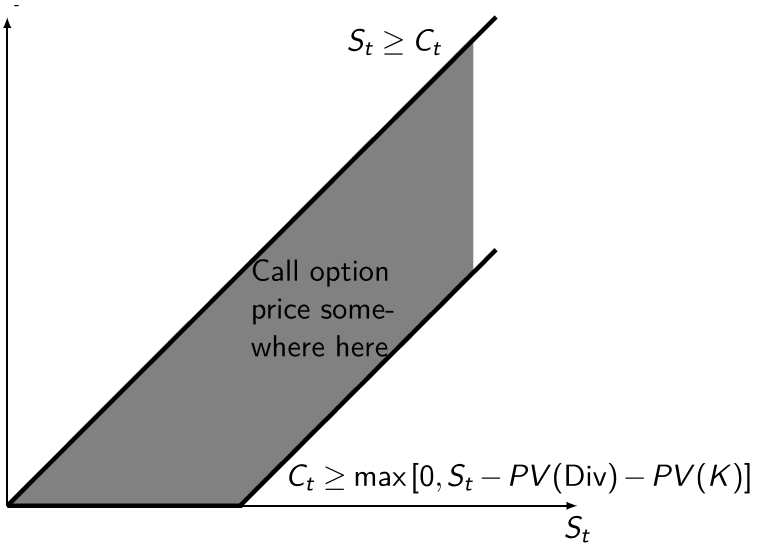


Figure 9: Call-price bounds: the shaded region lies below S_t and above $\max(0, S_t - PV(\text{Div}) - PV(K))$. The horizontal variable is today's stock price, not the terminal payoff variable.

Maximum and Minimum Option Prices: Put Price

Why it matters

The source drawing gives the loose upper bound $P_t \le K$. Its printed dividend term in the lower bound has a minus sign; for a European put with known dividends, put–call parity gives the corrected lower bound $P_t \ge \max\{0, PV(K) - S_t + PV(\text{Div})\}$. This correction follows from $p_t = c_t - S_t + PV(\text{Div}) + PV(K)$ and $c_t \ge 0$. Its K upper bound is looser than the European discounted-strike bound under nonnegative interest rates. For European

options with known dividends, the corresponding call lower bound is $C_t \geq \max\{0, S_t - PV(\text{Div}) - PV(K)\}$. Do not apply an exact European parity identity to American early-exercise rights.

Problem 8 (Minimum and Maximum Bounds, Arbitrage)

A one-month European put option on a non-dividend paying stock is currently selling for 2.50. The option has a strike of 50 and the underlying is currently worth 46. The interest rate is 10%.

Is there an arbitrage opportunity? If yes, show how you would implement it.

3.5 Build the payoff you need

Options can reshape an existing exposure. A long stock plus a put has terminal value $S_T + (K - S_T)^+ = \max(S_T, K)$: the put creates a *floor* before premium and financing. A short stock plus a call has terminal value $-S_T + (S_T - K)^+ = -\min(S_T, K)$, limiting the damage from a price rise before costs. A principal-protected equity-linked deposit combines repayment of principal with an option-like participation in upside. The historical course example invests \$10,000 and, after 5.5 years, promises $10{,}000[1 + 0.7\max(0, S_T/1300 - 1)]$; issuer credit risk and the opportunity cost of the locked-up principal are separate from this contractual payoff.

Common multi-option structures can be read from their legs. A *bull call spread* buys a lower-strike call and writes a higher-strike call: the sold upside reduces premium but caps the gain. A *straddle* buys a call and put at the same strike, seeking a large move in either direction; a *strangle* uses out-of-the-money strikes and is typically cheaper but needs a larger move to pay off. A *collar* can combine a protective put with a written call around a stock holding, placing a floor and a cap on its terminal value. A butterfly combines options to concentrate payoff near a middle strike. A ratio spread sells more options than it buys, so a low or zero initial premium does not imply low risk; its tail exposure must be inspected. Strategy names alone are insufficient: write down each signed leg's payoff before discussing profit.

The lecture's payoff-identification exercise could not be solved from OCR alone, so its source-page figure was recovered. It is a *long call butterfly* with strikes 90, 100 and 110: buy one 90-call, write two 100-calls, and buy one 110-call. The terminal payoff is

$$(S_T - 90)^+ - 2(S_T - 100)^+ + (S_T - 110)^+.$$

It is zero outside $[90, 110]$, rises with slope $+1$ from 90 to 100, reaches 10 at 100, then falls with slope -1 to 110. This identifies the positions from the graph, but it is a *payoff* only; option premiums are not supplied for a profit calculation. The quoted 2006 option-market activity findings are historical observations, not a forecast about current customer behavior.

Why it matters

For every option strategy, separate three questions: What rights and obligations exist? What is the terminal *payoff* in each price region? What *profit* remains after bringing premiums to the same date? Parity and bounds check the answer before any model-specific valuation.

Problem 9 (Building Payoffs)

Below is a payoff diagram for a position. All options have 1 year to maturity and the stock price today is \$100. The yearly interest rate (continuously compounded) is 8%. The underlying asset (the stock) is not paying any dividends.

Option	Call(90)	Call(100)	Call(110)
Position to identify	_____	_____	_____

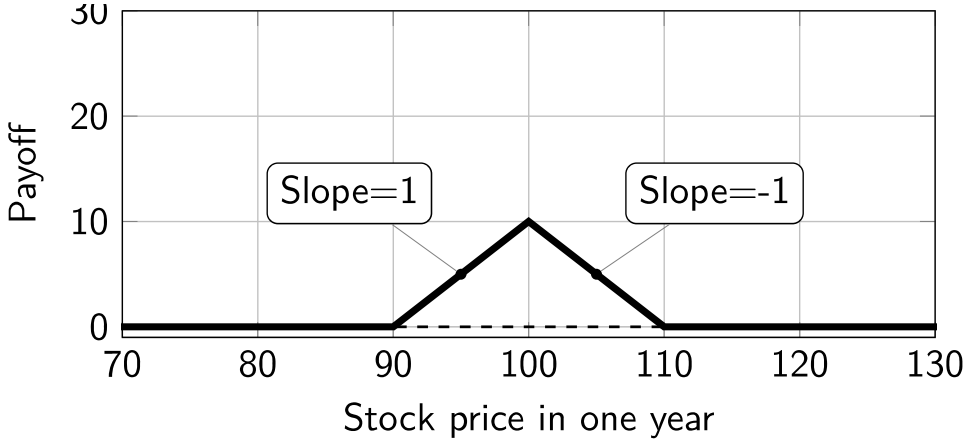


Figure 10: Identify the butterfly: slope $+1$ on $(90, 100)$, slope -1 on $(100, 110)$, peak 10 and zero payoff outside these strikes.

Example: Equity-Linked CDs

A 1,999 First Union National Bank CD promises to repay in 5.5 years initial invested amount and 70% of the gain in S&P 500 index (this is a principal protected equity-linked CD)

- Assume \$10,000 invested when S&P 500 = 1,300
  Final payoff is

$$\$10,000 \times \left(1 + 0.7 \times \max\left[0, \frac{S_{final}}{1300} - 1\right]\right)$$

  where S_{final} = value of the S&P 500 after 5.5 years.

Options are Insurance: Insuring a Long Position (Floors)

- A put option is combined with a position in the underlying asset
- Goal: to insure against a fall in the price of the underlying asset (when one has a long position in that asset)

Options are Insurance: Insuring a Short Position (Caps)

- A call option is combined with a position in the underlying asset
- Goal: to insure against an increase in the price of the underlying asset (when one has a short position in that asset)

Various Strategies: Payoffs

Various Strategies: Positions

The source strike-position grid is reconstructed below. A dash means no position; quantities refer to option contracts. The collar here contains only option legs: adding a long stock position gives a collared stock.

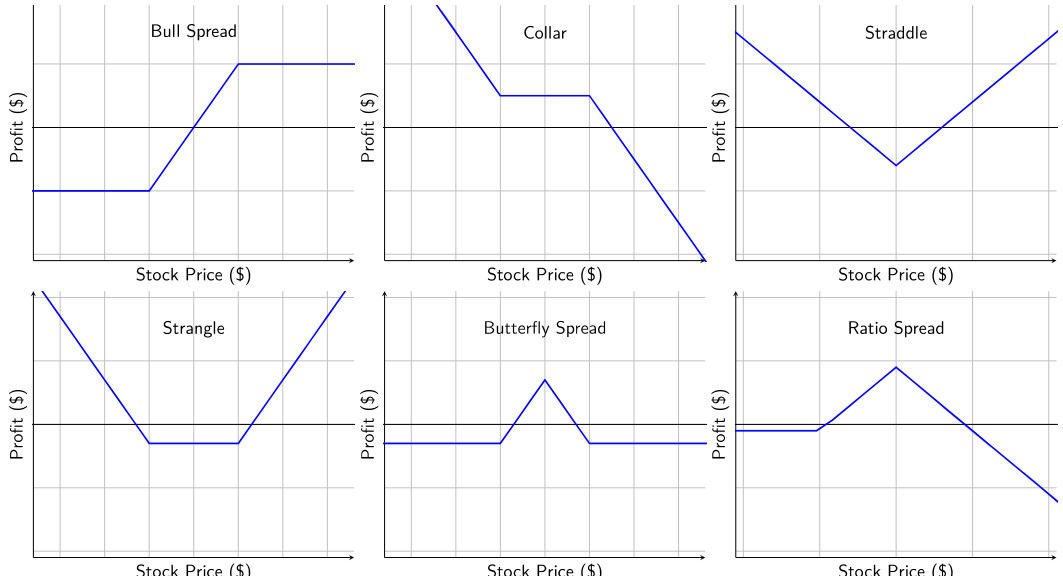


Figure 11: Strategy profit shapes. Read the signed option legs and the initial financed cost before using a strategy name; a zero premium does not remove tail risk.

Strategy	Option	K_{low}	K_{ATM}	K_{high}
Bull spread	Call	–	Buy	Sell
	Put	–	–	–
Collar	Call	–	–	Sell
	Put	Buy	–	–
Straddle	Call	–	Buy	–
	Put	–	Buy	–
Strangle	Call	–	–	Buy
	Put	Buy	–	–
Butterfly	Call	Buy	Sell (2)	Buy
	Put	–	–	–
Ratio spread	Call	–	Buy	Sell (n)
	Put	–	–	–

The source notes that alternative combinations can produce the same results. When invoking put–call parity, include any accompanying cash/bond position: two bare call/put spreads need not have identical initial premiums or absolute terminal cash flows.

Various Strategies: Rationales

Bull Spread You believe a stock will appreciate ⇒ buy a call option (forward position insured)

You can lower the cost if you are willing to reduce your profit should the stock appreciate ⇒ sell a call with higher strike

Surprisingly, you can achieve the same result by buying a low-strike put and selling a high-strike put

Opposite: bear spread

Collar A collar is fundamentally a short position (resembling a short forward contract)

Often used for insurance when we own a stock (collared stock)

The collared stock looks like a bull spread; however, it arises from a different set of transactions

Opposite: written collar

Straddle A straddle can profit from stock price moves in both directions
The disadvantage is that it has a high premium because it requires purchasing two options
Opposite: written straddle

Strangle To reduce the premium of a straddle, you can buy out-of-the-money options rather than at-the-money options.
Opposite: written strangle

Butterfly Spread A butterfly spread is a written straddle to which we add two options to safeguard the position: An out-of-the money put and an out-of-the money call.
A butterfly spread can be thought of as a written straddle for the timid (or for the prudent!)
- Opposite: long iron butterfly

Ratio Spread Ratio spreads involve buying one option and selling a greater quantity (n) of an option with a more out-of-the money strike
The ratio (i.e., "1 by n") is the number of short options divided by the number of long options
The options are either both calls or both puts
It is possible to construct ratio spreads with zero premium $\Rightarrow$ we can construct insurance that costs nothing if it is not needed!

Bloomberg: Products $\rightarrow$ Option Strategies

The dated strategy screen illustrates how legs combine into a portfolio payoff and how market inputs affect the initial cost.

Bloomberg: Products $\rightarrow$ Option Strategies

The dated strategy screen provides a second historical example of selecting and valuing option legs.

Lakonishok, Lee, Pearson, and Poteshman, Option Market Activity, The Review of Financial

Lakonishok, Lee, Pearson, and Poteshman, Option Market Activity, The Review of Financial Studies, 2006
- Stylized facts about option trading
- Written option positions are more common than purchased positions
- About 4 times more purchased calls than puts
- Main driver of option market activity is speculating on the direction of underlying stock movements
- Option trading strategies
- Small fraction of volatility trading strategies (straddles and strangles)
- Large fraction of covered-call strategies
- Option market activity during the stock market bubble of the late 1990s
- Call buying and put writing increased dramatically (mostly on growth stocks)
- Purchased puts less common (little appetite for betting against the bubble)

4 Cross-hedging an imperfectly matched exposure

4.1 Why a related future can still reduce risk

A *cross hedge* uses a derivative on one asset to hedge a different, related asset. An airline buying jet fuel, for example, may use heating-oil futures when an exact jet-fuel contract is unavailable or

illiquid. The hedge cannot generally cancel every price move because jet fuel and heating oil are not the same commodity. Let ΔS be the change in jet-fuel price per gallon and ΔF the change in the heating-oil futures price per gallon over the same horizon. Correlation measures how much of the exposure is shared; the volatility ratio tells us how many futures-price units offset one spot-price unit.

The *sign* of the position depends on the original exposure. Someone who owns an asset and fears a price fall normally sells futures. An airline that must *buy* fuel and fears a price rise normally buys futures, so futures gains offset a higher purchase cost. The addendum's generic expression $\Delta S + N\Delta F$ uses a signed N; the airline calculation below uses a positive number of long contracts and writes the residual cost as $\Delta S - h\Delta F$.

Cross Hedging

- Cross hedging = the use of a derivative on one asset to hedge another asset.

Example: Jet fuel futures do not exist (or they are very illiquid) in United States, but firms sometimes hedge jet fuel with crude oil futures, or heating oil futures.

If we own a quantity of jet fuel and hedge by holding N crude oil futures contracts, our mark-to-market profit depends on the change in the jet fuel price and the change in the futures price:

$$(S_t - S_{t-1}) + N(F_t - F_{t-1}) \tag{7}$$

where S_t is the price of jet fuel and F_t the heating oil futures price.

What is the optimal number of futures contracts for hedging, N^* ?

4.2 Choose the hedge ratio by minimizing variance

For one unit of future fuel consumption, write the hedged change in cost as $X(h) = \Delta S - h\Delta F$. Expanding its variance and differentiating gives

$$\begin{aligned} \mathrm{Var}[X(h)] &= \sigma_S^2 + h^2\sigma_F^2 - 2h\,\mathrm{Cov}(\Delta S, \Delta F), \\ 0 &= 2h\sigma_F^2 - 2\,\mathrm{Cov}(\Delta S, \Delta F), \\ h^* &= \frac{\mathrm{Cov}(\Delta S, \Delta F)}{\sigma_F^2} = \rho\frac{\sigma_S}{\sigma_F}. \end{aligned}$$

This is also the slope in a least-squares regression of ΔS on ΔF with an intercept. With a single explanatory variable, the regression's $R^2 = \rho^2$. Under the same variance assumptions, the minimum residual variance is $\sigma_S^2(1-\rho^2)$. Here R^2 measures variance reduction in the historical sample, not a guaranteed reduction in every future month.

Worked example

The 15 monthly price-change pairs in the addendum give, before rounding, $\sigma_F \approx 0.0313434$, $\sigma_S \approx 0.0262548$, and $\rho \approx 0.9283723$. Thus $h^* \approx 0.77765$, reported in the slides as 0.7777, while $R^2 \approx 0.8619$. If one instead substitutes the *printed rounded* standard deviations 0.0313 and 0.0263 into the formula, one gets about 0.7801. This small discrepancy is rounding of the inputs, not a different hedge rule; the unrounded 15 observations reproduce the slide's result.

Cross Hedging Example

An airline expects to purchase 2 million gallons of jet fuel in 1 month and decides to use heating oil futures for hedging.

- Define:

ΔS = change in the jet fuel price per gallon
ΔF = change in the heating oil futures price price per gallon

- Standard deviations and correlations are (data on next slide)

$$\sigma_F = 0.0313, \quad \sigma_S = 0.0263, \quad \text{and } \rho = 0.9284 \tag{8}$$

- Consider the linear relationship:

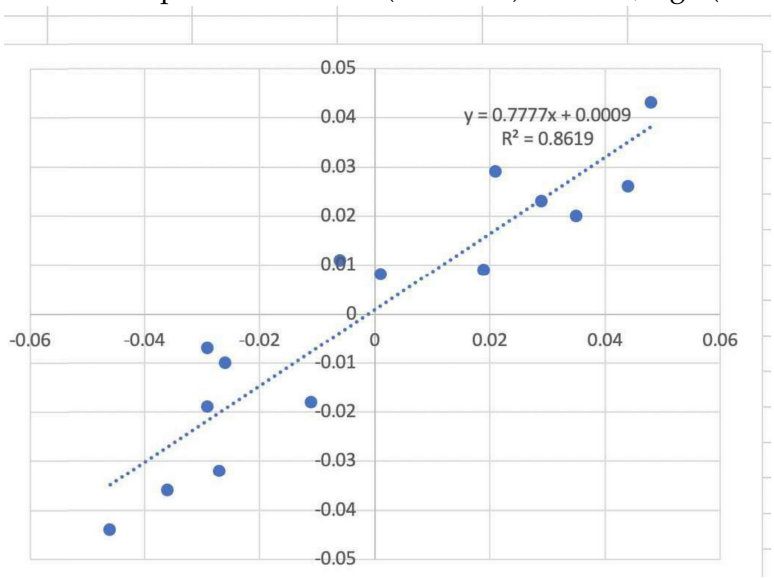
$$\Delta S = a + h^* \times \Delta F + \epsilon \tag{9}$$

$h^* = \rho \frac{\sigma_S}{\sigma_F}$ is the minimum variance hedge ratio
$R^2 = \rho^2$ is the hedge effectiveness
In this example, $h^* = 0.7777$ and $R^2 = 0.8619$

Cross Hedging Example

Month	ΔF	ΔS
1	0.021	0.029
2	0.035	0.02
3	-0.046	-0.044
4	0.001	0.008
5	0.044	0.026
6	-0.029	-0.019
7	-0.026	-0.01
8	-0.029	-0.007
9	0.048	0.043
10	-0.006	0.011
11	-0.036	-0.036
12	-0.011	-0.018
13	0.019	0.009
14	-0.027	-0.032
15	0.029	0.023

Printed sample statistics: rho (CORREL) = 0.9284; sigF (STDEV) = 0.0313; sigS (STDEV) = 0.0263.



4.3 Turn the ratio into tradable contracts

If the exposure is Q_A units and one futures contract represents Q_F units, the continuous optimum is $N^* = h^* Q_A / Q_F$. The airline expects to purchase 2 million gallons, and the historical heating-oil

contract in the example covers 42,000 gallons. Therefore

$$N^* \approx \frac{0.7777(2{,}000{,}000)}{42{,}000} = 37.03,$$

or 37 *long* contracts after whole-contract rounding. Contract rounding, imperfect correlation, and differences in location, quality or delivery date all leave basis risk. A futures hedge also requires liquidity for mark-to-market cash flows even if it reduces the variance of final purchase cost.

The addendum mentions *tailing the hedge*: adjusting the contract count slightly for the timing of daily futures settlement. No explicit tailing formula is supplied in these slides, so 37 is the untailored classroom answer rather than a claimed exact implementation for all settlement schedules.

Why it matters

The minimum-variance hedge is about covariance, not just correlation: direction follows the exposure, magnitude follows $\rho\sigma_S/\sigma_F$, and the contract count also needs both physical quantities. A strong historical fit reduces expected variance but never makes two different commodities identical.

Cross Hedging Example

To calculate the number of contracts that should be used in hedging, define:

- Q_A : Size of position being hedged (units)
- Q_F : Size of one futures contract (units)
- N^* : Optimal number of futures contracts for hedging

The futures contracts should be h^*Q_A units of the asset. The number of futures contracts required is therefore given by

$$N^* = \frac{h^*Q_A}{Q_F} \tag{10}$$

- Each heating oil contract traded on NYMEX is on 42,000 gallons of heating oil. Thus, the optimal number of contracts is

$$N^* = \frac{0.7777 \times 2{,}000{,}000}{42{,}000} = 37.03 \tag{11}$$

or, rounding to the nearest whole number, 37.

Cross Hedging

When futures are used for hedging, a small adjustment, known as tailing the hedge, can be made to allow for the impact of daily settlement.

This does not change the intuition, but only the result (slightly).

If you want to learn more about forwards and futures, read chapter 5, "Determination of forward and futures prices" in Hull (eight edition) and chapters 5, "Financial forwards and futures" and 6, "Commodity Forwards and Futures" in McDonald (third edition)

5 State prices and risk-neutral valuation

5.1 A two-state example fixes the intuition

Consider one period and two possible states tomorrow. The source slide's price vector was flattened by OCR; checking the original page shows it is $\boldsymbol{S} = (1, 100, c)^\top$. A risk-free asset costs 1

and pays 1.1 in either state. A risky asset costs 100 and pays 100 in state 1 or 150 in state 2. A call costs the unknown c and pays 0 or 50. In matrix form, with assets as rows and states as columns,

$$S = \begin{pmatrix} 1 \\ 100 \\ c \end{pmatrix}, \qquad Q = \begin{pmatrix} 1.1 & 1.1 \\ 100 & 150 \\ 0 & 50 \end{pmatrix}.$$

The question is not which state is objectively more likely. It is what price for the call is consistent with the two already traded assets.

Let ϕ_1, ϕ_2 be the current prices of claims paying one unit only in state 1 or state 2. The first two assets imply

$$1 = 1.1(\phi_1 + \phi_2), \qquad 100 = 100\phi_1 + 150\phi_2.$$

Solving gives $\phi_1 = 8/11 \approx 0.7273$ and $\phi_2 = 2/11 \approx 0.1818$. Thus $c = 0\phi_1 + 50\phi_2 = 100/11 \approx 9.09$. We can see the same price without naming state prices: one share of the risky asset minus a position that owes 100 tomorrow pays $(0, 50)$. The borrowing needed to owe 100 tomorrow brings in $100/1.1$ today, so the replicating strategy costs $100 - 100/1.1 = 100/11$.

Normalize the state prices by the gross risk-free return, 1.1: $p_1^* = 1.1\phi_1 = 0.8$ and $p_2^* = 1.1\phi_2 = 0.2$. These positive weights sum to one. The call price is the discounted expectation under them,

$$c = \frac{0(0.8) + 50(0.2)}{1.1} = \frac{100}{11}.$$

They are *risk-neutral probabilities*, pricing weights inferred from traded prices, not a claim that state 2 occurs in 20% of actual observations.

Example: Three Assets, Two States

Two states, three traded assets.

Prices today

$$S = \begin{pmatrix} 1 \\ 100 \\ c \end{pmatrix}.$$

Payoffs tomorrow

Asset	State $s = 1$	State $s = 2$
Risk-free asset	1.1	1.1
Risky asset	100	150
Call option	0	50

Question

What is the no-arbitrage value of c?

Step 1: Guess a Linear Pricing Rule

A no-arbitrage price should be a linear function of future payoffs.

$$\text{price of asset } i = \sum_{s=1}^{2} \phi(s) q_i(s), \quad \phi(s) > 0.$$

Interpretation

$\phi(s)$ is the value today of one unit of payoff delivered only in state s.

Step 2: Write the Pricing Equations

Apply the same pricing rule to all three assets:

$$\begin{cases} 1 = 1.1\phi(1) + 1.1\phi(2), \\ 100 = 100\phi(1) + 150\phi(2), \\ c = 0 \cdot \phi(1) + 50\phi(2). \end{cases}$$

Key idea

The first two traded assets pin down the two state prices. The third asset is then priced by the same rule.

Step 3: Use the Risk-Free Asset

The risk-free asset pays 1.1 in both states and costs 1.

$$1 = 1.1[\phi(1) + \phi(2)].$$

Hence

$$\phi(1) + \phi(2) = \frac{1}{1.1}.$$

Interpretation

The sum of state prices is the price of a claim to one unit in every state: the risk-free discount factor.

Step 4: Normalize into Risk-Neutral Probabilities

Define

$$\phi^\star(s) = 1.1\phi(s), \quad s = 1, 2.$$

Then

$$\phi^\star(1) + \phi^\star(2) = 1.1[\phi(1) + \phi(2)] = 1.$$

Meaning

$\phi^\star(1)$ and $\phi^\star(2)$ are probabilities. They are not objective probabilities; they are pricing weights normalized to sum to one.

Step 5: Solve for the State Prices

Start from the risk-free asset:

$$\phi(1) + \phi(2) = \frac{1}{1.1}.$$

Use the risky asset equation:

$$100 = 100\phi(1) + 150\phi(2).$$

Substitute

$$\phi(1) = \frac{1}{1.1} - \phi(2).$$

Step 5: Solve for the State Prices

After substitution,

$$100 = 100\left(\frac{1}{1.1} - \phi(2)\right) + 150\phi(2).$$

Therefore

$$\phi(2) = \frac{100 - 100/1.1}{50} = 0.1818.$$

Then

$$\phi(1) = \frac{1}{1.1} - 0.1818 = 0.7273.$$

Step 6: Price the Redundant Asset

The call payoff is 0 in state 1 and 50 in state 2.

$$\begin{aligned} c &= 0 \cdot \phi(1) + 50\phi(2) \\ &= 50 \times 0.1818 \\ &= 9.09. \end{aligned}$$

Conclusion
The no-arbitrage price of the call is

$$\boxed{c = 9.09}.$$

Same Price with Risk-Neutral Probabilities

The normalized weights are

$$\phi^*(1) = 1.1 \times 0.7273 = 0.8.$$

$$\phi^*(2) = 1.1 \times 0.1818 = 0.2.$$

The call price is the discounted risk-neutral expected payoff:

$$\begin{aligned} c &= \frac{1}{1.1}\left(0 \times 0.8 + 50 \times 0.2\right) \\ &= \frac{10}{1.1} = 9.09. \end{aligned}$$

What the Example Teaches

- No-arbitrage pricing does not require objective probabilities.
  It requires state prices, or equivalently risk-neutral probabilities plus a risk-free discount factor.
  The redundant asset is priced by replication: its payoff is already spanned by traded assets.
  Core formula
  price = discounted risk-neutral expected payoff.

5.2 The general finite-state framework

Suppose there are N traded securities and K mutually exclusive states. Let $S \in \mathbb{R}^N$ be today's price vector, and let $Q \in \mathbb{R}^{N\times K}$ contain their future payoffs, one asset per row. A portfolio $\theta \in \mathbb{R}^N$ costs $S^\top\theta$ and pays the state vector $Q^\top\theta$. This orientation matters: multiplying Q by state prices produces an N-vector of asset prices.

An arbitrage is a trading strategy with no possible loss, no positive initial investment, and a strict gain somewhere (either upfront or in at least one possible state). The lecture illustrates particularly clear cases: non-positive cost with strictly positive payoffs in every state, or strictly negative cost with non-negative payoffs everywhere. The more general wording also rules out a zero-cost payoff that is positive in some states and zero in the rest.

In the finite-state, frictionless setting of the lecture, absence of arbitrage is equivalent to the existence of a strictly positive state-price vector $\phi \in \mathbb{R}^K_{++}$ satisfying

$$S = Q\phi, \qquad S_i = \sum_{s=1}^{K} \phi_s q_i(s).$$

The equation prices all traded payoff rows with one linear rule. If two portfolios deliver exactly the same payoff in every state but cost different amounts, buying the cheaper and selling the dearer creates an arbitrage. Positivity prevents a positive state-contingent payoff from having a non-positive price under the stated assumptions. State prices may not be unique in an incomplete market, but any payoff replicated by traded assets has a unique no-arbitrage price.

Risk-Neutral Pricing: Assumptions

We work in a one-period, finite-state market.

- N securities and K states: $s = 1, \ldots, K$.
- Frictionless trading: no transaction costs, no taxes, unrestricted short sales.
- Securities are infinitely divisible.
- The number of states is finite.

Uncertainty is described by objective probabilities

$$\pi(1), \ldots, \pi(K), \qquad \pi(\boldsymbol{s}) > 0, \qquad \sum_{s=1}^{K} \pi(\boldsymbol{s}) = 1.$$

Securities and Payoffs

The price vector is

$$\boldsymbol{S} = \begin{bmatrix} S_1 & S_2 & \ldots & S_N \end{bmatrix}^\top > 0.$$

The payoff matrix is

$$\boldsymbol{Q} = \begin{bmatrix} q_1(1) & q_1(2) & \ldots & q_1(K) \\ q_2(1) & q_2(2) & \ldots & q_2(K) \\ \vdots & \vdots & \ddots & \vdots \\ q_N(1) & q_N(2) & \ldots & q_N(K) \end{bmatrix}.$$

Row i contains the state-contingent payoff of security i.

Portfolio Strategy

A portfolio strategy is a vector of holdings:

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_1 & \theta_2 & \ldots & \theta_N \end{bmatrix}^\top.$$

Cost today

$$\boldsymbol{S}^\top \boldsymbol{\theta} = \sum_{i=1}^{N} \theta_i \boldsymbol{S}_i.$$

Payoff in state s

$$X(s) = \sum_{i=1}^{N} \theta_i q_i(s).$$

In vector form, the payoff vector is

$$\boldsymbol{X} = \boldsymbol{Q}^\top \boldsymbol{\theta}.$$

Arbitrage

An arbitrage is a portfolio that produces a free lunch.

Type 1

$$\boldsymbol{S}^\top \boldsymbol{\theta} \leq 0, \quad \boldsymbol{X} > 0.$$

Non-positive cost and strictly positive payoff in every state.

Type 2

$$\boldsymbol{S}^\top \boldsymbol{\theta} < 0, \quad \boldsymbol{X} \geq 0.$$

Strictly negative cost and non-negative payoff in every state.

No arbitrage means that neither type exists.

First Fundamental Theorem of Asset Pricing

In a finite-state frictionless market,

No arbitrage $\iff \exists \phi \in \mathbb{R}^K_{++}$ such that

$$S = Q\phi.$$

Equivalently, for every asset i,

$$S_i = \sum_{s=1}^{K} \phi(s) q_i(s), \quad \phi(s) > 0.$$

Interpretation

The vector ϕ is a system of strictly positive state prices. It turns future state-contingent payoffs into current prices.

Linear Pricing Rule and No Arbitrage

The pricing rule is linear:

$$\text{price } = f(\text{ payoff }).$$

Under no arbitrage,

$$\begin{aligned} S_i &= \sum_{s=1}^{K} \phi(s) q_i(s) \\ &= \mathbb{E}[M q_i] \\ &= \frac{1}{1+R} \mathbb{E}^*[q_i]. \end{aligned}$$

Why Linearity Matters

If two portfolios have the same payoff in every state, no arbitrage requires the same price today.
Otherwise, investors could buy the cheaper portfolio and sell the more expensive one.
The payoff difference would be zero, but the initial cash flow would be positive.

5.3 From Arrow–Debreu prices to probability weights

An Arrow–Debreu claim for state s pays one unit there and zero elsewhere; its price is exactly ϕ_s. Buying one such claim for each state pays one unit for sure and costs $B_0 = \sum_s \phi_s$. Therefore its gross risk-free return is $1 + R = 1/B_0$. Define $p_s^* = (1 + R)\phi_s$; then $p_s^* > 0$ and $\sum_s p_s^* = 1$. The asset-pricing equation becomes

$$S_i = \frac{1}{1+R} \sum_{s=1}^{K} p_s^* q_i(s) = \frac{1}{1+R} \mathbb{E}^*[q_i].$$

This is not a separate model layered onto state prices: it is the same linear pricing rule written as a discount factor times an expectation. The objective probabilities π_s are not needed for this form. It applies directly to traded or replicated payoffs; an unspanned payoff in an incomplete market need not have a unique price.

Arrow-Debreu Securities

An Arrow-Debreu security for state s pays one unit in state s and zero otherwise.
With one Arrow-Debreu security for each state,

$$\boldsymbol{Q}^{AD} = I_K.$$

Since $S = Q\phi$, in the Arrow-Debreu case

$$\boldsymbol{S}^{AD} = I_K \phi = \phi.$$

Key point
State prices are literally the prices of Arrow-Debreu claims.

Build a Risk-Free Asset from Arrow-Debreu Claims

Buy one Arrow-Debreu claim for each state.
Payoff
1 in every state.
Price

$$B_0 = \sum_{s=1}^{K} \phi(s).$$

Therefore the gross risk-free return is

$$1 + R = \frac{1}{B_0} = \frac{1}{\sum_{s=1}^{K} \phi(s)}.$$

Normalize State Prices

Define risk-neutral probabilities by

$$\phi^\star(s) = (1+R)\phi(s), \quad s = 1, \ldots, K.$$

They are positive and sum to one:

$$\sum_{s=1}^{K} \phi^\star(s) = (1+R) \sum_{s=1}^{K} \phi(s) = 1.$$

Important
Risk-neutral probabilities are normalized state prices, not physical frequencies.

Rewrite Prices as Discounted Expectations

Start from the state-price formula:

$$S_i = \sum_{s=1}^{K} \phi(s) q_i(s).$$

Use $\phi(s) = \phi^\star(s)/(1+R)$:

$$\begin{aligned} S_i &= \frac{1}{1+R} \sum_{s=1}^{K} \phi^\star(s) q_i(s) \\ &= \frac{1}{1+R} \mathbb{E}^\star[q_i]. \end{aligned}$$

Risk-Neutral Pricing Formula

For every traded asset,

$$S_i = \frac{1}{1+R} \mathbb{E}^*[q_i]$$

- Time is handled by the risk-free discount factor $1/(1+R)$.
- Risk is handled by changing probabilities from objective probabilities to risk-neutral probabilities.
  The formula prices all payoffs spanned by traded assets.

5.4 The same rule under objective probabilities

If objective state probabilities $\pi_s > 0$ are given, define the *stochastic discount factor* $M_s = \phi_s/\pi_s$. Then

$$S_i = \sum_s \pi_s M_s q_i(s) = \mathbb{E}[M q_i].$$

Equivalently, the likelihood ratio $Z_s = p_s^*/\pi_s$ satisfies $M_s = Z_s/(1+R)$, yielding $\mathbb{E}[Mq_i] = (1+R)^{-1}\mathbb{E}^*[q_i]$. These formulas distinguish two roles: objective probabilities describe frequency beliefs, while state prices (or risk-neutral probabilities) encode time discounting and how the market values payoffs across states. A payoff in a bad state may command a high state price because it delivers when resources are especially valuable; the next chapter derives this link from marginal utility.

Why it matters

The practical sequence is: write the payoff matrix, find a positive state-price vector from traded prices, value any replicated claim, and only then normalize state prices into probabilities if the expectation form is useful. Never substitute empirical frequencies for risk-neutral weights without a pricing argument.

Objective Probabilities and the SDF

Suppose the objective probabilities are $\pi(s) > 0$.

Define the stochastic discount factor

$$M(s) = \frac{\phi(s)}{\pi(s)} > 0.$$

Then

$$\begin{aligned} S_i &= \sum_{s=1}^{K} \phi(s) q_i(s) \\ &= \sum_{s=1}^{K} \pi(s) M(s) q_i(s) \\ &= \mathbb{E}[M q_i]. \end{aligned}$$

Risk Adjustment as a Change of Probabilities

Equivalently, define

$$Z(s) = \frac{\phi^\star(s)}{\pi(s)}.$$

Since $\phi^\star(s) = (1+R)\phi(s)$,

$$M(s) = \frac{Z(s)}{1+R}.$$

Thus

$$S_i = \mathbb{E}\left[M q_i\right] = \frac{1}{1+R}\mathbb{E}\left[Z q_i\right] = \frac{1}{1+R}\mathbb{E}^\star[q_i].$$

Economic Interpretation

State prices combine two effects.

Time discounting

A payoff tomorrow is worth less than the same payoff today.

Risk adjustment

A payoff in a valuable state receives a higher weight.

For risk-averse investors, bad states are typically more valuable: payoffs there provide consumption when it is most needed.

6 Utility, the Euler equation and equilibrium

6.1 No-arbitrage prices versus an economic explanation

The previous chapter showed that, under no arbitrage, positive state prices ϕ_s can price traded payoffs. It did not explain why a payoff in one state might be worth more than the same payoff in another. A one-period consumption-investment problem supplies an economic interpretation through *marginal utility*: an additional unit of consumption is especially useful when consumption is scarce.

The lecture writes the investor's objective without an explicit time-preference multiplier:

$$\max_{c_0,\{c_s\},\theta} U(c_0) + \sum_{s=1}^{K} \pi_s U(c_s),$$

where $\pi_s > 0$ are objective state probabilities and U is increasing and concave. This exact classroom specification is used below; adding a separate discount factor would alter the formula. Current endowment is e_0, future state endowment is e_s, and the investor chooses holdings θ_i of securities priced S_i today and paying $q_i(s)$ tomorrow.

Utility Maximization and State Prices

No-arbitrage gives the pricing representation. Utility maximization explains where pricing weights can come from.

Goal

Show that, in a one-period consumption-investment problem,

$$S_i = \mathbb{E}\left[\frac{U'(c_s)}{U'(c_0)} q_i(s)\right].$$

The marginal rate of substitution becomes the stochastic discount factor.

One-Period Consumption-Investment Problem

The investor chooses current consumption, future consumption, and portfolio holdings.

$$\max_{c_0,\{c_s\},\theta} U(c_0) + \sum_{s=1}^{K} \pi(s) U(c_s).$$

- U is increasing and concave.
- Consumption is perishable.

6.2 Trace one extra unit of an asset through both budgets

With no waste and increasing utility, the budgets bind:

$$c_0 = e_0 - \sum_i \theta_i S_i, \qquad c_s = e_s + \sum_i \theta_i q_i(s).$$

Buying a little more of asset i reduces current consumption by S_i per unit and raises state-s future consumption by $q_i(s)$. For an interior, unconstrained optimum, differentiating expected utility with respect to θ_i yields

$$-S_i U'(c_0) + \sum_s \pi_s q_i(s) U'(c_s) = 0.$$

Rearrange and divide by positive current marginal utility:

$$S_i = \sum_s \pi_s \underbrace{\frac{U'(c_s)}{U'(c_0)}}_{M_s} q_i(s) = \mathbb{E}[M q_i].$$

This is the *asset-pricing Euler equation*. The random variable M_s is a stochastic discount factor (SDF), here equal to the investor's intertemporal marginal rate of substitution. Portfolio constraints, transaction costs or boundary choices can turn a first-order equality into a different condition; they are excluded from this derivation.

Budget Constraints

The investor starts with current endowment e_0 and future endowments e_s .

$$\begin{cases} \boldsymbol{c}_0 + \sum_{i=1}^{N} \theta_i \boldsymbol{S}_i \leq \boldsymbol{e}_0, \\ \boldsymbol{c}_s - \sum_{i=1}^{N} \theta_i \boldsymbol{q}_i(\boldsymbol{s}) \leq \boldsymbol{e}_s, \quad \boldsymbol{s} = 1, \ldots, K. \end{cases}$$

The first constraint is the date-0 budget. The second is the state-s future budget.

Step 1: Binding Constraints

Since U is increasing, more consumption is better. Therefore the budget constraints bind.

$$c_0 = e_0 - \sum_{i=1}^{N} \theta_i S_i.$$

$$c_s = e_s + \sum_{i=1}^{N} \theta_i q_i(s), \qquad s = 1, \ldots, K.$$

Step 1: Substitute into Utility

After imposing the binding constraints, the problem becomes a choice of portfolio holdings only.

$$\begin{array}{l} \max_{\theta} U\left(e_0 - \sum_{i=1}^{N} \theta_i S_i\right) \\ + \sum_{s=1}^{K} \pi(s) U\left(e_s + \sum_{i=1}^{N} \theta_i q_i(s)\right). \end{array}$$

Step 2: First-Order Condition

Differentiate the objective with respect to θ_i .

The marginal effect on current consumption is $-S_i$. The marginal effect on state-s future consumption is $q_i(s)$.

Therefore

$$0 = -S_i U'(c_0) + \sum_{s=1}^{K} \pi(s) q_i(s) U'(c_s).$$

Step 3: Euler Equation

Rearrange the first-order condition:

$$S_i U'(c_0) = \sum_{s=1}^{K} \pi(s) q_i(s) U'(c_s).$$

Divide by current marginal utility:

$$S_i = \sum_{s=1}^{K} \pi(s) \frac{U'(c_s)}{U'(c_0)} q_i(s).$$

This is the asset pricing Euler equation.

Step 4: Marginal Rate of Substitution

Define the intertemporal marginal rate of substitution:

$$\xi(s) = \frac{U'(c_s)}{U'(c_0)}.$$

Then

$$S_i = \mathbb{E}[\xi q_i].$$

Interpretation

$\xi(s)$ is the value today of one additional unit of consumption in future state s, measured in units of current marginal utility.

6.3 Connect the three pricing languages

Comparing the Euler equation to $S_i = \sum_s \phi_s q_i(s)$ identifies $\phi_s = \pi_s M_s$. If a one-unit risk-free payoff is traded, $\sum_s \phi_s = (1+R)^{-1}$, and risk-neutral probabilities are $p_s^* = (1+R)\phi_s$. Hence one and the same traded-asset price can be written in three equivalent ways:

$$\boxed{S_i = \mathbb{E}[Mq_i] = \sum_s \phi_s q_i(s) = \frac{1}{1+R}\mathbb{E}^*[q_i]}.$$

The expectation on the left uses objective probabilities π_s; the one on the right uses risk-neutral probabilities p_s^*. The latter do *not* assert that investors are risk neutral. They move the risk adjustment from the discount factor into the probability weights.

With concave utility, low consumption means high marginal utility. Thus a bad consumption state can have high M_s, and its state price $\phi_s = \pi_s M_s$ can be high relative to its objective likelihood. This is why an asset paying in bad states can be valuable insurance. A high state price depends on both likelihood and marginal utility; one should not infer it from low consumption alone without considering π_s.

MRS, State Prices, and Risk-Neutral Probabilities

The marginal rate of substitution is a stochastic discount factor:

$$M(s) = \xi(s) = \frac{U'(c_s)}{U'(c_0)}.$$

State prices are

$$\phi(s) = \pi(s)M(s).$$

Risk-neutral probabilities are

$$\phi^\star(s) = (1+R)\phi(s).$$

All Pricing Equations Are the Same

For every traded asset,

$$S_i = \mathbb{E}[Mq_i] = \sum_{s=1}^{K} \phi(s) q_i(s) = \frac{1}{1+R}\mathbb{E}^\star[q_i]$$

- $M(s)$: stochastic discount factor.
- $\phi(s)$: state price.
- $\phi^\star(s)$: risk-neutral probability.

Why Bad States Have High State Prices

If future consumption is low in state s, marginal utility is high:

$$c_s\text{low} \quad \Longrightarrow \quad U'(c_s)\text{high} \quad \Longrightarrow \quad M(s)\text{high}.$$

Because

$$\phi(s) = \pi(s)M(s),$$

that state receives a high state price.

Economic meaning

A payoff in a bad state is valuable because it delivers consumption exactly when consumption is scarce.

6.4 What equilibrium adds

Individual optimization supplies candidate pricing weights at the person's optimal consumption allocation. *Equilibrium* also requires market clearing: across investors, desired holdings must equal available securities, and consumption must be feasible given endowments and aggregate resources. Preferences, endowments, traded payoffs and supply jointly determine prices and allocations. The lecture's final diagrams summarize this path from fundamentals to state prices; they do not supply a separate numerical equilibrium model.

The hierarchy is useful when solving exercises. For a claim spanned by traded assets, replication and no arbitrage already fix a unique price. State prices and risk-neutral probabilities express that same price compactly. Utility and equilibrium explain why those pricing weights may take their observed values, but they are not required to compute the price of the two-state call in Section 5.

Why it matters

Ask two different questions separately: *What price avoids arbitrage given traded payoffs?* and *Why do investors value some states more?* The first is answered by replication and positive state prices; the second by marginal utility and market clearing.

Equilibrium

Equilibrium combines two ingredients:

Individual maximization

Investors choose consumption and portfolios optimally.

Market clearing

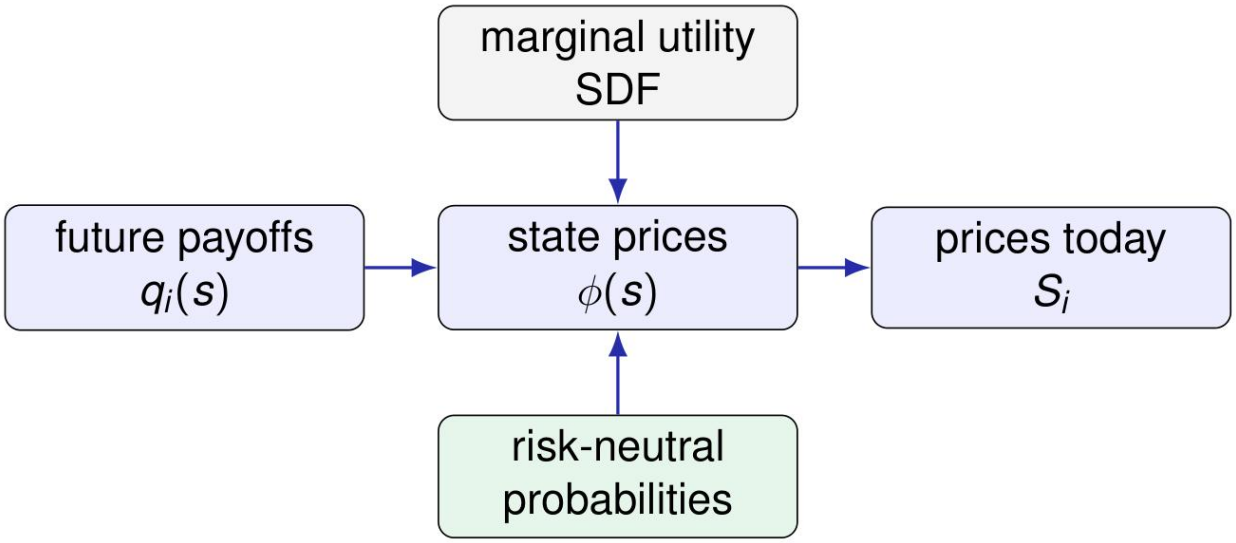
Aggregate demand equals aggregate supply.

Preferences, endowments, payoffs, and clearing

$$\Longrightarrow$$

equilibrium asset prices and state prices.

Big Picture



Risk-neutral pricing is not a statement that investors are risk neutral. It is a no-arbitrage way to price risky payoffs.

Takeaways

- A redundant asset has a unique no-arbitrage price if its payoff is spanned by existing assets.
- State prices convert state-contingent payoffs into current prices.
- Risk-neutral probabilities are normalized state prices.
- No-arbitrage pricing can be written with state prices, an SDF, or risk-neutral probabilities.
- Utility maximization gives an economic foundation for state prices through marginal utilities.

7 Forwards and futures: exercises with worked answers

The twelve supplied problems below retain their complete statements and alternatives. The worked answers are newly generated explanations, not a supplied answer key. Rate conventions and missing trading assumptions are stated in each answer.

Derivatives QFI 2023-24 Exercises & Solutions Series I Topic: Forwards and Futures

Problem 1.1

Stock XYZ has a current price of 100. The forward price for delivery of this stock in 1 year is 110. Unless otherwise indicated, the stock pays no dividends, and the annual effective risk-free interest rate is 10%. Determine which of the following statements is FALSE.

(A)
The time-1 profit diagram and the time-1 payoff diagram for long positions in this forward contract are identical.

(B)
The time-1 profit for a long position in this forward contract is exactly opposite to the time-1 profit for the corresponding short forward position.

(C)
There is no comparative advantage to investing in the stock versus investing in the forward contract.

(D)
If the 10% interest rate was continuously compounded instead of annual effective, then it would be more beneficial to invest in the stock, rather than the forward contract.

(E)
If there was a dividend of \$3.00 paid 6 months from now, then it would be more beneficial to invest in the stock, rather than the forward contract.

Worked answer (generated). With a 10% *effective annual* rate, $100(1.10) = 110$, exactly the quoted one-year forward price. A long forward has zero initial premium, so its time-1 payoff and profit are both $S_T - 110$; a short forward has the opposite profit. Buying the stock with borrowed funds gives the same time-1 profit. If 10% were instead *continuously* compounded, the fair delivery price would be $100e^{0.10} \approx 110.52$, making a long forward at 110 more favorable, not the stock. A dividend paid on the stock but not the forward does benefit direct ownership, all else equal.

Problem 1.2

A market maker in stock index forward contracts observes a 6-month forward price of 112 on the index. The index spot price is 110 and the continuously compounded dividend yield on the index is 2%. The continuously compounded risk-free interest rate is 5%. Describe actions the market maker could take to exploit an arbitrage opportunity and calculate the resulting profit (per index unit).

(A)
Buy observed forward, sell synthetic forward, Profit = 0.34

(B)
Buy observed forward, sell synthetic forward, Profit = 0.78

(C)
Buy observed forward, sell synthetic forward, Profit = 1.35

(D)
Sell observed forward, buy synthetic forward, Profit = 0.78

(E)
Sell observed forward, buy synthetic forward, Profit = 0.34

Worked answer (generated). With $S_0 = 110$, $r = 5\%$, $\delta = 2\%$, and $T = 0.5$, fair carry is $110e^{(0.05-0.02)0.5} \approx 111.6624$. The quoted forward at 112 is high by \$0.3376. Sell that forward and buy the synthetic forward (the dividend-reinvested index position financed at the risk-free rate), locking in about \$0.34 at delivery under the frictionless assumptions.

Problem 1.3

Judy decides to take a short position in 20 contracts of S&P 500 futures. Each contract is for the delivery of 250 units of the index at a price of 1500 per unit, exactly one month from now. The initial margin is 5% of the notional value, and the maintenance margin is 90% of the initial margin. Judy earns a continuously compounded risk-free interest rate of 4% on her margin balance. The position is marked-to-market on a daily basis. On the day of the first marking-to-market, the value of the index drops to 1498. On the day of the second marking-to-market, the value of the index is X and Judy is not required to add anything to the margin account. Calculate the largest possible value of X.

(A)
1490.50

(B)
1492.50
(C)
1500.50
(D)
1505.50
(E)
1507.50

Worked answer (generated). Twenty contracts at a \$250 multiplier give 5,000 index-point dollars. Initial margin is $0.05(5{,}000)(1{,}500) = \$375{,}000$; maintenance is \$337,500. Assuming 365 days per year and interest credited daily, a short gains $5{,}000(1{,}500 - 1{,}498) = \$10{,}000$ on day 1. Its balance is $M_1 = 375{,}000e^{0.04/365} + 10{,}000$. No top-up on day 2 requires $M_1 e^{0.04/365} + 5{,}000(1{,}498 - X) \geq 337{,}500$, hence $X \leq 1507.5167$. The nearest listed maximum is 1507.50. A different day-count convention changes only the last decimals.

Problem 1.4

The current price of a medical company's stock is 75. The expected value of the stock price in three years is 90 per share. The stock pays no dividends. You are also given: i) The risk-free interest rate is positive. ii) There are no transaction costs. iii) Investors require compensation for risk. The price of a three-year forward on a share of this stock is X, and at this price an investor is willing to enter into the forward. Determine what can be concluded about X.

(A)
X < 75
(B)
X = 75
(C)
75 < X < 90
(D)
90 < X

Worked answer (generated). No dividends and a positive risk-free rate imply a fair forward above today's 75. If the stock's expected terminal price is 90 and investors demand a positive expected risk premium, the expected stock return exceeds the risk-free return; hence the fair delivery price lies below 90. Therefore $75 < X < 90$. The expectation of 90 alone does not price the forward.

Problem 1.5

The ask price for a share of ABC company is 100.50 and the bid price is 100. Suppose an investor can borrow at an annual effective rate of 3.05% and lend (i.e., save) at an annual effective rate of 3%. Assume there are no transaction costs and no dividends. Determine which of the following strategies does not create an arbitrage opportunity.

(A)
Short sell one share, and enter into a long one-year forward contract on one share with a forward price of 102.50.
(B)
Short sell one share, and enter into a long one-year forward contract on one share with a forward price of 102.75.

(C)
Short sell one share, and enter into a long one-year forward contract on one share with a forward price of 103.00.

(D)
Purchase one share with borrowed money, and enter into a short one year forward contract on one share with a forward price of 103.60.

(E)
Purchase one share with borrowed money, and enter into a short one year forward contract on one share with a forward price of 103.75.

Worked answer (generated). The bid is 100 and lending grows it to $100(1.03) = 103$; a forward below 103 allows short-stock/long-forward arbitrage. The ask is 100.50 and borrowing it costs $100.50(1.0305) = 103.56525$ in one year; a forward above that allows buy-stock/short-forward arbitrage. Thus the no-arbitrage band is $103 \leq F \leq 103.56525$. Of the listed *strategies*, the one with forward price 103.00 produces zero, not a strictly positive arbitrage payoff. Prices 102.50, 102.75, 103.60 and 103.75 lie outside the band in the indicated trade direction.

Problem 1.6

Investors in a certain stock demand to be compensated for risk. The current stock price is 100. The stock pays dividends at a rate proportional to its price. The dividend yield is 2%. The continuously compounded risk-free interest rate is 5%. Assume there are no transaction costs. Let X represent the expected value of the stock price 2 years from today. Assume it is known that X is a whole number. Determine which of the following statements is true about X.

(A)
The only possible value of X is 105.

(B)
The largest possible value of X is 106.

(C)
The smallest possible value of X is 107.

(D)
The largest possible value of X is 110.

(E)
The smallest possible value of X is 111.

Worked answer (generated). With continuously compounded $r = 5\%$ and dividend yield $\delta = 2\%$, the two-year forward price is $100e^{(0.05-0.02)2} \approx 106.18$. Positive compensation for stock risk means its expected *total* return exceeds the risk-free return. Under the proportional-dividend model used in the course, expected terminal stock value must therefore exceed the forward benchmark. If X is an integer, its smallest possible value is 107. This conclusion uses the stated positive risk premium; the forward price itself does not forecast X.

Problem 1.7

An off-market forward contract is a foward where either you have to pay a premium or you receive a premium for entering into the contract. (With a standard forward contract, the premium is zero.) Suppose the effective annual interest rate is 10% and the S&R index is 1000. Consider 1-year forward contracts.

(a)
Verify that if the forward price is \$1100, the profit diagrams for the index and the 1-year forward

are the same.

(b)

Suppose you are offered a long forward contract at a forward price of \$1200. How much would you need to be paid to enter into this contract?

(c)

Suppose you are offered a long forward contract at a forward price of \$1000. How much would you be willing to pay to enter into this forward contract?

Worked answer (generated). At the 10% effective rate, a \$1,000 index investment financed for one year costs \$1,100 at maturity, matching a zero-premium long forward at $K = \$1{,}100$; both have time-1 profit $S_T - 1{,}100$. A long forward at $K = \$1{,}200$ is worth $1000 - 1200/1.1 = -\$90.91$ today, so the entrant must be *paid* \$90.91. At $K = \$1{,}000$, it is worth $1000 - 1000/1.1 = +\$90.91$, the most the entrant would pay under the model. Carrying those entry premiums forward reconciles the time-1 profit diagrams.

Problem 1.8

Suppose the stock price is \$35 and the continuously compounded interest rate is 5%.

(a)

What is the 6-month forward price, assuming dividends are zero?

(b)

If the 6-month forward price is \$35.50, what is the annualized forward premium?

(c)

If the forward price is \$35.50, what is the annualized continuous dividend yield?

Worked answer (generated). (a) The six-month no-dividend forward is $35e^{0.05(0.5)} \approx \$35.8860$. (b) For observed $F = \$35.50$, the annualized *continuously compounded* forward premium is $\ln(35.50/35)/0.5 \approx 2.8369\%$; if "forward premium" means a simple annualized percentage instead, it is $[(35.50/35) - 1]/0.5 \approx 2.8571\%$. (c) The continuous dividend yield implied by the observed forward is $\delta = 0.05 - \ln(35.50/35)/0.5 \approx 2.1631\%$. Distinguishing the two premium conventions prevents a silent mismatch.

Problem 1.9

The S&R spot price is 1100, the risk-free rate is 5%, and the dividend yield on the index is 0.

(a)

Suppose you observe a 6-month forward price of 1135. What arbitrage would you undertake?

(b)

Suppose you observe a 6-month forward price of 1115. What arbitrage would you undertake?

Worked answer (generated). Assuming the stated 5% rate is continuously compounded, the no-dividend six-month forward is $1100e^{0.05(0.5)} \approx 1127.85$. At 1135 the observed forward is high: short it and buy the stock/index financed with borrowing, yielding about \$7.15 per unit at maturity. At 1115 it is low: buy it and short the underlying, investing the short-sale proceeds, yielding about \$12.85. If 5% were effective annual instead, the exact gaps change but both trade directions remain the same.

Problem 1.10

The S&R index spot price is 1100 and the continuously compounded risk-free rate is 5%. You observe a 9-month forward price of 1129.257.

(a)
What dividend yield is implied by this forward price?

(b)
Suppose you believe the dividend yield over the next 9 months will be only 0.5%. What arbitrage would you undertake?

(c)
Suppose you believe the dividend yield will be 3% over the next 9 months. What arbitrage would you undertake?

Worked answer (generated). For $S_0 = 1100$, $F = 1129.257$, $r = 5\%$ continuous and $T = 0.75$, $\delta = r - \ln(F/S_0)/T \approx 1.5000\%$. If the future dividend yield were *known* to be 0.5%, fair carry would be about 1137.76: buy the underpriced observed forward and short its synthetic counterpart. At a known 3% yield, fair carry would be about 1116.62: sell the overpriced observed forward and buy the synthetic. The question says "believe" rather than "know". A forecast about uncertain dividends makes these directional trades, *not* riskless arbitrages, unless the dividend stream can be locked in and the replication assumptions hold.

Problem 1.11

Suppose the S&P 500 index is currently 950 and the initial margin is 10%. You wish to enter into 10 S&P 500 futures contracts.

(a)
What is the notional value of your position? What is the margin?

(b)
Suppose you earn a continuously compounded rate of 6% on your margin balance, your position is marked to market weekly, and the maintenance margin is 80% of the initial margin. What is the greatest S&P 500 index futures price 1 week from today at which you will receive a margin call?

Worked answer (generated). Using the lecture's historical \$250-per-index-point contract multiplier and interpreting "enter into 10 contracts" as a *long* position (the direction implied by asking for a low-price margin call), notional is $10(250)(950) = \$2{,}375{,}000$. Initial margin is \$237,500, maintenance is $0.8(237{,}500) = \$190{,}000$. After one week the account is $237{,}500e^{0.06/52} + 2500(F_1 - 950)$; it hits maintenance at $F_1 \approx 930.89$. A short position would instead face a call after a *rise*; the problem does not explicitly state the direction or multiplier, so both assumptions are recorded here.

Problem 1.12

A one-year forward contract on a stock has a price of \$75. The stock is expected to pay a dividend of \$1.50 at two future times, six months from now and one year from now, and the annual effective risk-free interest rate is 6%. Calculate the current stock price.

(A)
70.75

(B)
73.63

(C)
75.81
(D)
77.87
(E)
78.04

Worked answer (generated). With an effective annual 6% rate and \$1.50 dividends at 0.5 and 1 year, delivery price $F = 75$ implies $S_0 = 75/1.06 + 1.50/(1.06)^{1/2} + 1.50/1.06 \approx \$73.63$. This is the stock's current value: the stock price equals the present value of its future delivery price *plus* dividends received along the way.

8 Binomial pricing: replication before probabilities

8.1 Why a new pricing model is needed

Payoff diagrams, put–call parity and no-arbitrage bounds tell us important things about an option without determining its unique price from the underlying alone. Parity can price a call from a put or conversely because the other option is already an input. A binomial model supplies a simple description of future stock prices and traded assets with which to replicate the option directly.

Over a period of length h, a stock currently worth S_0 has two possible ex-dividend prices, uS_0 and dS_0, with $u > d > 0$. "Binomial" refers to these two movements, not to an assumption that real prices have only two possible values for all time. Repeated steps expand the possible terminal values. The method's central operation is to work backwards from a known terminal payoff.

8.2 The one-period example, including the financing leg

The source example has a non-dividend-paying stock priced at 41, a European call with $K = 40$, one year to maturity and continuously compounded $r = 8\%$. The stock will be 59.954 or 32.903, corresponding to $u = 1.4623$ and $d = 0.8025$. The call therefore pays

$$C_u = (59.954 - 40)^+ = 19.954, \qquad C_d = (32.903 - 40)^+ = 0.$$

Choose Δ shares and an amount B in the bank account at time zero. A negative B is borrowing. Replication requires the two equations

$$\begin{cases} 59.954\Delta + e^{0.08}B = 19.954, \\ 32.903\Delta + e^{0.08}B = 0. \end{cases}$$

Subtracting the equations first isolates the stock holding. The source's rounded solution is $\Delta = 0.738$ and $B = -22.405$. The portfolio costs

$$C_0 = \Delta S_0 + B = 0.738(41) - 22.405 = 7.839.$$

It has the same payoff as the call in both states, so the law of one price gives the call the same price. The shares cost money now, while borrowing finances part of that purchase; B is not the repayment at maturity. That repayment is approximately $22.405e^{0.08} = 24.271$.

Worked example

Mispricing creates an explicit trade. If the call sells for 8 instead of 7.839, write the call and buy its replicating stock-and-bank portfolio. Receive $8 - 7.839 = 0.161$ now, and the net risky payoff at maturity is zero:

Position at maturity	Down stock 32.903	Up stock 59.954
Written call	0	−19.954
Purchased 0.738 shares	24.271	44.225
Loan repayment	−24.271	−24.271
Net payoff	0	0

If the call instead costs 7.50, buy it, short the replication and receive $7.839 - 7.50 = 0.339$ now. The reverse terminal legs cancel in each state. These rounded tables retain the course example; small rounding differences are not new economic profits.

8.3 Dividend yield changes the replication, not the principle

Suppose the stock pays a continuously compounded dividend yield δ, with dividends reinvested. An initial share holding of Δ grows to $\Delta e^{\delta h}$ shares over the period, while uS_0 and dS_0 remain the *ex-dividend* end-of-period prices. Consequently,

$$\begin{cases} \Delta S_0 u e^{\delta h} + B e^{rh} = C_u, \\ \Delta S_0 d e^{\delta h} + B e^{rh} = C_d. \end{cases}$$

The solution is

$$\Delta = e^{-\delta h} \frac{C_u - C_d}{S_0(u-d)}, \qquad B = e^{-rh} \frac{uC_d - dC_u}{u-d} = e^{-rh}\left(C_u - \Delta S_0 u e^{\delta h}\right).$$

The option price can therefore be written without computing the hedge separately:

$$C_0 = \Delta S_0 + B = e^{-rh}\left[\frac{e^{(r-\delta)h} - d}{u-d} C_u + \frac{u - e^{(r-\delta)h}}{u-d} C_d\right].$$

These expressions apply to a general payoff, not only a call, when C_u, C_d are replaced by that payoff's state values.

Why it matters

No probability of an up move was needed to solve the two replication equations. In a complete two-state market, the option price is pinned down by the traded stock and bank account, so changing an investor's belief about up/down probabilities does not change the no-arbitrage replication cost. Beliefs remain relevant to investment decisions; they are not a missing input to this replicated price.

No-arbitrage requires

$$d < e^{(r-\delta)h} < u.$$

The stock's financed, dividend-adjusted growth must lie between its two possible returns. If it were on or outside the interval, a combination of the stock and bank account would generate an arbitrage or a degenerate boundary inconsistent with the strictly positive state probabilities used here.

8.4 Delta is a holding and a local sensitivity

Δ is the number of shares needed in the hedge, adjusted for reinvested dividends in the convention above. It is also the sensitivity of the model option value to a small change in the underlying price.

The stock price and the node-specific hedge change over time, so replication over several periods is dynamic. The source's Bloomberg delta screenshot supplies no additional derivation; its teaching point is expressed here without an unexplained terminal image.

For the source's special case $\delta = 0, h = 1$,

$$\Delta = \frac{C_u - C_d}{S_0(u-d)}, \quad B = e^{-r}(C_u - \Delta S_0 u), \quad C_0 = e^{-r}\left[\frac{e^r - d}{u-d}C_u + \frac{u - e^r}{u-d}C_d\right].$$

Worked example

Lecture Problem 1, preserved. Let $S_0 = 100, K = 105, r = 8\%$ continuously compounded, $T = 0.5$, $\delta = 0, u = 1.3, d = 0.8$ and one binomial period. Find the call premium, Δ and B. Here the period is $h = T = 0.5$, not one year. Terminal payoffs are 25 and 0; $\Delta = 25/(130 - 80) = 0.5$, $B = -40e^{-0.04} = -38.4316$ and $C_0 = 50 - 38.4316 = 11.5684$.

8.5 Two periods: calculate terminal payoffs, then step back

Return to $S_0 = 41, K = 40, r = 0.08, \delta = 0$, now with two one-year periods and the same u, d. The intermediate prices are 59.954 and 32.903, and the terminal recombining prices are approximately 87.669, 48.114, 26.405. The call terminal payoffs are 47.669, 8.114, 0.

The up-then-down path and down-then-up path reach the same stock price because $ud = du$; a terminal payoff depending only on that price is consequently the same at the shared node. Starting at time 1, replicate the next period's option values separately at each node:

Node	Call value	Shares Δ	Bank investment B
Time 1, up	23.029	1	−36.925
Time 1, down	3.187	0.374	−9.111
Time 0	10.737	0.734	−19.337

For example, the first row is $23.029 = 59.954 - 36.925$. Rolling the two time-1 values back gives the initial call value 10.737. The source's complete labelled tree is retained as a diagram below because it connects prices, payoffs and the hedge at each node, rather than merely showing a screen.

8.6 Three periods and self-financing rebalancing

Adding a third period produces terminal stock prices approximately 128.198, 70.356, 38.612, 21.191 and call payoffs 88.198, 30.356, 0, 0. Backward induction yields these source node values:

Node	Call value	Shares	Bank investment
Time 2, up-up	50.745	1	−36.925
Time 2, mixed	11.925	0.956	−34.086
Time 2, down-down	0	0	0
Time 1, up	26.258	0.981	−32.580
Time 1, down	4.685	0.549	−13.390
Time 0	12.799	0.798	−19.899

These are rounded readings of the source tree; the precise values follow from the same recursion. At a node, the arriving hedge has the resources required to buy the new hedge: one rebalances the stock and bank positions without an external cash injection. This is the *self-financing* property. Keeping the original time-zero delta unchanged for all future nodes would not be the replication strategy described by the tree.

Figure 12: The arrows connect each current hedge to the next state. The middle terminal price is shared by up–down and down–up paths.

9 Puts, path dependence and the information a tree must retain

9.1 The put uses the same stock tree

For a European put, change the terminal condition to $(K - S_T)^+$; the stock tree and rollback rule stay the same. In the two-year $S_0 = 41, K = 40$ example, terminal put payoffs are $0, 0, 13.595$. At the up node the put is zero with a zero hedge. At the down node its value is 7.209, with approximately $\Delta = -0.626, B = 27.814$. At time zero the put costs 3.823, with $\Delta = -0.266, B = 14.749$.

For a non-dividend-paying stock, put–call parity applies at every node:

$$C_t - P_t = S_t - Ke^{-r(T-t)}.$$

Thus the 10.737 two-year call price and the corresponding stock and discounted strike imply the same 3.823 put price. Applying parity at intermediate nodes is a useful check on the entire calculation, not merely a shortcut at the initial date.

Worked example

Lecture Problem 2, preserved. With $S_0 = 100, K = 105, r = 8\%$ continuously compounded, $T = 0.5, \delta = 0, u = 1.3, d = 0.8$ and one period, find the put premium, Δ and B. Put payoffs are 0 and 25; $\Delta = (0 - 25)/(130 - 80) = -0.5$, $B = 65e^{-0.04} = 62.4513$ and $P_0 = -50 + 62.4513 = 12.4513$. The source's call-minus-put identity provides an independent check.

9.2 The three supplied interview questions

1. How many nodes are there in a recombining binomial tree with N time steps?

2. How many nodes are there in a non-recombining binomial tree with N time steps?
3. Assume no dividends, stock price 100, annual continuously compounded risk-free rate 5%, and a one-year European call struck at the money. If volatility is zero, what is the call worth?

Generated explanations. At terminal time there are $N+1$ recombining nodes and 2^N non-recombining nodes. If "nodes in a tree" means all dates including the root, the counts are $(N+1)(N+2)/2$ and $2^{N+1}-1$ respectively; state the interpretation rather than confusing the terminal count with the whole-tree count. With zero volatility and no dividends, $S_T = 100e^{0.05}$ is deterministic, so the call price is $e^{-0.05}(100e^{0.05} - 100) = 100(1 - e^{-0.05}) \simeq 4.8771$. Zero volatility does not make an at-the-money call worthless when deterministic growth moves the terminal stock above the strike.

9.3 An Asian payoff cannot forget the path

For an Asian option, an average of observed prices enters the payoff. Two paths can have the same terminal stock price but different intermediate prices and hence different averages. Recombining solely on the terminal stock price would merge economically different states. A full non-recombining tree retains the path but grows exponentially, with 2^n states at date n instead of $n+1$. Other state representations may compress information, but must still retain what the payoff needs.

The arithmetic-average source example uses $S_0 = 41, K = 40, u = 1.4623, d = 0.8025, r = 0.08, T = 2, h = 1$ and observes all three dates, including time zero:

$$G_T = \left(\frac{1}{3}\sum_{k=0}^{2} S_k - 40\right)^+.$$

The source spreadsheet's economically meaningful information is reconstructed as a table, rather than left as a screenshot:

Path	Terminal stock	Arithmetic average	Call payoff
up-up	87.6712	62.8752	22.8752
up-down	48.1133	49.6892	9.6892
down-up	48.1133	40.6719	0.6719
down-down	26.4043	33.4356	0

The intermediate up-node has stock 59.9543, current average 50.4772, option value 14.1243, $\Delta = 0.3333$ and $B = -5.8605$. The down-node has stock 32.9025, average 36.9513, option value 0.2640, $\Delta = 0.0310$ and $B = -0.7544$. At the root, stock and current average both equal 41, the option is worth 5.6886 and the hedge has approximately $\Delta = 0.5124, B = -15.3182$. In particular, the up-down and down-up rows cannot be merged even though the terminal stock prices match.

The source also sets a geometric-average exercise:

$$G_T = \left[\left(\prod_{k=0}^{2} S_k\right)^{1/3} - 40\right]^+,$$

and refers to McDonald, Chapter 14.2. Preserve the three observation dates and replace the arithmetic average by the geometric average; do not price a geometric-average contract using the arithmetic-average terminal values. The reference is optional further reading on this specific topic rather than an unexplained external link.

10 Volatility and risk-neutral weights

10.1 Square-root-of-time scaling and the forward tree

The source measures uncertainty by the annualized standard deviation σ of continuously compounded stock returns. If period returns are uncorrelated in the stated model, a period of length h has standard deviation

$$\sigma_h = \sigma\sqrt{h}.$$

With no uncertainty, stock growth must satisfy $S_{t+h} = S_t e^{(r-\delta)h}$. Uncertainty is introduced with

$$u = e^{(r-\delta)h+\sigma\sqrt{h}}, \qquad d = e^{(r-\delta)h-\sigma\sqrt{h}}.$$

The deterministic drift and the two opposite shocks have different roles. Setting $\sigma = 0$ fixes the future price in advance, but does not make prices constant through time. This *forward tree* construction places $e^{(r-\delta)h}$ between d and u when $\sigma > 0$. In the original one-year example, $r = 0.08, \delta = 0, \sigma = 0.3$ gives $u = e^{0.38} = 1.4623$ and $d = e^{-0.22} = 0.8025$.

Worked example

Lecture Problem 3, preserved. A stock index has $S_0 = 100, \sigma = 30\%, r = 5\%, \delta = 3\%, T = 3$ and three periods. Find European call and put prices at strike 95. Using $h = 1$ and the forward-tree parameters, the source reports call 24.0058 and put 14.3799. The dividend-adjusted parity check is $C_0 - P_0 = 100e^{-0.03(3)} - 95e^{-0.05(3)}$. This checks both the dividend convention and the maturity.

10.2 The pricing weights are risk-neutral probabilities

Under the strict no-arbitrage inequality, both coefficients inside the replication formula are positive and add to one. Define

$$p^* = \frac{e^{(r-\delta)h} - d}{u - d}, \qquad 1 - p^* = \frac{u - e^{(r-\delta)h}}{u - d}.$$

Then the one-step value is

$$V_t = e^{-rh}\left[p^* V^u_{t+h} + (1 - p^*)V^d_{t+h}\right].$$

This is a discounted expectation under the risk-neutral pricing measure. It is an interpretation of the replicating price, not an estimate of the empirical frequency of up moves. A risk-neutral investor is indifferent between a sure payment and a risky payment with the same expected amount; a risk-averse investor prefers the sure payment at that same real-world expectation. Changing to pricing probabilities does not assert that actual investors become risk-neutral.

For the one-year 41 stock example, $p^* = 0.4256$ and $1 - p^* = 0.5744$, so $C_0 = e^{-0.08}[0.4256(19.954) + 0.5744(0)] = 7.839$. For two periods, the terminal call payoffs and probabilities are

State	Call payoff	Probability
up-up	47.669	$(p^*)^2 = 0.1811$
one up, one down	8.114	$2p^*(1 - p^*) = 0.4889$
down-down	0	$(1 - p^*)^2 = 0.3300$

Their discounted weighted sum is 10.737. The factor two counts two different sequences reaching the shared middle stock node. For three periods, the source payoffs $88.198, 30.356, 0, 0$ have probabilities approximately $0.0771, 0.3121, 0.4213, 0.1896$ and give the price 12.799. These probabilities sum to one; their products count paths rather than expressing a new price model.

10.3 The n-period European formula

With constant u, d, p^* and a payoff depending only on terminal stock price,

$$C_0 = e^{-rT} \sum_{k=0}^{n} \binom{n}{k} (p^*)^k (1-p^*)^{n-k} \left(S_0 u^k d^{n-k} - K\right)^+, \quad T = nh.$$

For a put, replace the final positive part by $(K - S_0 u^k d^{n-k})^+$. This one-step terminal summation matches backward induction for European terminal-price payoffs. It does not eliminate the need to retain path information for an Asian option or to compare early exercise for an American option. As the discretization is refined in the specified diffusion limit, the binomial method connects to Black–Scholes; a finite tree is an approximation, not the continuous model itself.

Worked example

Lecture Problem 4, preserved. Let $S_0 = 100, \sigma = 0.3, r = 0.08, T = 1, \delta = 0$ and $n = 3$. Use the terminal formula and plot the risk-neutral distribution of the year-1 stock price. The source gives $u = 1.2212, d = 0.8637, p^* = 0.4568$ and this distribution:

Up moves	Stock price	Probability
3	182.14	0.0953
2	128.81	0.3400
1	91.10	0.4044
0	64.43	0.1603

The probabilities are masses on four discrete prices, not the heights of a continuous probability density. The supplied exercise does not identify a strike for a unique numerical call price; the formula above describes how to price a chosen terminal payoff.

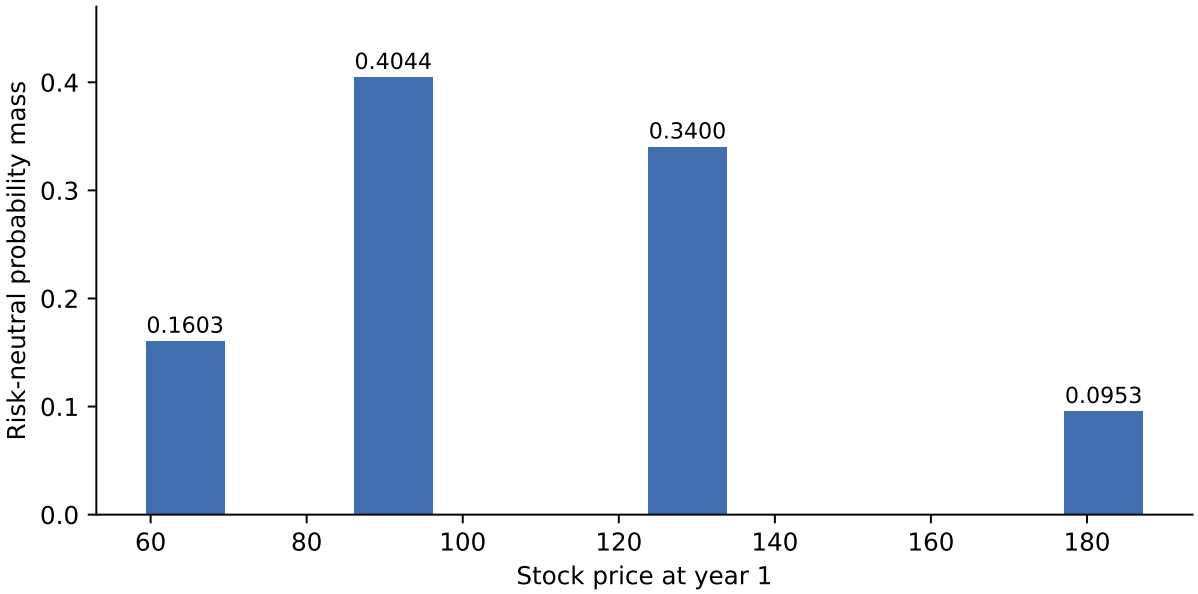


Figure 13: Discrete risk-neutral probability masses for Lecture Problem 4. The labels correspond to the table, not observed empirical frequencies.

11 American exercise and real investment options

11.1 Exercise value versus continuation value

An American option can be exercised before maturity and must be at least as valuable as the corresponding European option:

$$C_{\text{Am}} \geq C_{\text{Eur}}, \qquad P_{\text{Am}} \geq P_{\text{Eur}}.$$

The source combines this with the loose bounds $S \geq C_{\text{Am}} \geq C_{\text{Eur}} \geq \max(0, S - PV(\text{Div}) - PV(K))$ and $K \geq P_{\text{Am}} \geq P_{\text{Eur}} \geq \max(0, PV(K) - S + PV(\text{Div}))$. Their applicability depends on the dividend and rate assumptions developed in the bounds chapter.

At each permitted exercise node, compute discounted continuation and compare it with immediate exercise:

$$V_t^{\text{Am}} = \max\left\{ g(S_t), e^{-rh}\left[p^* V_{t+h}^u + (1-p^*) V_{t+h}^d \right] \right\}.$$

Use $g(S) = (S-K)^+$ for a call and $(K-S)^+$ for a put. The terminal values are the same European-style intrinsic payoffs; the extra maximization occurs before expiry.

For a non-dividend-paying stock under the course's nonnegative-rate assumptions, early exercise of an American call is not optimal. The holder would pay the strike earlier and surrender the remaining insurance/time value without gaining dividends. With dividends, exercise can become optimal: receiving the stock gives access to dividends that an unexercised call does not receive. The source modifies its original example to a continuously compounded yield of 6.5% and illustrates a call exercise node at stock price 76.982. Its labelled exercise/continuation comparison is retained in a graph rather than a whole slide.

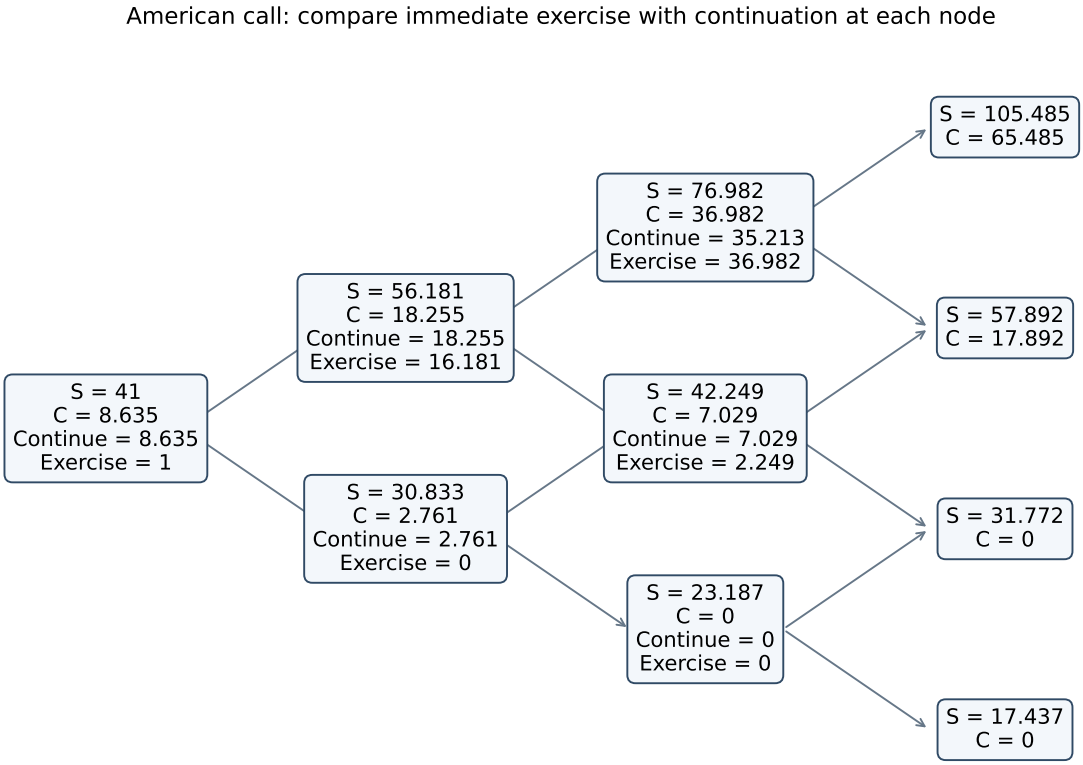


Figure 14: Source call tree reconstructed as a diagram: at $S = 76.982$, exercise 36.982 exceeds continuation 35.213.

An American put may optimally be exercised early even with no dividends. If the stock approaches zero, the European put is approximately worth $PV(K)$ while immediate exercise pays K; at a positive rate, earlier receipt of the strike can be worth more than the lost protection. The source's dividend-yield example identifies an early-exercise node at stock 23.187. The graph distinguishes the value if exercised, P_{EX}, from continuation, P_{NO}.

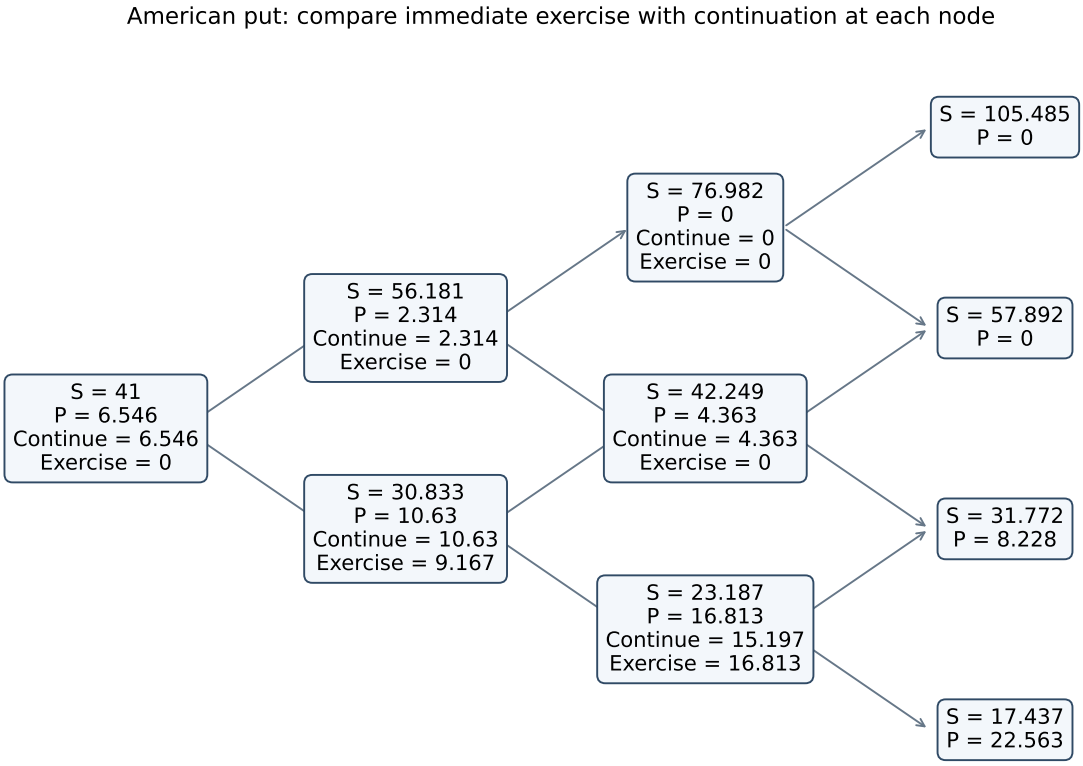


Figure 15: Source put tree reconstructed: at $S = 23.187$, exercise 16.813 exceeds continuation 15.197. Other nodes retain the option when continuation is higher.

Worked example

Lecture Problem 5, preserved. For $S_0 = 100, \sigma = 30\%, r = 5\%, \delta = 3\%, T = 3, n = 3$, price the American call and put struck at 95. The source reports call 24.1650 and put 15.2593, both above their European counterparts 24.0058 and 14.3799. The call also benefits from exercise flexibility here because the example has dividends; the non-dividend call result is not applicable.

The costs and benefits of early exercise are asymmetric. A call holder gains the stock/dividends, but pays the strike sooner and loses insurance against a later fall. A put holder receives the strike sooner, but loses dividends on a stock position surrendered through exercise and loses insurance against a later rise. Compare values at each node; "in the money" alone does not determine optimal early exercise.

11.2 A real option is a right to take a business decision

A real option applies contingent-decision logic to an investment project rather than a traded option contract. Paying an investment cost to receive a project's future cash flows resembles paying a call strike to receive an asset. The mapping is investment cost $\leftrightarrow$ strike, present value of project

$\leftrightarrow$ underlying asset value. Waiting, expansion, shutdown and abandonment change the feasible future decisions.

The source project requires an initial investment of 100 and is expected to produce a perpetual cash flow of 18 in year 1, growing at 3% annually thereafter. With the stated 15% discount rate for the growing perpetuity, $PV = 18/(0.15 - 0.03) = 150$ and the static NPV is $150 - 100 = 50$. Its cash-flow yield is 18/150, represented continuously as $\delta = \log(1 + 18/150) = 0.1133$. These project-discounting inputs are not the risk-free rate subsequently used in the option-pricing tree.

The source uses continuously compounded risk-free $r = 6.766\%$ and annual project-return volatility 50%. A forward tree with $u = 1.5751, d = 0.5795$ produces the following cash flows and project values:

Node	Project cash flow	Present value of project
Initial	18.00	150.00
Year 1, up	28.35	236.27
Year 1, down	10.43	86.92
Year 2, up-up	44.66	372.15
Year 2, mixed	16.43	136.91
Year 2, down-down	6.04	50.37

The risk-neutral probability is approximately 0.3775 at each step. Investment can occur at time 0, 1 or 2. At the final date, the exercise values are 272.15, 36.91 and 0. At the year-1 up node, continuation is 117.50 but exercise is $236.27 - 100 = 136.27$, so invest. At the down node, continuation is 13.02 while immediate exercise has value zero, so wait. At time zero, continuation is 55.66 versus immediate investment NPV of 50.

The ability to wait adds value by allowing rejection of bad future investments without giving up the opportunity to invest in good states. Investing immediately abandons the remaining exercise flexibility, not merely a five-unit accounting item. The source also asks the reader to repeat the calculation with 30% volatility and discuss the change; that exercise is retained as an invitation to compare the whole decision tree, not answered by simply assuming that static NPV changes with volatility.

11.3 The supplied expansion exercise

Desautels Industries considers a plant costing 140 million initially. Cash flows at year 2 are 200 million after two good years, 150 million after one good and one bad year, and 100 million after two bad years. After year 1 the company can double capacity by investing another 140 million. The annual risk-free rate is 6% and the risk-neutral probability of a good year is 0.53. Compute the project value with and without the option to expand. These exact supplied inputs are the exercise; the timing and incremental cash flows must be used consistently.

Generated solution, for the stated two-year terminal cash-flow interpretation. Write $q = 0.53$ and use the source's 6% annual risk-free rate. Without expansion,

$$NPV_0 = \frac{q^2(200) + 2q(1-q)(150) + (1-q)^2(100)}{(1.06)^2} - 140.$$

At year 1 after a good year, the extra capacity has incremental discounted expected cash flow $[q(200) + (1-q)(150)]/1.06$ and costs 140. After a bad year, substitute $[q(150) + (1-q)(100)]/1.06$. Take the positive part at each node and discount the risk-neutral weighted incremental exercise values to time zero. The option value is added to the no-expansion NPV. This generated calculation assumes effective annual compounding for the source's unqualified "6% per year"; a continuous interpretation would use different discount factors and must be stated explicitly.

Numerically, the no-expansion NPV is -3.8305 million. The incremental year-1 NPVs are 26.5094 after a good year and -20.6604 after a bad year, so expand only after a good year. The capacity option is worth 13.2547 million initially; the project NPV including expansion is 9.4242 million.

11.4 Other real options and limitations of the model

The source examples classify when to invest as a call option; shutting down, restarting or permanently abandoning a project as put-like rights; an investment that generates later opportunities as a compound option; and flexibility to switch inputs, outputs or technologies as a rainbow option. The economic payoff and permitted decision times must be specified before using these names.

The final slide cautions that the model's assumptions simplify reality. Volatility need not be constant; stock prices may jump; returns may exhibit short-run momentum and longer-run negative correlation rather than time independence. Dividends are often discrete quarterly or annual payments, and their amount can be uncertain, unlike a known continuous yield. The source notes that a binomial tree can be adapted for discrete dividends and refers to McDonald, Chapter 11.4. Model assumptions are conditions for a calculation, not properties guaranteed by a fitted tree.

12 Options: supplied exercise series II

Problem 2.1

Happy Jalapenos, LLC has an exclusive contract to supply jalapeno peppers to the organizers of the annual jalapeno eating contest. The contract states that the contest organizers will take delivery of 10,000 jalapenos in one year at the market price. It will cost Happy Jalapenos 1,000 to provide 10,000 jalapenos and today's market price is 0.12 for one jalapeno. The continuously compounded risk-free interest rate is 6%.

Happy Jalapenos has decided to hedge as follows: Buy 10,000 0.12-strike put options for 84.30 and sell 10,000 0.14-stike call options for 74.80. Both op- tions are one-year European. Happy Jalapenos believes the market price in one year will be somewhere between 0.10 and 0.15 per jalapeno.

Determine which of the following intervals represents the range of possible profit one year from now for Happy Jalapenos.

(A) - 200 to 100
(B) -110 to 190
(C) -100 to 200
(D) 190 to 390
(E) 200 to 400

Worked example

Generated solution. At expiry, the collar replaces the selling price by $\min(0.14, \max(S_T, 0.12))$. The net premium paid now is $84.30 - 74.80 = 9.50$. Profit at the common terminal date is

$$\Pi_T = 10000 \min(0.14, \max(S_T, 0.12)) - 1000 - 9.50e^{0.06}.$$

For the supplied price range, profit runs from 189.91 to 389.91, rounded to 190–390. Answer D. The financed premium must be subtracted even when the collar payoff is zero.

Problem 2.2

The PS index has the following characteristics:
1. One share of the PS index currently sells for 1,000.
2. The PS index does not pay dividends.

Sam wants to lock in the ability to buy this index in one year for a price of 1,025. He can do this by buying or selling European put and call options with a strike price of 1,025. The annual effective risk-free interest rate is 5%.

Determine which of the following gives the hedging strategy that will achieve Sam's objective and also gives the cost today of establishing this position.

(A) Buy the put and sell the call, receive 23.81
(B) Buy the put and sell the call, spend 23.81
(C) Buy the put and sell the call, no cost
(D) Buy the call and sell the put, receive 23.81
(E) Buy the call and sell the put, spend 23.81

Worked example

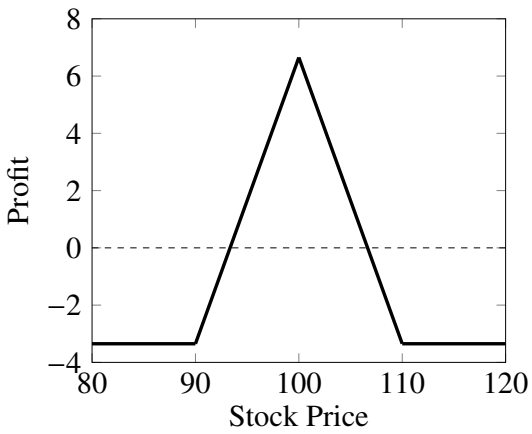
Generated solution. Buy the call and sell the put at strike 1025. The combined terminal payoff is $S_T - 1025$, so together with payment of 1025 it delivers the index. Put–call parity gives $C_0 - P_0 = 1000 - 1025/1.05 = 23.8095$. The position costs 23.81 now. Answer E.

Problem 2.3

Stock ABC has the following characteristics:
1. The current price to buy one share is 100.
2. The stock does not pay dividends.
3. European options on one share expiring in one year have the following prices:

Strike price	Call option price	Put option price
90	14.63	0.24
100	6.80	1.93
110	2.17	6.81

A butterfly spread on this stock has the following profit diagram.
The continuously compounded risk-free interest rate is 5%.
Determine which of the following will NOT produce this profit diagram.
(A) Buy a 90 put, buy a 110 put, sell two 100 puts
(B) Buy a 90 call, buy a 110 call, sell two 100 calls
(C) Buy a 90 put, sell a 100 put, sell a 100 call, buy a 110 call
(D) Buy one share of the stock, buy a 90 call, buy a 110 put, sell two 100 puts
(E) Buy one share of the stock, buy a 90 put, buy a 110 call, sell two 100 calls.



The source profit curve has flat negative tails and a peak around $S_T = 100$.

Worked example

Generated solution. Answer D. The call butterfly in B has payoff $(S_T - 90)^+ - 2(S_T - 100)^+ + (S_T - 110)^+$, a tent of height 10; A has the same payoff. C has that payoff minus 10; E has that payoff plus 90. Their corresponding financed costs offset these constant differences (up to rounding in the quoted premiums). D instead has payoff $2S_T - 90$ below 90 and above 110, and so cannot produce a flat tail. For B the premium is $14.63 + 2.17 - 2(6.80) = 3.20$, and the tails of its profit diagram are $-3.20e^{0.05} \simeq -3.364$. Always compare profit with profit, accounting for the initial stock position as well as option premiums.

Problem 2.4

A trader shorts one share of a stock index for 50 and buys a 60 strike European call option on that stock that expires in 2 years for 10. Assume the annual effective risk-free interest rate is 3%. The stock index increases to 75 after 2 years. Calculate the profit on your combined position, and determine an alternative name for this combined position.

(A) Profit: -22.64 Name: Floor
(B) Profit: -17.56 Name: Floor
(C) Profit: -22.64 Name: Cap
(D) Profit: -17.56 Name: Cap
(E) Profit: -22.64 Name: "Written" covered call

Worked example

Generated solution. Short stock plus long call is a cap on the short-stock exposure. The initial receipt is $50 - 10 = 40$. At expiry the position pays $-75 + (75 - 60) = -60$, hence $\Pi_T = 40(1.03)^2 - 60 = -17.564$. Answer D: profit -17.56, cap.

Problem 2.5

The current price of a non-dividend paying stock is 40 and the continuously compounded risk-free interest rate is 8%. You are given that the price of a 35-strike call option is 3.35 higher than the price of a 40 strike call option, where both options expire in 3 months.

Calculate the amount by which the price of an otherwise equivalent 40-strike put option exceeds the price of an otherwise equivalent 35-strike put option.

(A) 1.55
(B) 1.65
(C) 1.75
(D) 3.25
(E) 3.75

Worked example

Generated solution. Subtract put–call parity at the two strikes:

$$P_{40} - P_{35} = C_{40} - C_{35} + 5e^{-0.08(3/12)} = -3.35 + 4.90099 = 1.55099.$$

Answer A.

Problem 2.6

The current price of a non-dividend paying stock is 40 and the continuously compounded risk-free interest rate is 8%. You enter into a short position on 3 call options, each with 3 months to maturity, a strike price of 35, and an option premium of 6.13. Simultaneously, you enter into a long position on 5 call options, each with 3 months to maturity, a strike price of 40, and an option premium of 2.78. All 8 options are held until maturity.

Calculate the maximum possible profit and the maximum possible loss for the entire option portfolio.

(A) Maximum profit: 3.42 Maximum loss: 4.58
(B) Maximum profit: 4.58 Maximum loss: 10.42
(C) Maximum profit: Unlimited Maximum loss: 10.42
(D) Maximum profit: 4.58 Maximum loss: Unlimited
(E) Maximum profit: Unlimited Maximum loss: Unlimited

Worked example

Generated solution. The initial receipt is $3(6.13) - 5(2.78) = 4.49$, whose future value is $4.49e^{0.02} = 4.5807$. The payoff is $-3(S_T - 35)^+ + 5(S_T - 40)^+$: zero below 35; $-3(S_T - 35)$ between 35 and 40; $2S_T - 95$ above 40. Its minimum is -15 at 40. Maximum loss is therefore $15 - 4.5807 = 10.4193$; the upper tail is unbounded. Answer C.

Problem 2.7

An investor is analyzing the costs of two-year, European options for aluminum and zinc at a particular strike price. For each ton of aluminum, the two-year forward price is 1400, a call option costs 700, and a put option costs 550. For each ton of zinc, the two-year forward price is 1600 and a put option costs 550. The annual effective risk-free interest rate is 6%.

Calculate the cost of a call option per ton of zinc.

(A) 522
(B) 800
(C) 878
(D) 900
(E) 1231

Worked example

Generated solution. Use forward-form parity $C - P = (F - K)/(1.06)^2$. For aluminium, $700 - 550 = 150$, giving $K = 1400 - 150(1.06)^2 = 1231.46$. Zinc then has $C = 550 + (1600 - 1231.46)/(1.06)^2 = 877.999$. Answer C: 878.

Problem 2.8

You are given:

i) An investor short-sells a non-dividend paying stock that has a current price of 44 per share.
ii) This investor also writes a collar on this stock consisting of a 40-strike European put option and a 50-strike European call option. Both options expire in one year.
iii) The prices of the options on this stock are:

Strike	Call	Put
40	8.42	2.47
50	3.86	7.42

iv) The continuously compounded risk-free interest rate is 5%.
v) Assume there are no transaction costs.

Calculate the maximum profit for the overall position at expiration.

(A) 2.61
(B) 3.37
(C) 4.79
(D) 5.21
(E) 7.39

Worked example

Generated solution. Writing the collar means short the 40-put and long the 50-call, combined with short stock. The initial cash receipt is $44 + 2.47 - 3.86 = 42.61$. Terminal payoff is $-S_T - (40 - S_T)^+ + (S_T - 50)^+$: -40 below 40, $-S_T$ between 40 and 50, and -50 above 50. Maximum profit is $42.61e^{0.05} - 40 = 4.7947$. Answer C.

Problem 2.9

An insurance company sells single premium deferred annuity contracts with return linked to a stock index, the time-t value of one unit of which is denoted by S (t). The contracts offer a minimum guarantee return rate of g%. At time 0, a single premium of amount π is paid by the policyholder, and $\pi \times$ y% is deducted by the insurance company. Thus, at the contract maturity date, T, the insurance company will pay the policyholder

$$\pi(1 - y\%) \max\left[\frac{S(T)}{S(0)}, (1 + g\%)^T\right].$$

You are given the following information:

(i) The contract will mature in one year.
(ii) The minimum guarantee rate of return, g%, is 3%.
(iii) Dividends are incorporated in the stock index. That is, the stock index is constructed with all stock dividends reinvested.
(iv) S (0) = 100
(v) The price of a one-year European put option, with strike price \$103, on the stock index is \$15.21.

Determine y%, so that the insurance company does not make or lose money on this contract.

(A) 12.8%
(B) 13.0%
(C) 13.2%
(D) 13.4%
(E) 13.6%

Worked example

Generated solution. For $T = 1$, the payoff is $\pi(1 - y)\max(S_T, 103)/100$, where y is the deduction as a decimal fraction. Since $\max(S_T, 103) = S_T + (103 - S_T)^+$, the replicating stock-plus-put costs $100 + 15.21 = 115.21$ per 100 of participation. Fairness requires $(1 - y)115.21/100 = 1$, so $y = 15.21/115.21 = 0.132019$. Answer C: 13.2%. Dividends are reinvested in the index, as specified in the question.

Problem 2.10

Consider a chooser option (also known as an as-you-like-it op- tion) on a nondividend-paying stock. At time 1, its holder will choose whether it becomes a European call option or a European

put option, each of which will expire at time 3 with a strike price of $100. The chooser option price is $20 at time t = 0. The stock price is $95 at time t = 0. Let C(T) denote the price of a European call option at time t = 0 on the stock expiring at time T, T > 0, with a strike price of $100.

You are given:

(i) The risk-free interest rate is 0.

(ii) C(1) = $4.

Determine C(3)

(A) $9

(B) $11

(C) $13

(D) $15

(E) $17

Worked example

Generated solution. At time 1, maturity-3 parity with $r = 0$ gives $P_{1,3} = C_{1,3} + 100 - S_1$. Thus the chooser is worth $C_{1,3} + (100 - S_1)^+$, replicated by a maturity-3 call plus a maturity-1 put of strike 100. At time zero the latter put costs $P(1) = C(1) + 100 - S_0 = 4 + 5 = 9$. Hence $20 = C(3) + 9$ and $C(3) = 11$. Answer B.

Problem 2.11

The current price of a non-dividend paying stock is 40 and the continuously compounded risk-free interest rate is 8%. The following table shows call and put option premiums for three-month European of various ex- ercise prices:

Strike	Call	Put
35	6.13	0.44
40	2.78	1.99
45	0.97	5.08

A trader interested in speculating on volatility in the stock price is consider- ing two investment strategies. The first is a 40-strike straddle. The second is a strangle consisting of a 35-strike put and a 45-strike call.

Determine the range of stock prices in 3 months for which the strangle out- performs the straddle.

(A) The strangle never outperforms the straddle

(B) 33.56 < S T < 46.44

(C) 35.13 < S T < 44.87

(D) 36.57 < S T < 43.43

(E) The strangle always outperforms the straddle.

Worked example

Generated solution. The straddle costs $2.78 + 1.99 = 4.77$; the strangle costs $0.44 + 0.97 = 1.41$. The strangle saves $3.36e^{0.02} = 3.42788$ at expiry. For $35 < S_T < 40$, its profit advantage is $S_T - 40 + 3.42788$; for $40 < S_T < 45$, it is $40 - S_T + 3.42788$. Outside these intervals it loses five units of payoff relative to the straddle. The positive-advantage interval is $36.5721 < S_T < 43.4279$. Answer D.

Problem 2.12

For each ton of a certain type of rice commodity, the four-year forward price is 300. A four-year 400-strike European call option costs 110. The continuously compounded risk-free interest rate is 6.5%.

Calculate the cost of a four-year 400-strike European put option for this rice commodity

(A) 10.00
(B) 32.89
(C) 118.42
(D) 187.11
(E) 210.00

Worked example

Generated solution. Forward-form parity gives $P = C + e^{-rT}(K - F) = 110 + 100e^{-0.065(4)} = 187.1052$. Answer D: 187.11.

Problem 2.13

Suppose call and put prices are given by

Strike	50	55
Call premium	9	10
Put premium	7	6

What no-arbitrage property is violated? What spread position would you use to effect arbitrage? Demonstrate that the spread position is an arbitrage.

Worked example

Generated solution. Calls must be nonincreasing and puts nondecreasing in strike for equal maturities and otherwise identical contracts. Both inequalities are violated. Buy the 50-call for 9 and sell the 55-call for 10: receive 1 immediately and obtain $(S_T - 50)^+ - (S_T - 55)^+ \geq 0$ later. Alternatively buy the 55-put for 6 and sell the 50-put for 7: receive 1 immediately and obtain $(55 - S_T)^+ - (50 - S_T)^+ \geq 0$. For the call spread the terminal payoff is 0, $S_T - 50$, or 5 in the regions below 50, between 50 and 55, or above 55. For the put spread it is 5, $55 - S_T$, or 0 in those regions. Neither strategy requires a forecast.

13 Binomial model: complete exercise set

The supplied exercise statements are retained below. The worked solutions are **generated study explanations**, independently calculated from their inputs. Unless explicit up/down multipliers are given, the course uses the forward-tree convention

$$h = T/n, \quad u = e^{(r-\delta)h+\sigma\sqrt{h}}, \quad d = e^{(r-\delta)h-\sigma\sqrt{h}}, \quad p^* = \frac{e^{(r-\delta)h} - d}{u - d}.$$

These u, d differ from the drift-free CRR convention; changing conventions changes finite-step numerical prices. American values are obtained by comparing continuation with immediate exercise at every node.

13.1 Problem 3.1: replication versus immediate exercise

Statement. A stock is initially 50 and after one year is either 125 or 50. The annual *effective* risk-free rate is 50%. Consider a European put with strike 150. (a) What are the payoffs? (b) Which stock and risk-free-bond portfolio replicates the option? (c) What is its price? (d) Verify using risk-neutral probabilities. (e) Would an American put have the same price?

Generated solution. The up/down payoffs are 25 and 100. With B the current bank-account investment,

$$125\Delta + 1.5B = 25, \quad 50\Delta + 1.5B = 100 \quad \Longrightarrow \quad \Delta = -1,\ B = 100.$$

The replicating portfolio costs $-50 + 100 = 50$. The risk-neutral up probability is $p^* = (1.5 - 1)/(2.5 - 1) = 1/3$, so $(25/3 + 200/3)/1.5 = 50$. Immediate exercise gives $150 - 50 = 100$. The American put is worth 100 and is exercised now. An effective 50% rate means a growth factor 1.5, not $e^{0.5}$.

13.2 Problem 3.2: American call in a two-period tree

Statement. Each period is one year. A non-dividend-paying stock has $S_0 = 20$, $u = 1.2840$ and $d = 0.8607$ (gross up/down capital-gain factors). The continuously compounded risk-free rate is 5%. Calculate a two-period American call with strike 22. Choices: (A) 0; (B) 1; (C) 2; (D) 3; (E) 4.

Generated solution. $p^* = (e^{.05} - .8607)/(1.2840 - .8607) = .4502034$. Terminal payoffs in down-down, mixed and up-up states are $0, .102776, 10.973120$. The one-period continuation values are .0440135 and 4.7529527, and the root is 2.0584547. Exercise never improves the call's value: choose **C**, the nearest listed value.

13.3 Problem 3.3: call and put hedge quantities

Statement. $S_0 = 100, K = 105, r = 8\%, T = .5, \delta = 0, u = 1.3, d = .8, n = 1$. Find (a) the premium, Δ and B for a European call; (b) the corresponding quantities for a European put. Rates are continuously compounded.

Generated solution. The call payoffs are 25 and 0, the put payoffs 0 and 25. Therefore

$$\Delta_C = .5,\ B_C = -40e^{-.04} = -38.431578,\ C_0 = 11.568422;$$

$$\Delta_P = -.5,\ B_P = 65e^{-.04} = 62.451314,\ P_0 = 12.451314.$$

The negative call bank position is borrowing. Check $C_0 - P_0 = 100 - 105e^{-.04}$. Here B denotes the amount invested today, rather than its terminal repayment.

13.4 Problem 3.4: dividends and early exercise

Statement. A one-period model has $h = 1, S_0 = 100, r = 0, \sigma = 30\%, \delta = .08$. Compute American calls for strikes 70, 80, 90 and 100. (a) Which strikes lead to early exercise? (b) Use put–call parity to explain why early exercise does not occur at the higher strikes and occurs for all strikes lower than those identified in (a).

Generated solution. $u = e^{.22} = 1.2460767$, $d = e^{-.38} = .6838614$ and $p^* = .4255575$. Compare each European continuation value with $100 - K$:

K	Continuation	Immediate exercise	American price
70	23.2387	30	30
80	18.9831	20	20
90	14.7276	10	14.7276
100	10.4720	0	10.4720

Exercise occurs at 70 and 80. European parity with $r = 0$ gives $C_E - (S_0 - K) = P_E - S_0(1 - e^{-\delta})$. The lost dividend value is fixed at 7.688365, while the European put increases with K. At 80 the put is still below that amount, so every lower strike also favours exercise; at 90 and 100 it exceeds that amount. This is an application of *European* parity to the continuation value; American prices do not obey the same equality.

13.5 Problem 3.5: three-year American put

Statement. A stock pays a continuous dividend yield of 6.5%. Each period is one year and the continuously compounded risk-free rate is 10%. The three-period stock tree is

Time	Stock prices (highest to lowest)
0	300
1	375, 210
2	468.75, 262.5, 147
3	585.9375, 328.125, 183.75, 102.9

Find a three-year at-the-money American put. Choices: (A) 15.86; (B) 27.40; (C) 32.60; (D) 39.73; (E) 57.49.

Generated solution. $K = 300, u = 1.25, d = .7$ and $p^* = (e^{.035} - .7)/.55 = .6102177$. Terminal payoffs from highest to lowest are $0, 0, 116.25, 197.10$. American values at time 2 are $0, 41.000172, 153$ and at time 1 are $14.460336, 90$. Early exercise occurs at $S = 147$ at time 2 and at $S = 210$ at time 1. The root is 39.726311: choose **D**.

13.6 Problem 3.6: the first integer strike that should be exercised

Statement. Use the usual one-period tree for a non-dividend-paying stock. The period is 3 months, $S_0 = 100$, volatility 30%, and continuously compounded risk-free rate 4%. An investor owns an American put expiring at the period's end. Find the smallest integer strike for which exercise at the beginning is optimal. Choices: (A) 114; (B) 115; (C) 116; (D) 117; (E) 118.

Generated solution. $h = .25, u = e^{.16} = 1.1735109, d = e^{-.14} = .8693582, p^* = .4625702$. In the relevant strike range $Sd < K < Su$, only the down payoff is positive. Immediate exercise dominates when

$$K - S_0 \geq e^{-rh}(1 - p^*)(K - S_0 d) \quad \Longleftrightarrow \quad K \geq \frac{S_0[1 - e^{-rh}(1 - p^*)d]}{1 - e^{-rh}(1 - p^*)}.$$

At 114 continuation is $14.400370 > 14$; at 115 it is approximately $14.932452 < 15$. The first integer is **115 (B)**.

13.7 Problem 3.7: full node values and parity

Statement. The annual effective riskless rate is 25% per period. A stock initially worth 100 doubles or halves each period: prices at time 1 are 200 and 50, and at time 2 are 400, 100 and 25. Consider European and American options with $K = 75$ and maturity two periods away. (a) Find risk-neutral probabilities. (b) Find European put values at each node. (c) Find American put values at each node. (d) Find the European call using parity.

Generated solution. $p^* = (1.25 - .5)/(2 - .5) = .5$, so up and down are equally weighted. Put payoffs at 400, 100, 25 are $0, 0, 50$. European values at 200 and 50 are 0 and 20, giving $P_E = 8$. The American values at those nodes are 0 and 25, since exercise at 50 gives 25; thus $P_A = 10$. Terminal values coincide for both exercise styles. The European call is

$$C_E = P_E + 100 - 75/(1.25)^2 = 8 + 100 - 48 = 60.$$

13.8 Problem 3.8: two-step stock tree

Statement. $S_0 = 40, K = 40, r = 8\%$ continuously compounded, $\sigma = 30\%, \delta = 0, T = .5, n = 2$. (a) Construct the stock tree and give u, d. (b) Show the European call is 4.110. (c) Compute European and American puts.

Generated solution. $h = .25$, $u = e^{.17} = 1.1853049$, $d = e^{-.13} = .8780954$, $p^* = .4625702$. Prices at time 1 are $47.412194, 35.123817$, and terminal prices are $56.197904, 41.632431, 30.842063$. Backward induction gives $C_E = 4.109801$, $P_E = 2.541379$, $P_A = 2.568715$. At the down node the American put exercises for $40 - 35.123817 = 4.876183$, versus continuation 4.824291. European parity checks $C_E - P_E = 40 - 40e^{-.04}$.

13.9 Problem 3.9: three-step values and the exercise decision

Statement. $S_0 = 100, K = 95, r = 8\%$ continuously compounded, $\sigma = 30\%, \delta = 0, T = 1, n = 3$. (a) Verify the American call is 18.283, and that it never exercises early so the European call is equal. (b) Show the European put is 5.979 and verify parity. (c) Verify the American put is 6.678.

Generated solution. $h = 1/3$, $u = 1.2212461, d = .8636926, p^* = .4568067$. The call continuation values always dominate exercise at every pre-expiry node, giving $C_A = C_E = 18.282552$. The European put is 5.978605 and

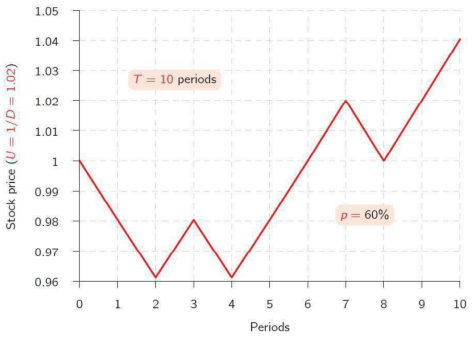
$$18.282552 - 5.978605 = 100 - 95e^{-.08} = 12.303947.$$

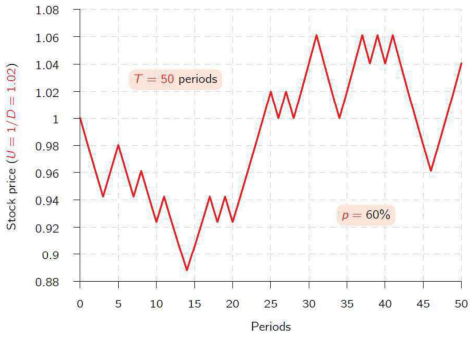
The American put is 6.677901. At the lowest time-2 node $S = 74.596483$, exercise pays 20.403517, exceeding European continuation 17.903663. This benefit is propagated backwards to the root.

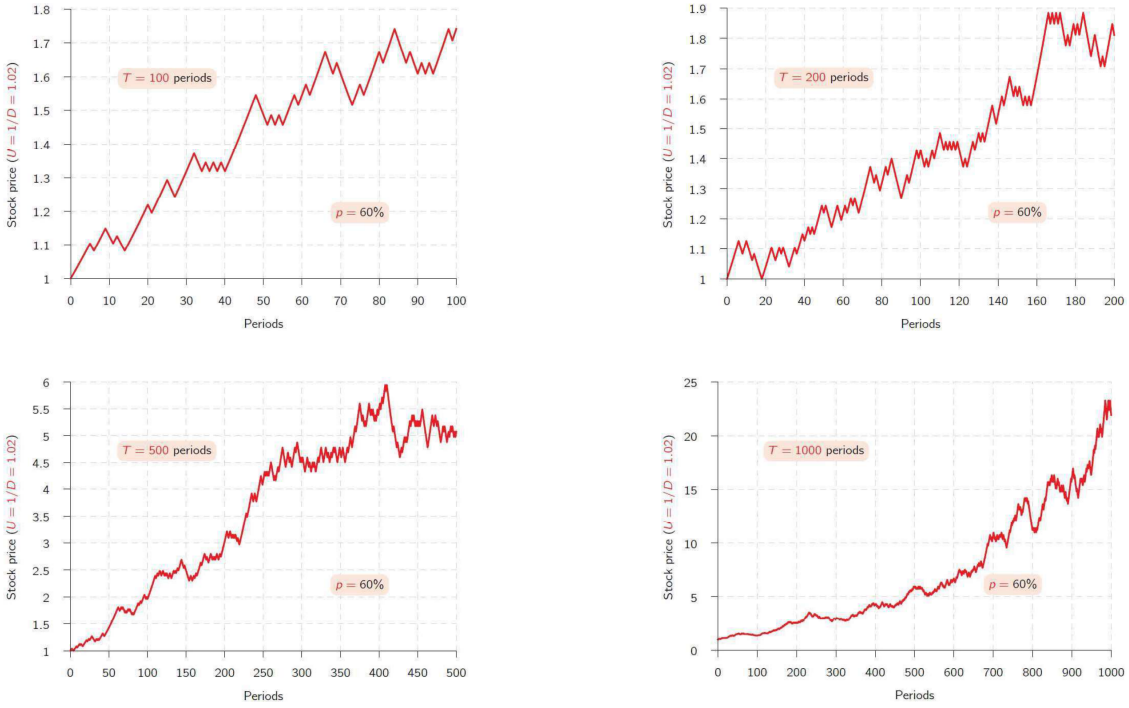
14 From binomial trees to Black–Scholes

14.1 Discrete steps and the continuous-time limit

A binomial model evolves at distinct dates. Continuous time is the limit in which the interval between those dates becomes arbitrarily small. This changes the model's time structure: it does not mean that a finite tree is literally a continuous path. The successive simulated stock paths show what happens when 10, 50, 100, 200, 500 and 1000 increments are used; the illustrated up-move probability is 60%. These paths demonstrate increasing time resolution, rather than supplying a risk-neutral probability estimate.







The axes identify the number of periods and normalised stock price. Increasing the number of draws changes the realised path too, so these examples should not be read as successive observations of the same underlying path.

14.2 Refining the tree changes both the outcomes and their weights

A coarse tree gives only a few terminal prices. Divide the time to maturity T into n periods, $h = T/n$, and recompute

$$u = e^{(r-\delta)h+\sigma\sqrt{h}}, \quad d = e^{(r-\delta)h-\sigma\sqrt{h}}, \quad p^* = \frac{e^{(r-\delta)h} - d}{u - d}.$$

For the original one-year European call $S_0 = 41, K = 40, r = .08, \sigma = .3, \delta = 0$, three periods give $h = 1/3, u = 1.2212, d = .8637, p^* = .4568$, rather than the one-period multipliers 1.4623, .8025. The terminal payoffs and probabilities, with k the number of up moves, are

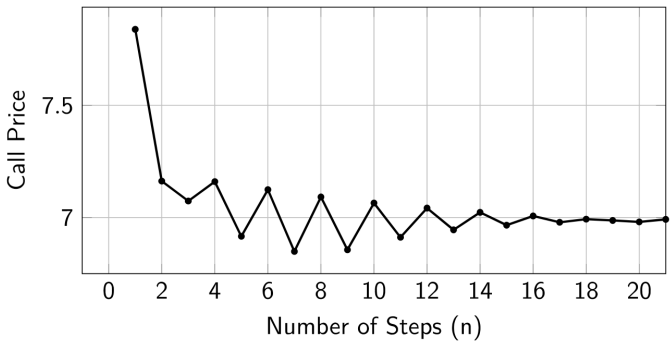
k	Payoff	Probability
3	34.678	$(p^*)^3 = .0953$
2	12.814	$3(p^*)^2(1-p^*) = .3400$
1	0	$3p^*(1-p^*)^2 = .4044$
0	0	$(1-p^*)^3 = .1603$

Thus $C_0 = e^{-.08}[.0953(34.678) + .3400(12.814)] \simeq 7.0739$. For general n,

$$C_0 = e^{-rT} \sum_{k=0}^{n} \binom{n}{k} (p^*)^k (1-p^*)^{n-k} \max\{S_0 u^k d^{n-k} - K, 0\}.$$

The binomial coefficient counts paths reaching the same recombining terminal state. Holding u, d, p^* fixed while increasing n would change the experiment rather than refine its time discretisation.

Number of steps	Call price
1	7.839
2	7.162
3	7.074
4	7.160
10	7.065
50	6.969
100	6.966
500	6.960
∞	6.961

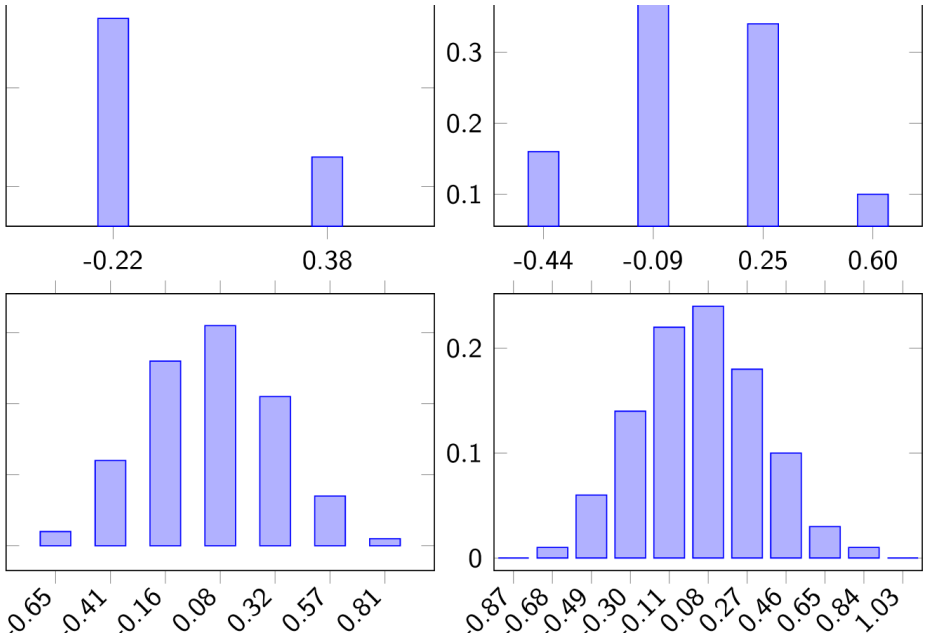


The oscillation around the limiting value is caused by the finite tree's terminal grid relative to the strike. Convergence is not monotone.

Problem 12.2, retained with its results. Use the binomial function for a call with $S_0 = 41, K = 40, \sigma = .3, r = .08, T = .25$ (three months), $\delta = 0$. Calculate $n = 8, 9, 10, 11, 12$ and describe the approximation's behaviour. The respective prices are $3.464, 3.361, 3.454, 3.348, 3.446$. They slowly approach the Black–Scholes price 3.399, with odd/even oscillations. These values were independently recomputed with the course's forward-tree convention.

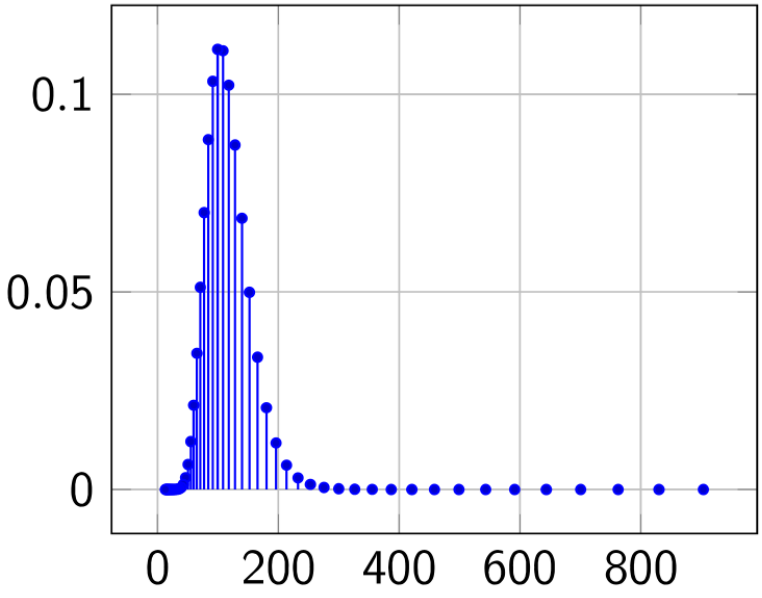
14.3 Normal log returns and lognormal stock prices

The lognormal distribution arises when *continuously compounded returns* are normally distributed. As the tree is refined, the distribution of accumulated log returns approaches a normal distribution. The four panels compare $n = 2, 3, 6, 10$; ten steps already give a recognisable bell-shaped log-return distribution.



Stock prices themselves are exponentials of log returns, so their limiting distribution is lognormal, with positive support and a right tail. A normal stock-price model and a normal log-return model are different assumptions.

Problem 11.13, retained. Let $S_0 = 100, \sigma = .30, r = .08, T = 1, \delta = 0$. Use the terminal binomial formula to calculate each terminal price and its probability, plot the year-1 risk-neutral stock distribution for $n = 50$, and find $\Pr(S_1 < 80)$ and $\Pr(S_1 > 120)$. With $u = 1.0450, d = .9600, p^* = .4894$, each state $S_0 u^k d^{50-k}$ has probability $\binom{50}{k}(p^*)^k(1-p^*)^{50-k}$. Sum the states satisfying the strict inequalities: $\Pr(S_1 < 80) = .2006$ and $\Pr(S_1 > 120) = .2829$.



These are pricing-measure probabilities, not historical forecasts. The unequal tails result from the distribution and the chosen thresholds, even though both thresholds are 20 currency units from the initial price.

14.4 Model assumptions and inputs

- Continuously compounded stock returns are normal and independent over disjoint time intervals; volatility is known and constant.
- Future dividends are known, either as cash amounts in an appropriate treatment or as a fixed yield. The displayed formulas below specifically use a continuous proportional dividend yield δ.
- The risk-free interest rate is known and constant.
- There are no transaction costs or taxes, short selling is costless, and borrowing is possible at the risk-free rate.

The six Black–Scholes inputs are current stock price S_0, strike K, volatility σ, continuously compounded risk-free rate r, maturity T, and dividend yield δ. The finite binomial model has one extra numerical input: number of periods n.

14.5 The limiting derivation: two binomial tails

An option is in the money when $S_0 u^k d^{n-k} > K$. Writing $h = T/n$, this means $k > \alpha$, where

$$\alpha = \frac{n}{2} - \frac{\log(S_0/K) + (r-\delta)T}{2\sigma\sqrt{T/n}}.$$

Separate the payoff into its stock and strike components:

$$\begin{aligned} C_0 &= e^{-rT} \sum_{k>\alpha} \binom{n}{k} (p^*)^k (1-p^*)^{n-k} (S_0 u^k d^{n-k} - K) \\ &= e^{-rT}(S_0 U_1 - K U_2), \\ U_2 &= \sum_{k>\alpha} \binom{n}{k} (p^*)^k (1-p^*)^{n-k}, \\ U_1 &= \sum_{k>\alpha} \binom{n}{k} (p^* u)^k [(1-p^*)d]^{n-k}. \end{aligned}$$

U_2 is the risk-neutral in-the-money probability. The binomial normal-tail approximation is $U_2 \simeq N((np^* - \alpha)/\sqrt{np^*(1-p^*)})$. Substitution of the time-step parameters gives the limit

$$U_2 \longrightarrow N\left(\frac{\log(S_0/K) + (r - \delta - \sigma^2/2)T}{\sigma\sqrt{T}}\right) = N(d_2).$$

For U_1, use the stock-weighted probability

$$q = \frac{p^* u}{p^* u + (1-p^*)d} = \frac{p^* u}{e^{(r-\delta)T/n}}, \qquad 1 - q = \frac{(1-p^*)d}{e^{(r-\delta)T/n}}.$$

Then

$$U_1 = e^{(r-\delta)T} \sum_{k>\alpha} \binom{n}{k} q^k (1-q)^{n-k} \longrightarrow e^{(r-\delta)T} N(d_1),$$

where

$$d_1 = \frac{\log(S_0/K) + (r - \delta + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

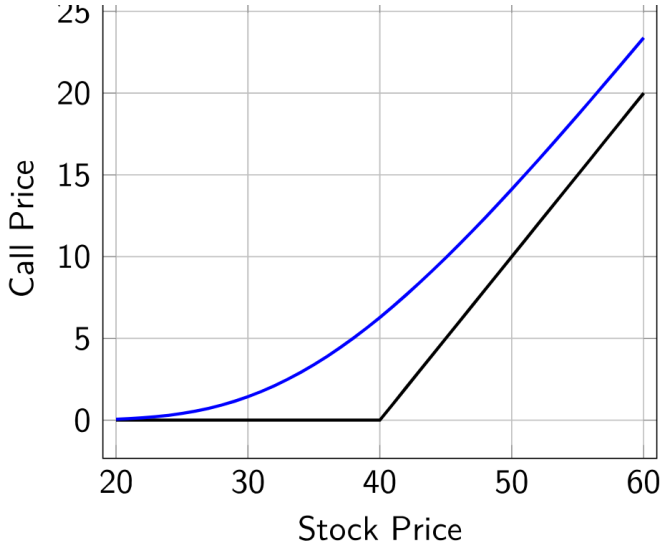
The stock-tail approximation is $U_1 \simeq e^{(r-\delta)T} N((nq - \alpha)/\sqrt{nq(1-q)})$. The factor $e^{(r-\delta)T}$ follows by factoring $[p^* u + (1-p^*)d]^n$. Combining both limiting tails gives

$$\boxed{C_0 = S_0 e^{-\delta T} N(d_1) - K e^{-rT} N(d_2).}$$

$N(x) = \Pr(Z \le x)$ for a standard normal Z. The risk-neutral exercise probability is $N(d_2)$; $N(d_1)$ comes from the stock-weighted tail and is not the same probability under the same measure.

14.6 European call, put, replication and beta

For the original inputs $41, 40, .3, .08, 1, 0$, $d_1 = .4998, d_2 = .1998$, $N(d_1) = .6911, N(d_2) = .5789$ and $C_0 = 6.961$. The stock-price plot compares the smooth pre-expiry call value with its terminal payoff.



The historical expected return on the stock does not enter the formula. The hedge is

$$C_0 = \Delta S_0 + B, \quad \Delta = e^{-\delta T} N(d_1), \quad B = -Ke^{-rT} N(d_2).$$

If the stock has beta β_S and the financing asset has zero beta, the option's beta is

$$\beta_{\text{option}} = \frac{\Delta S_0}{\Delta S_0 + B}\, \beta_S.$$

A larger stock beta raises the option's expected payoff and its required expected-return discount rate. In this idealised replication setting the effects offset: beta is irrelevant to the arbitrage price. A higher historical expected return alone does not justify substituting that return for r in the pricing formula.

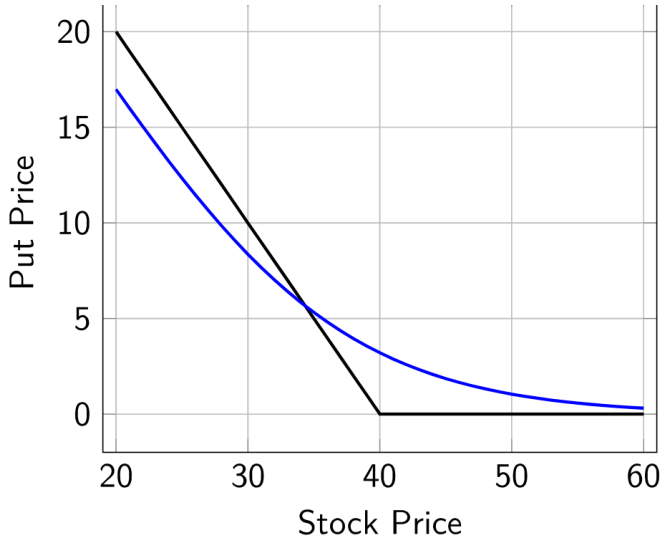
The European put is

$$\boxed{P_0 = -S_0 e^{-\delta T} N(-d_1) + Ke^{-rT} N(-d_2).}$$

Since $N(-x) = 1 - N(x)$,

$$P_0 = C_0 + Ke^{-rT} - S_0 e^{-\delta T}.$$

For the same 41 stock example, $P_0 = 2.886$; a 500-step tree gives approximately 2.885. Parity independently gives $6.961 + 40e^{-.08} - 41 = 2.886$.



Problem 12.20, retained. (a) Price a call with $S_0 = 100, K = 90, \sigma = 30\%, r = 8\%, \delta = 5\%, T = 1$. (b) Price a put with $S_0 = 90, K = 100, \sigma = 30\%, r = 5\%, \delta = 8\%, T = 1$. The supplied spreadsheet outputs are both 17.70; independent evaluation gives 17.698814 for each. Swapping S_0, K and r, δ relates the two formulas; the equality is not evidence that calls and puts with identical inputs generally cost the same.

14.7 Precursors and the replication breakthrough

Before organised options markets developed, many option-valuations concerned warrants and OTC claims. The historical introduction dates modern listed options to April 1973 and mentions a Genesis narrative as an early option analogy. It names Castelli (1877), Bachelier (1900), Sprenkle (1961), Ayres (1965), Boness (1964), Samuelson (1965), Baumol, Malkiel and Quandt (1966), and Chen (1970). The subsequent comparison also includes Samuelson and Merton (1969) and Thorp and Kassouf (1967). Their limitations show why selecting a suitable expected-return discount rate was difficult; the shared missing ingredient is a self-financing riskless hedge.

Bachelier. The printed historical expression, with the strike denoted by K, is

$$C(S,T) = S\Phi\left(\frac{S-K}{\sigma\sqrt{T}}\right) - K\Phi\left(\frac{S-K}{\sigma\sqrt{T}}\right) - \sigma\sqrt{T}\left(\frac{K-S}{\sigma\sqrt{T}}\right).$$

Typography/formula issue. The last term is printed without a normal-density symbol and with the sign shown above. It cannot be silently treated as a verified pricing identity: at $S = K$ the displayed expression gives zero despite positive volatility. The accompanying prose says that negative underlying and option prices are possible and that the time value of money is omitted; the claim about negative *option* prices does not hold for the expectation of a nonnegative call payoff.

Generated clarification. If one specifically assumes $S_T = S + \sigma\sqrt{T}Z$ with Z standard normal and no discounting, directly integrating $(S_T - K)^+$ gives

$$C = S\,\Phi\left(\frac{S-K}{\sigma\sqrt{T}}\right) - K\,\Phi\left(\frac{S-K}{\sigma\sqrt{T}}\right) + \sigma\sqrt{T}\,\phi\left(\frac{K-S}{\sigma\sqrt{T}}\right).$$

Here σ is a price-change volatility, Φ the standard normal CDF and ϕ its density. This generated expectation is nonnegative; it is not a claim that the ambiguous printed last term has been transcribed correctly by replacing it. Normal price levels can be negative. The volatility convention differs from Black–Scholes return volatility, and discounting has been omitted. The handwritten comment credits Bachelier's ideas as ahead of their time and names his *Théorie de la spéculation*.

Sprenkle. The lecture presents

$$C(S,T) = e^{\rho T} S\Phi(d_1) - (1-A)K\Phi(d_2), \quad d_1 = \frac{\log(e^{\rho T}S/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

ρ is expected share growth and A represents risk aversion. Riskless discounting is not included in this expression; the resulting premium is described as too large.

Boness and Samuelson. Boness writes $C = S\Phi(d_1) - Ke^{-\rho T}\Phi(d_2)$ with $d_1 = [\log(S/K) + (\rho + \sigma^2/2)T]/(\sigma\sqrt{T})$. This accounts for time value but still needs the stock's expected return. Samuelson distinguishes stock expected return ρ and the option's expected return α:

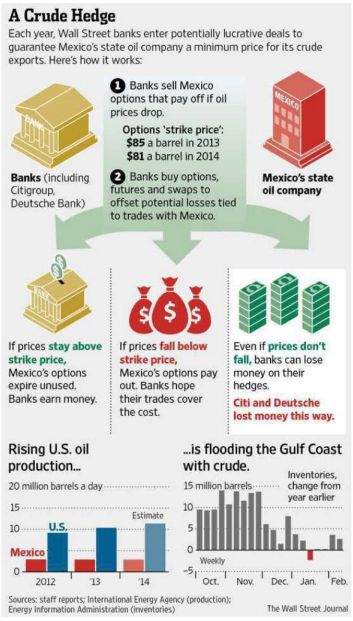
$$C = Se^{(\rho-\alpha)T}\Phi(d_1) - Ke^{-\alpha T}\Phi(d_2), \quad d_1 = \frac{\log(S/K) + (\rho + \sigma^2/2)T}{\sigma\sqrt{T}}.$$

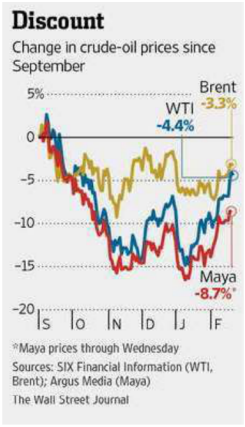
The printed historical expression places an additional minus sign before $(\rho - \alpha)T$. The displayed version here is a **generated algebraic clarification**: expected stock growth $e^{\rho T}$ discounted at $e^{-\alpha T}$ gives $e^{(\rho-\alpha)T}$. The original printed sign is retained in the faithful Notes and flagged rather than silently accepted as a verified identity. The two expected-return inputs remain unknown. Samuelson and Merton derive an option return as a function of stock price using a representative investor's utility, with many conditional assumptions. Thorp and Kassouf approximate warrant prices using linear regression for an exponent; the lecture notes that the hedged position's no-arbitrage expected return should be the riskless rate and that their approach lacked a direct volatility measure.

Black, Scholes and Merton. The breakthrough identifies the continuously rebalanced stock-and-bond replicating portfolio, whose riskless hedged return must match the riskless rate. Merton emphasises the *self-financing* condition and the arbitrage argument. The resulting value no longer depends on the stock's expected return. The lecture writes the familiar no-dividend form $C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2)$ with $d_1 = [\log(S/K) + (r + \sigma^2/2)T]/(\sigma\sqrt{T})$. The historical commentary distinguishes the original two-security hedge from Merton's three-security portfolio involving the option, stock and bank account, $w_C C + w_S S + w_B B$, with self-financing explicitly imposed. It credits Merton's help in the original paper's footnote and a direct comment from Robert Merton to the authors of this historical discussion. Scholes and Merton later received the economics prize; the printed year 1996 is a typo: the official Nobel record gives 1997. Black had died before the award.

14.8 Historical oil hedges: basis risk and feedback

The supplied 2014 news examples discuss bank hedges of Mexican oil-company put protection, crude-price spreads, and the difference between the client's exposure and the futures used to hedge it. The two visuals preserve the contract/hedge chain and the Brent–WTI discount comparison.





The Reuters example describes producers buying put protection with strikes around \$75–85. As oil prices fell toward those strikes, banks sold futures to reduce the changing exposure of their written puts. Traders cited this delta-hedging flow as one factor behind a rapid price fall. The teaching points are the sign and changing size of the hedge, basis risk between different oils, and potential feedback when many hedgers trade together. These historical observations do not establish that hedge trades were the sole cause of the price move.

15 A measure-change derivation of Black–Scholes

15.1 Historical dynamics and the discounted stock

On a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0\le t\le T}, \mathbb{P})$ carrying a standard Brownian motion $W^{\mathbb{P}}$, suppose the non-dividend-paying stock satisfies

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t^{\mathbb{P}}, \qquad \sigma > 0.$$

For constant risk-free rate r, the discount factor is $\beta(t) = e^{-rt}$, so $d\beta(t) = -r\beta(t)dt$. Define $Y_t = \beta(t)S_t$, the stock price in discounted units. The product rule gives

$$\begin{aligned} dY_t &= d(\beta_t S_t) = \beta_t\,dS_t + S_t\,d\beta_t + d[\beta, S]_t \\ &= \beta_t(\mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}) - r\beta_t S_t dt \\ &= (\mu - r)Y_t dt + \sigma Y_t dW_t^{\mathbb{P}}. \end{aligned}$$

The quadratic-covariation term vanishes because β has finite variation. Under $\mathbb{P}$, discounted stock still has drift $\mu - r$.

15.2 Eliminating the drift changes the measure

Set $W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \int_0^t \gamma_s ds$. Under the historical measure this shifted process is generally *not* a martingale; adding a drift does not by itself create another Brownian motion. Substitution gives

$$\begin{aligned} Y_t &= Y_0 + \int_0^t (\mu - r)Y_s ds + \int_0^t \sigma Y_s dW_s^{\mathbb{P}} \\ &= Y_0 + \int_0^t (\mu - r - \sigma\gamma_s)Y_s ds + \int_0^t \sigma Y_s dW_s^{\mathbb{Q}}. \end{aligned}$$

Choosing $\gamma_s = (\mu - r)/\sigma$ removes the drift. We must also find an equivalent measure under which $W^{\mathbb{Q}}$ is a standard Brownian motion. The measure change makes the stochastic-integral representation a martingale representation under the appropriate integrability conditions.

15.3 Radon–Nikodym and Cameron–Martin–Girsanov

For equivalent measures $\mathbb{P}, \mathbb{Q}$, the Radon–Nikodym theorem provides a positive random variable $Z = d\mathbb{Q}/d\mathbb{P}$ with

$$\mathbb{Q}(A) = \mathbb{E}^{\mathbb{P}}[Z\mathbf{1}_A], \qquad \mathbb{E}^{\mathbb{P}}[Z] = 1.$$

Equivalent measures agree on which events have probability zero; they need not assign the same positive probability to other events.

Let γ be predictable and satisfy the sufficient Novikov condition

$$\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{1}{2}\int_0^T \gamma_t^2 dt\right)\right] < \infty.$$

Then the exponential density

$$Z_T = \exp\left[-\int_0^T \gamma_t \, dW_t^{\mathbb{P}} - \frac{1}{2}\int_0^T \gamma_t^2 dt\right]$$

defines a measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ on $\mathcal{F}_T$, and

$$\widetilde{W}_t = W_t^{\mathbb{P}} + \int_0^t \gamma_s ds$$

is a Brownian motion under $(\mathcal{F}_t, \mathbb{Q})$. The sign in the density and the sign of the Brownian drift shift must match this convention. For constant $\gamma = (\mu - r)/\sigma$ over a finite horizon, Novikov is satisfied. Consequently

$$dY_t = \sigma Y_t \, dW_t^{\mathbb{Q}}, \qquad dS_t = rS_t \, dt + \sigma S_t \, dW_t^{\mathbb{Q}}.$$

The stock's risk-neutral drift is r in this *no-dividend* derivation. With a continuous dividend yield the ex-dividend drift is $r - \delta$ instead.

15.4 Computing the call expectation explicitly

Under $\mathbb{Q}$, the solution is

$$S_T = S_0 \exp\left[(r - \tfrac{1}{2}\sigma^2)T + \sigma W_T^{\mathbb{Q}}\right], \qquad W_T^{\mathbb{Q}} \sim \mathcal{N}(0, T).$$

Therefore

$$\begin{aligned} C_0 &= \mathbb{E}^{\mathbb{Q}}[e^{-rT}(S_T - K)^+] \\ &= \mathbb{E}^{\mathbb{Q}}\left[\left(S_0 e^{-\sigma^2 T/2 + \sigma W_T^{\mathbb{Q}}} - Ke^{-rT}\right)^+\right] \\ &= \int_{-\infty}^{\infty} \left(S_0 e^{-\sigma^2 T/2 + \sigma z} - Ke^{-rT}\right)^+ f_T(z)\, dz, \end{aligned}$$

where $f_T(z) = (2\pi T)^{-1/2} e^{-z^2/(2T)}$. The positive-payoff region satisfies

$$\begin{aligned} S_0 e^{-\sigma^2 T/2 + \sigma z} > Ke^{-rT} &\iff \sigma z > \log K - \log S_0 - rT + \tfrac{1}{2}\sigma^2 T \\ &\iff z > -d_2\sqrt{T}, \end{aligned}$$

with $d_2 = [\log(S_0/K) + (r - \sigma^2/2)T]/(\sigma\sqrt{T})$. Thus

$$C_0 = S_0 \int_{-d_2\sqrt{T}}^{\infty} \frac{1}{\sqrt{2\pi T}} \exp\left[-\frac{z^2 - 2T\sigma z + \sigma^2 T^2}{2T}\right] dz$$
$$- Ke^{-rT} \int_{-d_2\sqrt{T}}^{\infty} \frac{1}{\sqrt{2\pi T}} e^{-z^2/(2T)} dz.$$

Complete the square in the first integrand: $z^2 - 2T\sigma z + \sigma^2 T^2 = (z - \sigma T)^2$. Set $u = (z - \sigma T)/\sqrt{T}$ in the first integral and $v = z/\sqrt{T}$ in the second:

$$\begin{aligned} C_0 &= S_0 \int_{-d_2-\sigma\sqrt{T}}^{\infty} \phi(u)du - Ke^{-rT} \int_{-d_2}^{\infty} \phi(v)dv \\ &= S_0[1 - N(-d_2 - \sigma\sqrt{T})] - Ke^{-rT}[1 - N(-d_2)] \\ &= S_0 N(d_1) - Ke^{-rT} N(d_2), \qquad d_1 = d_2 + \sigma\sqrt{T}. \end{aligned}$$

This retains the density theorem, integrability condition, drift-removal calculation, payoff threshold and both changes of integration variable. It also explains why the historical drift μ disappears from the option price.

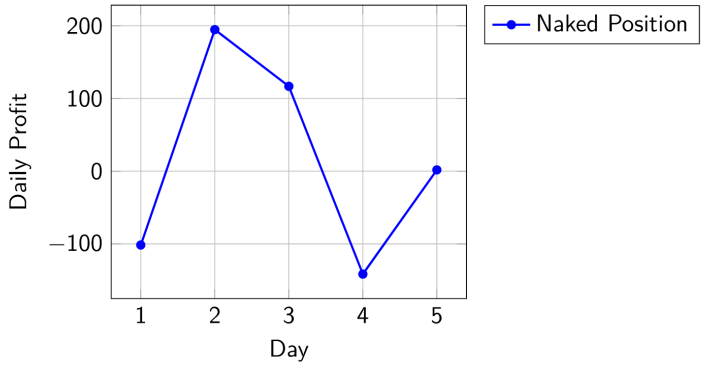
16 Greeks, market-making and dynamic hedging

16.1 An unhedged dealer has an arbitrary position

A market-maker sells to buyers and buys from sellers. The position generated by customer orders can have uncontrolled risk, so the dealer trades to hedge it. Consider selling 100 European calls with $S_0 = 41, K = 40, \sigma = .3, r = .08, \delta = 0$ and 91 days to expiry, initially $\tau = 91/365 = .249$. With no hedge, the stock path and daily option-position changes are:

Day	Stock	Marked value of 100 short calls	Daily profit
0	41	-339.47	—
1	42.5	-441.04	-101.57
2	39.5	-246.31	194.73
3	37	-129.49	116.82
4	40	-271.04	-141.55
5	40	-269.27	1.77

Profit is measured by marking the position to market, as if it were liquidated each day. It is not the terminal exercise payoff. A price increase hurts the short-call seller and a fall helps; the sign follows the short position.

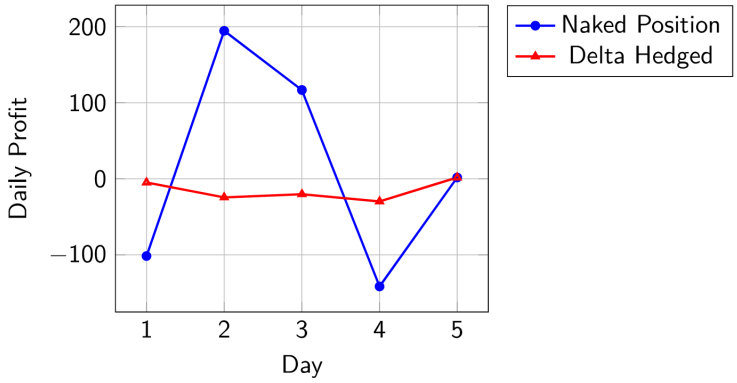


16.2 Delta neutrality and daily rebalancing

Initially one call has delta about .645, so a one-dollar stock move changes its value by about .645. The short 100-call position is offset by approximately 64.54 shares. The delta changes as the stock and remaining life change; the hedge is recalculated and rebalanced each day.

Day	Stock	Call delta	Shares	Call profit	Share profit	Total
0	41	.645	64.54	—	—	—
1	42.5	.730	73.03	-101.57	96.81	-4.77
2	39.5	.548	54.81	194.73	-219.10	-24.38
3	37	.373	37.27	116.82	-137.02	-20.20
4	40	.581	58.06	-141.55	111.82	-29.73
5	40	.580	58.01	1.77	0.00	1.77

The share profit over a day uses the *previous day's* hedge: for example $64.54(42.5 - 41) = 96.81$. Rebalancing at day 1 then sets the shares held over the next day. The displayed totals use underlying unrounded values. Interest expenses are ignored in this illustrative table, as in the supplied example.



Delta hedging offsets the first-order response to *small* stock moves. For larger changes, the change in delta matters. The remaining losses are consistent with a short-gamma position and discrete rebalancing; the table is not a proof of exact continuous-time replication.

16.3 Definitions, signs and reporting units

Greeks are partial derivatives of an option price: vary one input while holding the others fixed. They quantify exposures of market-makers, index-option portfolios and individual investors. Let $\tau = T - t$, and denote a price by V.

Input	Greek	Meaning and convention
S	$\Delta = \partial_S V$	Currency change for a unit stock-price increase.
S	$\Gamma = \partial_{SS} V$	Change in delta for a unit stock-price increase.
S	$\Omega = (S/V)\Delta$	Percentage price change per percentage stock-price change.
σ	Vega $= \partial_\sigma V$	Divide by 100 to report a one percentage point volatility increase.
t	$\Theta = \partial_t V$	Calendar-time derivative with maturity fixed; divide by 365 for per-day theta.
r	$\rho = \partial_r V$	Divide by 100 for a percentage point, by 10,000 for a basis point.
δ	$\Psi = \partial_\delta V$	Divide by 100 for a percentage point dividend-yield increase.

The lecture's table uses daily theta and per-percentage-point vega, rho and psi. Check a terminal or spreadsheet's units before multiplying a Greek by an input change; a decimal-rate derivative and a displayed percentage-point sensitivity differ by a factor of 100.

For a standard European call and put with $\sigma > 0, \tau > 0$:

Input change	Call response	Put response
S rises	Increases, $\Delta_C > 0$	Decreases, $\Delta_P < 0$
S changes	$\Gamma_C > 0$	$\Gamma_P > 0$
S percentage rises	$\Omega_C \geq 1$ for usual r, δ inputs	$\Omega_P \leq 0$
σ rises	Increases, Vega > 0	Increases, Vega > 0
t rises (τ falls)	Generally decreases	Sign can be either
r rises	Increases, $\rho_C > 0$	Decreases, $\rho_P < 0$
δ rises	Decreases, $\Psi_C < 0$	Increases, $\Psi_P > 0$

Higher volatility raises the value of the option's convex payoff. A stock-price increase makes a call more likely to finish in the money and a put less likely. Higher r reduces the discounted strike paid by the call holder and received by the put holder. A call owner receives no stock dividends before exercise, so increasing the dividend yield reduces call value and raises put value. Time decay is *not universal*: a deep-in-the-money call on a stock with a high dividend yield can have positive theta, and a deep-in-the-money put can also gain from the passage of time. The lecture's signs are qualitative for ordinary inputs, rather than a claim that every option always loses value as calendar time advances.

16.4 Analytic formulas

Let

$$d_1 = \frac{\log(S/K) + (r - \delta + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}, \quad \phi(x) = N'(x).$$

$$\begin{aligned}
\Delta_C &= e^{-\delta\tau}N(d_1), & \Delta_P &= -e^{-\delta\tau}N(-d_1),\\
\Gamma_C &= \frac{e^{-\delta\tau}\phi(d_1)}{S\sigma\sqrt{\tau}}, & \Gamma_P &= \Gamma_C,\\
\Omega_C &= \frac{S\Delta_C}{C}, & \Omega_P &= \frac{S\Delta_P}{P},\\
\text{Vega}_C &= Se^{-\delta\tau}\phi(d_1)\sqrt{\tau}, & \text{Vega}_P &= \text{Vega}_C,\\
\rho_C &= K\tau e^{-r\tau}N(d_2), & \rho_P &= -K\tau e^{-r\tau}N(-d_2),\\
\Psi_C &= -\tau S e^{-\delta\tau}N(d_1), & \Psi_P &= \tau S e^{-\delta\tau}N(-d_1).
\end{aligned}$$

The theta expressions, *before* division by 365, are

$$\begin{aligned}
\Theta_C &= \delta S e^{-\delta\tau}N(d_1) - rKe^{-r\tau}N(d_2) - \frac{Se^{-\delta\tau}\phi(d_1)\sigma}{2\sqrt{\tau}},\\
\Theta_P &= \Theta_C + rKe^{-r\tau} - \delta S e^{-\delta\tau}.
\end{aligned}$$

The last equality follows by differentiating put–call parity with respect to calendar time while keeping T fixed. Using time *to maturity* instead would reverse the derivative's sign. Elasticity requires a nonzero option price. The lecture writes the diffusion term in theta as $-Ke^{-r\tau}\phi(d_2)\sigma/(2\sqrt{\tau})$. This is the same term because $Se^{-\delta\tau}\phi(d_1) = Ke^{-r\tau}\phi(d_2)$; both forms retain the same dividend and rate assumptions.

16.5 The complete numerical Greek example

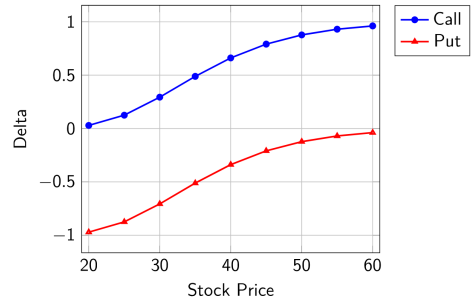
For $S = 41, K = 40, \sigma = .3, r = .08, \delta = 0, \tau = 1$, the displayed example reports:

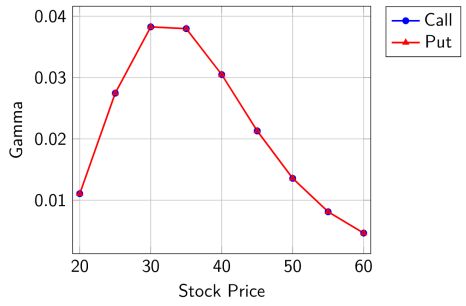
Greek and display unit	Call	Put
Delta	.691	-.309
Gamma	.029	.029
Elasticity	4.071	-4.389
Vega, per percentage point	.144	.144
Theta, per day	-.011	-.003
Rho, per percentage point	.214	-.156
Psi, per percentage point	-.283	.127

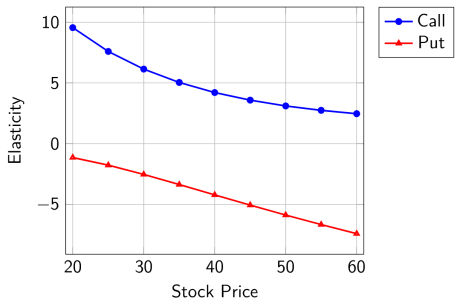
These values preserve the lecture's units and rounding. In particular, Δ is not gamma, and vega is a derivative with respect to *volatility*, not variance.

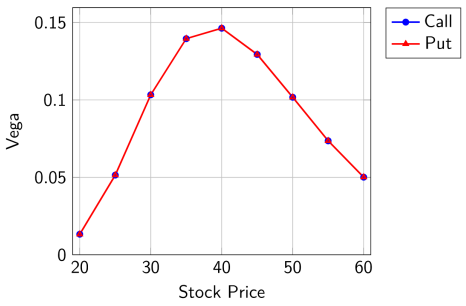
16.6 How the Greek curves change with moneyness

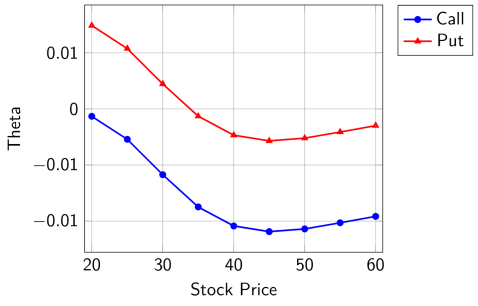
The seven original plots below compare call and put sensitivities as stock price runs from 20 to 60, with the other inputs fixed. Delta rises with S; gamma and vega are largest near the central region; put elasticity is negative; theta can change sign; rho and psi have opposite call/put signs.

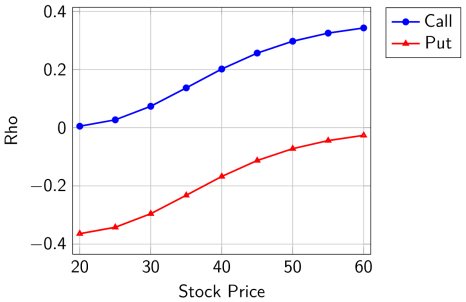


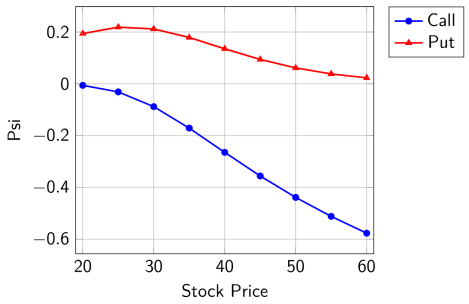












The figures' scales are the display conventions explained above. The Bloomberg scenario illustrations distinguish unhedged, delta-hedged and delta–gamma-hedged exposures; the formulas and tables make their economic content explicit here. No additional terminal screenshot is needed to calculate the hedge.

16.7 Gamma neutrality requires another option

The gamma of a stock holding is zero because its delta is constant at one. Thus shares alone cannot neutralise option gamma. Return to the 91-day, 40-strike written call and choose a second, 120-day call with strike 45. The gamma ratio is

$$\frac{\Gamma_{40,91}}{\Gamma_{45,120}} \simeq 1.1213.$$

Buy approximately 1.1213 of the 45-strike calls for each 40-strike call sold. This ratio is obtained from unrounded gammas; the printed rounded components .0606/.0540 do not reproduce its last digits exactly.

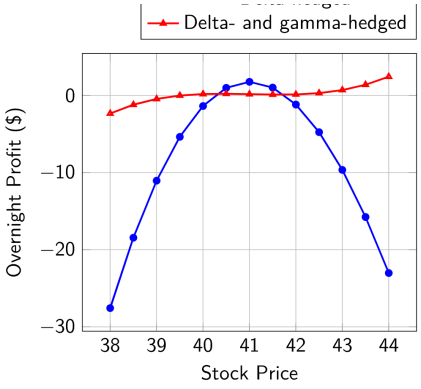
	40-strike call	45-strike call	Combined option position
Price	3.395	1.707	-1.481
Delta	.645	.381	-.218
Gamma	.061	.054	0
Theta per day	-.018	-.014	.002

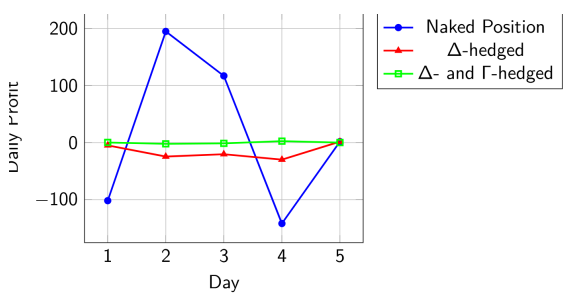
The combined position is one short 40-strike call and 1.1213 long 45-strike calls. Its residual delta is about $-.218$, so add .218 shares per original call, or 21.8 shares for 100 calls, to make both delta and gamma zero at the current inputs.

$$\Gamma_{\text{portfolio}} = -\Gamma_{40} + w\Gamma_{45} = 0, \qquad \Delta_{\text{portfolio}} = -\Delta_{40} + w\Delta_{45} + x = 0.$$

Local Taylor expansion explains the improvement:

$$\Delta V \simeq \Delta_{\text{portfolio}}\Delta S + \tfrac{1}{2}\Gamma_{\text{portfolio}}(\Delta S)^2 + \Theta_{\text{portfolio}}\Delta t.$$



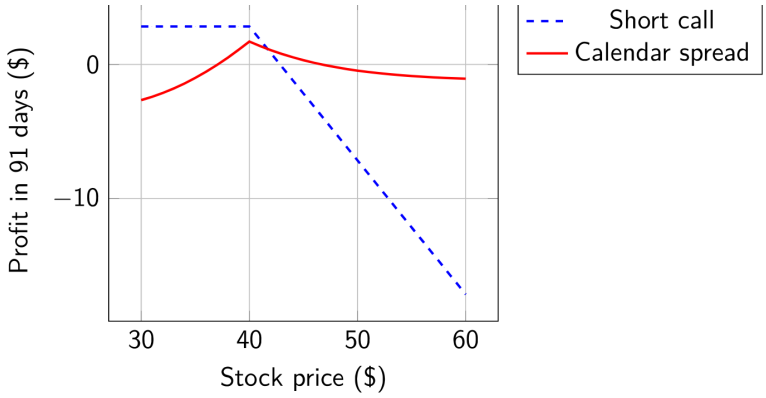


The first graph compares one-day profit as S changes; the second compares profits along the five-day stock path. The delta–gamma hedge is less sensitive to large moves than the delta-only hedge. Its zero sensitivities are *local*; gamma neutrality does not eliminate every finite-move, volatility, funding or model risk. Further rebalancing is still necessary as inputs change.

16.8 Calendar spreads and unequal time decay

To protect against a stock-price increase when selling a call, one may buy another call with the same strike but a later expiry. The short near-expiry call generally loses time value faster than the longer-lived call; with no stock move this difference can favour the spread.

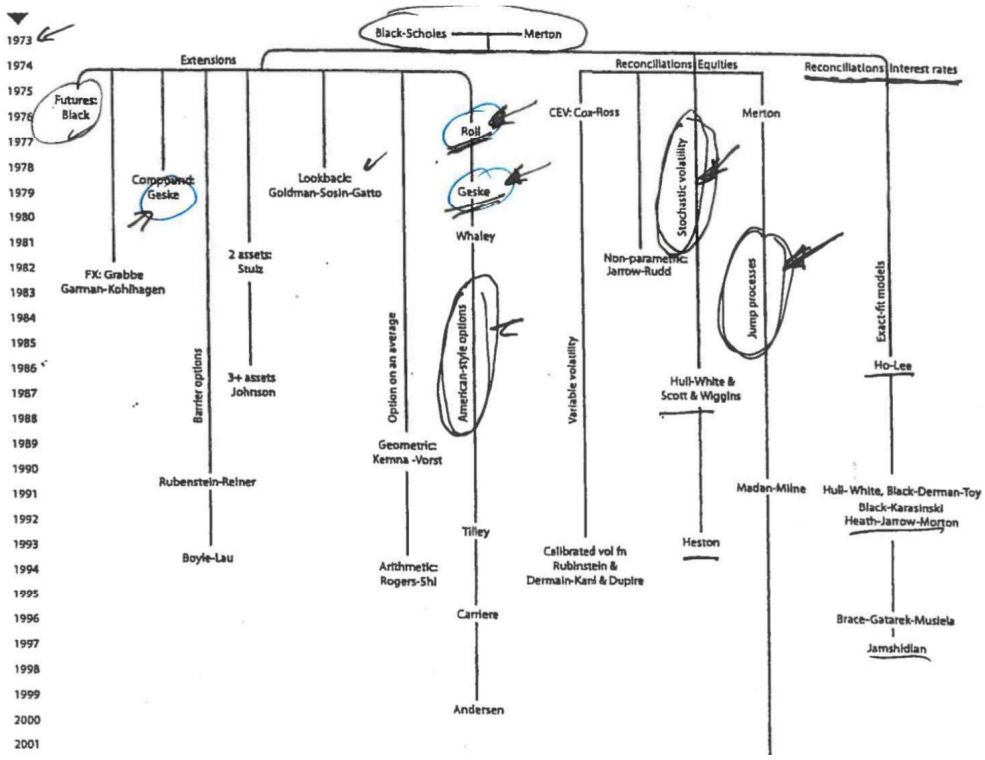
The supplied example sells a 40-strike call with 91 days left and buys a 40-strike one-year call. At $S = 40, r = 8\%, \sigma = 30\%, \delta = 0$, premiums are 2.78 and 6.28, so the initial net debit is about 3.50. Daily thetas are $-.0173$ and $-.0104$ respectively. The spread theta is $+.0069$ per share per day at these inputs: if the stock does not move over one day, the short option loses more value than the long option.



The graph evaluates profit over 91 days as a function of the stock price then. The short call expires at that horizon, but the long call still has remaining life and must be *valued*, not replaced by its final payoff. The "profitable if the stock does not move" explanation concerns time decay at the illustrated inputs; it does not make every calendar spread a guaranteed-profit position.

16.9 Developments after Black–Scholes

The historical diagram places Black–Scholes and Merton alongside later work on stochastic volatility, interest-rate models, jump models and more complex instruments. It is a map of modelling extensions rather than an extra pricing formula.



The original model's constant volatility, frictionless continuous hedge and simple rate/dividend assumptions are the reference point. Extensions change those assumptions and therefore change the hedge and valuation problem.

Part II
Code

17 Reading the MATLAB payoff explorer

The supplied `code1.txt` defines `TradingStrategyGUI`, a MATLAB graphical interface for building portfolios of at most eight positions from stock, a zero-coupon bond, a forward, a European call and a European put. The code is preserved verbatim in the project as `code/TradingStrategyGUI.m`; this section explains its logic rather than replacing it with a new program. It was inspected as source but *not executed in MATLAB* on this PC, so numerical and GUI compatibility are not claimed as tested.

17.1 Inputs, positions and the terminal-payoff curve

The window asks for a stock-price plotting range, today's stock price S_0, continuously compounded interest rate R, volatility σ, and time to expiry T. Each row selects a security type, a strike or bond face value K, and signed units. Positive units buy a position; negative units write or short it. The arrays `pT` and `p0` are the central translation table:

Position	`pT`: payoff at T	`p0`: negative model cost at 0
Stock	S_T	$-S_0$
Bond	K	$-Ke^{-RT}$
Forward	$S_T - K$	$Ke^{-RT} - S_0$
Call	$(S_T - K)^+$	$-c_{BS}(S_0, K, R, T, \sigma)$
Put	$(K - S_T)^+$	$-p_{BS}(S_0, K, R, T, \sigma)$

For each filled row, `evalSchedule` multiplies these expressions by signed units and adds them to two strings. It then replaces symbols with the GUI inputs. The payoff buttons evaluate the combined expression over 100 stock prices between the selected minimum and maximum. The vertical value therefore depends on a hypothetical S_T; the horizontal axis is not time.

17.2 Why the second curve is payoff minus financed cost

In the source code `p0` is negative of each position's model cost. The routine multiplies it by e^{RT}, so its `e` expression is $-e^{RT}C_0$ for total initial cost C_0. The `Cost` and `Payoff & Cost` buttons plot $p + e$:

$$\text{terminal profit in the code's convention} = \text{terminal payoff} - e^{RT} \times \text{initial model cost.}$$

This puts the payoff and premium on the same future date, as in the option examples of Section 3. It is not merely the option's intrinsic payoff. For a multi-leg strategy, signed costs and payoffs of all rows are summed before plotting.

Worked example

To reproduce the long call butterfly in Problem 9, choose a common T and enter three rows: $+1$ call with $K = 90$, -2 calls with $K = 100$, and $+1$ call with $K = 110$. The `Payoff` curve is $(S_T - 90)^+ - 2(S_T - 100)^+ + (S_T - 110)^+$. The adjusted curve subtracts the future value of the net Black–Scholes cost $c_{90} - 2c_{100} + c_{110}$. A payoff diagram can be identified without σ, but the priced-profit curve cannot be drawn without the valuation inputs.

17.3 Pricing and implementation limits

The helper functions `blscall` and `blsput` implement European Black–Scholes formulas using `normcdf`. In particular, they model a non-dividend-paying stock with fixed R, σ and T. The GUI's positive-input callback merely beeps on some invalid values; it does not enforce $S_0, K, T, \sigma > 0$, which the formulas need. Expression strings are evaluated with MATLAB `eval`, so inputs should be trusted and numeric. The code does not model bid–ask spreads, commissions, discrete dividends, early exercise, margin calls or cross-hedge basis risk.

The program's actual on-screen behavior, availability of `normcdf` and required MATLAB toolbox should be checked in a MATLAB installation before treating it as a validated teaching tool. The unmodified 225-line source is included as `code/TradingStrategyGUI.m` in the accompanying Overleaf project and is also reproduced in the source-faithful `Derivatives_Notes`; the study PDF keeps the explanation rather than four pages of tiny code text.

18 Efficient binomial implementation: European and American puts

The PDF listings use $S = 8, K = 10, T = 1, r = 0.03, \sigma = 0.3, M = 256$; the separately supplied script starts from $S = 5, r = 0.06$. These are different numerical examples, not conflicting descriptions of one run. Source code is preserved; the discussion below does not modify its behaviour.

18.1 From backward induction to a terminal weighted sum

Let $h = T/M$, $u = e^{rh+\sigma\sqrt{h}}$, $d = e^{rh-\sigma\sqrt{h}}$ and $p = (e^{rh} - d)/(u - d)$ for the non-dividend example. Number the terminal nodes by $j = 0, \ldots, M$ upward moves:

$$S_{M,j} = S_0 d^{M-j} u^j, \qquad W_j = (K - S_{M,j})^+, \qquad V_0 = e^{-rT} \sum_{j=0}^{M} \binom{M}{j} p^j (1-p)^{M-j} W_j.$$

Code 1 builds the terminal prices by nested updates and rolls option values back with nested loops. Code 2 computes terminal prices directly. Code 3 vectorizes terminal payoffs. Code 4 moves the common discount factor outside the backward recursion for a European payoff. Code 5 vectorizes each rollback. This algebraic movement of discounting uses a constant deterministic rate; do not carry it over indiscriminately to early-exercise decisions. Code 6 evaluates binomial coefficients with `nchoosek`; the source disables warnings and identifies inefficiency. The following implementations avoid forming large factorials.

18.2 Recursive and logarithmic probabilities

Define $a_j = \binom{M}{j} p^j (1-p)^{M-j}$. Then

$$a_0 = (1-p)^M, \qquad a_j = a_{j-1} \frac{M-j+1}{j} \frac{p}{1-p}, \quad j = 1, \ldots, M.$$

Code 7 uses `cumprod`. Code 8 computes

$$\log a_j = \log(M!) - \log(j!) - \log((M-j)!) + j \log p + (M-j) \log(1-p)$$

with cumulative sums of logarithms, reducing overflow/underflow problems associated with factorials. The addendum uses $k = j+1$; its initial binomial coefficient should be read as $\binom{M}{0} = 1$, rather than the printed $\binom{M}{1} = 1$. For Code 9, nonzero put payoffs satisfy $S_0 d^{M-j} u^j < K$. At $u > d > 0$ this is

$$j < \frac{\log(K/(S_0 d^M))}{\log(u/d)}.$$

Only these nodes need to be summed. At exact equality the payoff is zero. The source cutoff expression and clamping are preserved in its listing; no claim is made that every numerical edge case of that implementation is validated.

18.3 Black–Scholes comparison and early exercise

Code 10 provides the European put formula:

$$P_0 = K e^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1), \quad d_1 = \frac{\log(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

The binomial value converges towards the continuous-time value as the discretization is refined under the stated assumptions; finite-M equality is not required. Code 11 values an American put by taking the maximum of immediate exercise and discounted continuation at each rollback. It precomputes powers of u, d and discounted probability weights, but retains the early-exercise comparison.

18.4 Reported computational times

These are the source's historical times, not benchmarks measured on this PC:

Code	1	2	3	4	5	6	7	8	9	10	11
Seconds	.0102	.0073	.0062	.0042	.0068	.1232	.0039	.0036	.0033	.0006	.0092

Codes 1–10 refer to the European put and Code 11 to the American put. Vectorization changes implementation cost; it does not change the payoff or the no-arbitrage pricing rule. The original PDF and separate MATLAB script are the provenance for these examples.

18.5 Supplied comparison script

```
clc

% Parameters
S = 5;
K = 10;
T = 1;
r = 0.06;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%

% CODE 1 ==========================================================
tic

W = zeros(M+1,1);
W(1) = S;

% Asset prices at time T
for i = 1:M
for n = i:-1:1
W(n+1) = u*W(n);
end
W(1) = d*W(1);
end

% Option values at time T
for n = 1:M+1
W(n) = max(K-W(n),0);
end

% Re-trace to get option value at time zero
for i = M:-1:1
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end

t1 = toc
disp('Option value is'), disp(W(1))
%==========================================================================
clear W

% CODE 2 ==========================================================
tic

W = zeros(M+1,1);
% Asset prices at time T

for n = 1:M+1
W(n) = S*(d^(M+1-n))*(u^(n-1));
end

% Option values at time T
for n = 1:M+1
W(n) = max(K-W(n),0);
end

% Re-trace to get option value at time zero
for i = M:-1:1
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end

t2 = toc
disp('Option value is'), disp(W(1))
%==========================================================================
clear W
```

```
% CODE 3 ============================================================
tic

% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% Re-trace to get option value at time zero
for i = M:-1:1
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end

t3 = toc
disp('Option value is'), disp(W(1))
%=====================================================================
clear W

% CODE 4 ============================================================
tic

% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% Re-trace to get option value at time zero
q = 1-p;
for i = M:-1:1
for n = 1:i
W(n) = p*W(n+1) + q*W(n);
end
end
W(1) = exp(-r*T)*W(1);

t4 = toc
disp('Option value is'), disp(W(1))
%=====================================================================
clear W

% CODE 5 ============================================================
tic

% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% Re-trace to get option value at time zero
q = 1-p;
for i = M:-1:1
W = p*W(2:i+1) + q*W(1:i);
end
W(1) = exp(-r*T)*W(1);

t5 = toc
disp('Option value is'), disp(W(1))
%=====================================================================
clear W

% CODE 6 ============================================================
tic

% nchoosek produces numerical warning
warning off

% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% Binomial coefficients ( nchoosek =  (n!)/(k!(n-k)! )
v = zeros(M+1,1);
for k = 1:M+1
v(k) = nchoosek(M,k-1);
end

value = exp(-r*T)*sum(v.*(p.^([0:M]')).*((1-p).^([M:-1:0]')).*W)';

t6 = toc
```

```
disp('Option value is'), disp(value)
%=========================================================================
clear W v value

% CODE 7 ==================================================================
tic

% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% Recursion for coefficients
a = cumprod([(1-p)^M,(p/(1-p))*[M:-1:1]./[1:M]]);

value = exp(-r*T)*sum(a'.*W);

t7 = toc
disp('Option value is'), disp(value)
%=========================================================================
clear W a value

% CODE 8 ==================================================================
tic

% Option values at time T
W = max(K-S*d.^([M:-1:0]').*u.^([0:M]'),0);

% log/cumsum version
csl = cumsum(log([1;[1:M]']));
tmp = csl(M+1) - csl - csl(M+1:-1:1) + log(p)*([0:M]') + log(1-p)*([M:-1:0]');

value = exp(-r*T)*sum(exp(tmp).*W);

t8 = toc
disp('Option value is'), disp(value)
%=========================================================================
clear W csl tmp value

% CODE 9 ==================================================================
tic

% cut-off index (for nodes > z, option payoff is zero)
z = max(1,min(M+1,floor( log((K*u)/(S*d^(M+1)))/(log(u/d)) )));

% Option values at time T
W = K - S*d.^([M:-1:M-z+1]').*u.^([0:z-1]');

% log/cumsum version using cut-off index z
tmp1 = cumsum(log([1;[M:-1:M-z+2]'])) - cumsum(log([1;[1:z-1]']));
tmp2 = tmp1 + log(p)*([0:z-1]') + log(1-p)*([M:-1:M-z+1]');

value = exp(-r*T)*sum(exp(tmp2).*W);

t9 = toc
disp('Option value is'), disp(value)
%=========================================================================
clear W tmp1 tmp2 value

% CODE 10 =================================================================
tic

d1 = (log(S/K) + (r + 0.5*sigma^2)*T)/(sigma*sqrt(T));
d2 = d1 - sigma*sqrt(T);
N1 = 0.5*(1+erf(-d1/sqrt(2)));
N2 = 0.5*(1+erf(-d2/sqrt(2)));

value = K*exp(-r*T)*N2 - S*N1;

t10 = toc
disp('Option value is'), disp(value)
%=========================================================================
clear d1 d2 N1 N2 value

% CODE 11 =================================================================
tic
```

```
% Re-usable computations
dpowers = d.^([M:-1:0]');
upowers = u.^([0:M]');
scale1 = p*exp(-r*dt);
scale2 = (1-p)*exp(-r*dt);

% Option values at time T
W = max(K - S*dpowers.*upowers,0);

% Re-trace to get option value at time zero
for i = M:-1:1
Si = S*dpowers(M-i+2:M+1).*upowers(1:i);
% Check for early exercise
W = max(max(K - Si, 0), scale1*W(2:i+1) + scale2*W(1:i));
end

t11 = toc
disp('Option value is'), disp(W)
%=========================================================================
clear dpowers upowers scale1 scale2 Si W


%=========================================================================
tt = [t1 t2 t3 t4 t5 t6 t7 t8 t9 t10];
fprintf('\n\nComputational time in seconds for a European put option:\n\n');
for i=1:10,
    fprintf('code %2.0d:\t %2.4f\n',[i tt(i)]);
end
fprintf('\n\nComputational time in seconds for an American put option:\n\n');
fprintf('code 11:\t %2.4f\n',t11);
```

18.6 All eleven source addendum listings

The PDF listings below remove displayed line numbers, replace typographic apostrophes by ASCII apostrophes and visible space glyphs by spaces. The original PDF is the authority for any extraction doubt; the separate `BinomialComparison.m` remains byte-identical to `code2.txt`.

Addendum Code 1

```
% Code 1: Binomial method for a European put.
%
% Unvectorized version.
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
W = zeros(M+1,1);
W(1) = S;
% Asset prices at time T
for i = 1:M
for n = i:-1:1
W(n+1) = u*W(n);
end
W(1) = d*W(1);
end
% Option values at time T
for n = 1:M+1
W(n) = max(K-W(n) ,0);
end
% Re -trace to get option value at time zero
for i = M:-1:1
```

```
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end
disp('Option value is'), disp(W(1))
```

Addendum Code 2

```
% Code 2: Binomial method for a European put.
%
% Initial tree computation improved.
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
W = zeros(M+1,1);
% Asset prices at time T
for n = 1:M+1
W(n) = S*(d^(M+1-n))*(u^(n-1));
end
% Option values at time T
for n = 1:M+1
W(n) = max(K-W(n) ,0);
end
% Re-trace to get option value at time zero
for i = M:-1:1
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end
disp('Option value is'), disp(W(1))
```

Addendum Code 3

```
% Code 3: Binomial method for a European put.
%
% Partially vectorized
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Option values at time T
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% Re-trace to get option value at time zero
for i = M:-1:1
for n = 1:i
W(n) = exp(-r*dt)*(p*W(n+1) + (1-p)*W(n));
end
end
disp('Option value is'), disp(W(1))
```

Addendum Code 4

```
% Code 4: Binomial method for a European put.
%
% Partially vectorized , improved
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Option values at time T
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% Re -trace to get option value at time zero
q = 1-p;
for i = M:-1:1
for n = 1:i
W(n) = p*W(n+1) + q*W(n);
end
end
W(1) = exp(-r*T)*W(1);
disp('Option value is'), disp(W(1))
```

Addendum Code 5

```
% Code 5: Binomial method for a European put.
%
% Partially vectorized , improved
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Option values at time T
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% Re -trace to get option value at time zero
q = 1-p;
for i = M:-1:1
W = p*W(2:i+1) + q*W(1:i);
end
W(1) = exp(-r*T)*W(1);
disp('Option value is'), disp(W(1))
```

Addendum Code 6

```
% Code 6: Binomial method for a European put.
%
% Partially vectorized
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
```

```
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% nchoosek produces numerical warning
warning off
% Option values at time T
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% Binomial coefficients ( nchoosek =  (n!)/(k!(n-k)! )
v = zeros(M+1,1);
for k = 1:M+1
v(k) = nchoosek(M,k-1);
end
value = exp(-r*T)*sum(v.*(p.^([0:M]')).*((1 -p).^([M: -1:0] ')).*W) ';
disp('Option value is'), disp(value)
```

Addendum Code 7

```
% Code 7: Binomial method for a European put.
%
% Vectorized (recursive coefficients )
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Option values at time T
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% Recursion for coefficients
a = cumprod ([(1-p)^M,(p/(1-p))*[M: -1:1]./[1:M]]);
value = exp(-r*T)*sum(a'.*W);
disp('Option value is'), disp(value)
```

Addendum Code 8

```
% Code 8: Binomial method for a European put.
%
% Vectorized (log recursive coefficients )
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Option values at time T
W = max(K-S*d.^([M: -1:0] ').*u.^([0:M]') ,0);
% log/cumsum version
csl = cumsum(log ([1;[1:M]']));
tmp = csl(M+1) - csl - csl(M+1: -1:1) + log(p)*([0:M]') + log(1-p)*([M: -1:0] ');
value = exp(-r*T)*sum(exp(tmp).*W);
disp('Option value is'), disp(value)
```

Addendum Code 9

```
% Code 9: Binomial method for a European put.
%
% Vectorized (log recursive coefficients and null terms)
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% cut -off index (for nodes > z, option payoff is zero)
z = max(1,min(M+1, floor( log((K*u)/(S*d^(M+1)))/(log(u/d)) )));
% Option values at time T
W = K - S*d.^([M:-1:M-z+1]').*u.^([0:z-1] ');
% log/cumsum version using cut -off index z
tmp1 = cumsum(log ([1;[M:-1:M-z+2] ’])) - cumsum(log ([1;[1:z-1] ’]));
tmp2 = tmp1 + log(p)*([0:z-1]’) + log(1-p)*([M:-1:M-z+1] ’);
value = exp(-r*T)*sum(exp(tmp2).*W);
disp(’Option value is’), disp(value)
```

Addendum Code 10

```
% Code 10: Black -Scholes formula for a European put.
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
%
d1 = (log(S/K) + (r + 0.5* sigma ^2)*T)/( sigma*sqrt(T));
d2 = d1 - sigma*sqrt(T);
N1 = 0.5*(1+ erf(-d1/sqrt (2)));
N2 = 0.5*(1+ erf(-d2/sqrt (2)));
value = K*exp(-r*T)*N2 - S*N1;
disp(’Option value is’), disp(value)
```

Addendum Code 11

```
% Code 11: Binomial method for an American put.
%
% Parameters
S = 8;
K = 10;
T = 1;
r = 0.03;
sigma = 0.3;
M = 256;
dt = T/M;
u = exp(r*dt + sigma*sqrt(dt));
d = exp(r*dt - sigma*sqrt(dt));
p = (exp(r*dt)-d)/(u-d);
%
% Re -usable computations
dpowers = d.^([M: -1:0] ’);
upowers = u.^([0:M]’);
scale1 = p*exp(-r*dt);
scale2 = (1-p)*exp(-r*dt);
% Option values at time T
W = max(K - S*dpowers .* upowers ,0);
```

```
% Re -trace to get option value at time zero
for i = M:-1:1
Si = S*dpowers(M-i+2:M+1).* upowers (1:i);
% Check for early exercise
W = max(max(K - Si , 0), scale1*W(2:i+1) + scale2*W(1:i));
end
disp('Option value is'), disp(W)
```

A Course references

Class materials:

Slides

Textbooks:

1. Robert McDonald, Derivatives Markets, Pearson Addison Wesley
2. John Hull, Options, Futures, and Other Derivatives, Prentice Hall

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Historical market statistics

The BIS series provide historical context on exchange-traded and OTC markets. Contract count, notional amount and market value measure different objects. For optional historical data, see `http://www.bis.org/statistics/extderiv.htm` and `http://www.bis.org/statistics/derstats.htm`.